

Game of Flows: Algal functional groups & their response to the presence and absence of grazers in Texas intermittent riverine pools

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Intermittent rivers and ephemeral streams (IRES) in Texas are experiencing frequent and prolonged droughts due to climate change, altering flow regimes and creating environmental stress for aquatic communities. IRES rely on algal assemblages to support food webs within remnant pools that persist through dry periods. Algae use various strategies (e.g., mucilage production, filamentous growth, and resting stages) to help withstand hydric stress. Intermittent flows and associated remnant pools also shape the distribution and abundance of fish and macroinvertebrates, which can further influence the composition and abundance of algal assemblages. This collaborative project employed a fish-grazer exclusion experiment to investigate how fish grazing dynamics and intermittent hydrology influence algal functional groups in Texas riverine pools. We hypothesized that transitions between flowing and drying phases would favor assemblages dominated by cyanobacteria or diatoms with traits like mucilage production. Alternatively, lower water levels, which concentrate nutrients during transitional phases, may favor taxa tolerant of higher nutrient levels or shifts in nutrient ratios. Release from grazing pressure in the presence of invertebrate-grazing could increase algal abundance, or grazing pressure in the presence of macroinvertebrates could reduce algal abundance, especially for nutritious groups like diatoms, and increase in grazer-resistant algae (cyanobacteria). We sampled epilithic algae in remnant pools in the presence and absence of fish grazers during the wet (July) and dry (October) periods in 2024. We used light microscopy and FlowCam® imaging to characterize assemblage structure, functional groups (e.g., motile, low or high profile, filaments), and the relative abundance of major taxa (cyanobacteria, diatoms, green algae). Preliminary results indicate that mucilage-rich cyanobacterial filaments and diatoms dominated across all grazer treatments and hydrologic phases. The relative abundance of diatoms increased in the presence of fish, which may indicate reduced macroinvertebrate grazing pressure. Prolonged drying conditions showed increased surface filamentous green algae. Findings suggest that algal assemblages in Texas intermittent rivers are adapted to fluctuating hydrology but remain sensitive to shifts in grazer presence and pool connectivity. Together, this work enhances our understanding of how climate-driven changes in flow and food-web interactions influence the formation of algal functional groups and ecosystem dynamics in ephemeral waters.

Benthic Macroinvertebrates and Juvenile Winter Run Chinook Salmon in Sacramento River Side Channels: Indicators of Juvenile Salmonid Habitat Quality

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The Sacramento River has had a series of restoration projects in recent years poised at restoring habitat for salmon spawning and juvenile rearing. This project focuses on one of those restoration sites, the Kapusta island complex. This complex is a series of interconnected human made and naturally carved channels in the Sacramento River, restored for the purpose of Winter Run Chinook salmon conservation. However, scant information exists on the quality of these sites, namely that of benthic macroinvertebrates. Thus, we compared community structures of restored channels versus non-restored channels on the island and outlined methods for improving habitat. Samples were collected following a modified Surface Water Ambient Monitoring Program (SWAMP) protocol accounting for micro-habitat. 600 benthic macroinvertebrates were subsampled from samples and taken to their farthest suitable epithet, usually genus or species. Preliminary data suggests higher abundances of Glossomatidae and Chironomidae in restored channels. Non-restored channels contain higher counts of oligochaetes and gastropods. Plecoptera have been infrequent in either channel with one *Claassenia sabulosa* found in the restored channel. We project restored channels to hold a starkly different environment from non-restored channels. While either channel type presents habitat for juvenile salmonids, restored channels provide a higher quality habitat. Neither channel presents suitable communities for juvenile salmonids. Changes to test in future studies may include higher diversification of substrates, increased introduction of wood debris, and increased channel restriction points to improve dissolved oxygen levels.

The effects of heavy metal contaminants on negative phototaxis behaviors in a common mayfly

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I investigated water quality in an ecological context at two locations along the New River in southwest Virginia with similar land use, with the exception of being downstream and upstream of the Radford Army Ammunition Plant (RAAP). Freshwater macroinvertebrates at a low trophic level, such as Ephemeroptera (mayflies), are sensitive to environmental changes, making them known bioindicators, especially of heavy metals. I predicted that 1) water samples collected downstream of the RAAP will have greater concentrations of heavy metals and 2) nymphal mayflies collected downstream will exhibit decreased negative phototaxis responses and have a smaller length-mass ratio than nymphal mayflies collected upstream of RAAP. There is evidence to support that increased concentrations of some metals, such as iron and cadmium, can be correlated to changes in bioturbation and escape behaviors in mayflies. Additionally, it has been found that increased ion concentration may decrease growth rates in mayflies. Water samples were collected and analyzed for metal ion content using Inductively Coupled Plasma Mass Spectrometry (ICP-MS), revealing increased downstream concentrations of Li, Mg, Ca, S, and Sr. I collected Heptageniidae, a common mayfly

scraper, from upstream and downstream locations, then measured the negative phototaxis response by using a green light source to create a light intensity gradient in shallow water in an enclosed space in a lab environment. I measured the net distance that individual mayflies traveled along this gradient using manual video tracking. I found that sampling location was not a significant factor in distance traveled. The length and mass of the mayflies were measured and showed that mayflies from upstream were less massive per length and I found regression constants using a power law model. Relationships from both locations agreed with published length-mass relationships. I failed to reject some parts of the alternative hypothesis, but showed differences in length-mass relationship that may suggest lower biomass in the downstream location. The study suggests greater investigation into the physiological and ecological effects of manufacturing discharge on key aquatic macroinvertebrates.

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Unravelling environmental influences on relative biological integrity across national freshwater assessments using interpretable machine learning

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Reliable assessment of freshwater ecosystem condition requires a biological framework that remains interpretable across space and time without dependence on regional calibration. This study develops a universal and ecoregion-independent relative bioassessment using National Rivers and Streams Assessment (NRSA) data. In this approach, eight widely macroinvertebrate-metrics were applied, each providing complementary structural and functional ecological signals. A percentile-based Index of Biotic Integrity (IBI) was constructed to support the relative bioassessment, avoiding predefined regional indices and absolute ecological thresholds. All sites were classified as good (top 25%), average (middle 75%) and poor (bottom 25%). The study employed year-specific Generalized Linear Models, Random Forests and eXtreme Gradient Boosting (XGboost) models that consistently identified non-linear relationships between key physical and water-chemistry drivers and IBI. Furthermore, Shapley Additive Explanations (SHAP)-based attribution from XGBoost models identified a stable set of influential stressors that persisted across years despite regional heterogeneity. To quantify uncertainty and sensitivity by Monte Carlo simulation, an XGBoost model was trained exclusively with physical and water-chemistry drivers to predict IBI. The prediction performed moderately due to the intentional exclusion of responsive biotic metrics from the model. Meanwhile, scenario-based simulations revealed a clear leftward shift of IBI distribution under realistic degraded conditions compared with baseline variability. Importantly, this framework enables the identification of regionally agnostic driver ranges beyond which biological condition exhibits measurable decline, independent of predefined ecological thresholds. Thus, the model framework provides a transferable and policy-relevant approach for interpreting biological change under real-world stressors without depending on absolute thresholds.

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Spring Into Summer, Crawl Into Fall: How Seasonality Affects the Results of Macroinvertebrate Bioassessments

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This assesses the correlation between seasonality in macroinvertebrate assemblage structure and its implications for rapid bioassessment results. Changes have been observed in the composition of the macroinvertebrate assemblage across seasons at a long-term monitoring site in North Alabama, which inspired this project. This has led to varying results of stream conditions (e.g., excellent vs. good vs. fair) across seasons at that site. Due to funding restraints, most biomonitoring programs can only afford to take one sample per site per year, which could lead to improper classification of stream quality. By understanding the effects of seasonality on macroinvertebrate bioassessments in our region, biologists would have more information to choose the best time of year to conduct sampling efforts. This study will be conducted in the Uchee Creek Watershed, a tributary to the Chattahoochee River and a Strategic Habitat Unit for Alabama which contains many rare, threatened, and endangered aquatic species. We collected samples during the summer, fall, and winter seasons from 10 sites in first and second order streams downstream of road-stream crossings. Twenty sampling efforts were completed using a D-Net to survey all available habitat types over a 100 meter reach. Collected macroinvertebrates were preserved in 70% ethanol and identified to Order in the laboratory. The taxon richness, percentage of the Dominant Taxon, Bioassessment Score, and other metrics were calculated for each sample. We used ordinations such as nonmetric multidimensional scaling to examine the differences between macroinvertebrate assemblages across the seasons.

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Bioassessment post-introduction of endangered darters in the Muskingum River Watershed (Ohio, USA)

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Freshwater ecosystems face significant biodiversity losses, with fish species such as the Tippecanoe darter (*Etheostoma tippecanoe*) being particularly vulnerable due to their sensitivity to habitat degradation and pollution. Reintroduction efforts are increasingly employed to restore extirpated populations, but their success depends on the suitability of receiving habitats. This study evaluates environmental and biological parameters at reintroduction sites within the Muskingum River drainage in Ohio (USA) to determine habitat suitability for Tippecanoe darters. Six sites across four rivers (Kokosing, Walhonding, Muskingum, and Tuscarawas) were sampled for water chemistry, algae, macrophytes, macroinvertebrates, and fish communities. Multivariate analyses (CCA, NMDS, MRPP, ISA) revealed two primary site clusters. Group A (Kokosing and Walhonding Rivers) exhibited

significantly higher dissolved oxygen, algal richness and density, and Tippecanoe darter abundance, alongside lower turbidity and macrophyte cover compared to Group B (Muskingum and Tuscarawas Rivers). Indicator taxa such as Hydropsychidae and Isonychiidae, known for their sensitivity and value as prey for darters, were also more prevalent in Group A. These findings suggest that smaller, cooler, and higher-quality tributaries, such as the Kokosing and Walhonding Rivers, provide more suitable conditions for Tippecanoe darter survival and reintroduction. Future analyses will include further incorporation of fish community data. This research supports the use of comprehensive bioassessment in identifying viable habitats for conservation and offers key insights for future darter reintroduction strategies.

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A dual-framework approach to stream bioassessment: morphology and DNA metabarcoding

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Streams are vital components of landscapes, providing essential ecosystem services such as nutrient cycling, biodiversity support, and habitat connectivity. However, agricultural intensification, land-use change, and climate variability continue to threaten their ecological integrity. Effective biomonitoring is therefore critical for maintaining water quality and sustaining aquatic biodiversity. Traditional morphology-based identification of macroinvertebrates (e.g., Ephemeroptera, Plecoptera, and Trichoptera) remains central to freshwater bioassessment but is constrained by the need for taxonomic expertise, time-intensive processing, and limited detection sensitivity. High-throughput sequencing technologies, including DNA metabarcoding of bulk tissue and environmental DNA (eDNA), can overcome many of these limitations. This ongoing study integrates morphology-based identification with DNA metabarcoding to evaluate stream health across impacted and reference sites within the Ochlockonee River Basin in southwest Georgia and northwest Florida. Both bulk macroinvertebrate and eDNA water samples were analyzed using the cytochrome oxidase I (COI) gene primers (mICOLintF/jgHCO2198). Physicochemical parameters (dissolved oxygen, turbidity, pH, among others) were also measured at each site. Preliminary results indicate the detection of 8 families and 17 genera of Ephemeroptera (5 families, 9 genera), Plecoptera (1 family, 5 genera), and Trichoptera (2 families, 3 genera) across study sites. Morphology-based identification recovered a greater number of taxa than DNA metabarcoding, while molecular approaches provided faster processing times and finer taxonomic resolution, revealing community differences between impacted and reference streams. Leptoceridae were detected exclusively at reference sites, whereas Leptophlebiidae were observed only at impacted sites, highlighting biodiversity patterns associated with stream condition. These findings highlight the value of integrating DNA-based and traditional approaches in freshwater biomonitoring. Combining molecular precision with established morphology-based frameworks enhances accuracy, reduces processing time, and uncovers hidden biodiversity, strengthening data-driven water quality management and aquatic conservation efforts.

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Characterizing fluorescence signatures of grass and silver carp milt for environmental detection

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Invasive Asian carp have posed persistent ecological and management challenges in U.S. waterways since their introduction in the 1970s. Species such as grass and silver carp alter aquatic ecosystems by disrupting food webs through sediment resuspension and competition with native organisms. Effective management would benefit from improved tools for detecting carp presence and reproductive activity in near real time. Fluorescent spectrophotometry, widely used to characterize dissolved organic matter and contaminants in aquatic systems, may provide a novel approach for detecting carp milt as an indicator of spawning activity. However, the fluorescent properties of fish milt and its detectability relative to background water remain largely unexplored.

This study evaluated whether invasive carp milt exhibits distinct fluorescence signatures that can be differentiated from well water. Grass and silver carp milt samples were provided by the U.S. Geological Survey Central Midwest Water Science Center and analyzed at multiple dilutions using a Cary Eclipse fluorescence spectrometer. Synchronous excitation–emission scans were collected and processed in RStudio to generate three-dimensional excitation–emission contour plots. Samples were reanalyzed after 24 hours to evaluate changes in fluorescence intensity over time.

Results indicate that carp milt fluorescence is distinguishable from well water and that grass and silver carp display different relative fluorescence patterns. Fluorescence intensity decreased after samples were left overnight, suggesting temporal degradation of detectable signals. While these findings are preliminary, they demonstrate the potential for fluorescence-based detection of carp biological material. If reproducible, this approach could support the development of in-situ sensors to improve monitoring and management of invasive carp populations.

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Advancing wetland forecasts for restoration planning: Can integrating flow better predict food web performance targets?

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Ecological forecasting provides a framework for evaluating ecosystem restoration by simulating alternative management scenarios and predicting their ecological consequences. In wetlands, restoration efforts have focused on hydroperiods, but recent studies have shown that water flow has been an overlooked driver of nutrient availability and consumer production. Yet, the ability of flow to predict consequences of management changes has never been assessed. The Florida Everglades is a managed ecosystem that is undergoing the largest restoration project in the world. Current objectives of Everglades restoration include increasing wet season discharges across the southern portion of the ecosystem to match historic wet-season water velocities (2–5 cm/s). One indicator of restoration success is the production of small fish and crayfish that are the preferred prey items of wading bird populations in south Florida. We used a long-term monitoring data set (2018–2025) to model the densities of total fish and the two of the most common species in Everglades National Park (Bluefin Killifish, Least Killifish) using antecedent hydrological covariates. We used a cross-validation procedure with a least absolute shrinkage and selection operator (LASSO) regression to compare the change in fish density predictive performance (mean squared error) when including discharge to a baseline hydrologic model. Including discharge reduced mean-squared error by 8 – 10% for all fish density response variables and discharge had a positive effect in all cases. Using the final LASSO model that included discharge, we predicted fish densities to a simulated baseline hydrologic scenario and two restoration scenarios. The two restoration scenarios we assessed differed by the spatial extent of an underground seepage wall along the eastern boundary of ecosystem (including Everglades National Park [ENP]). We calculated the relative change in fish density between the baseline scenario and the two restoration scenarios. Both restoration scenarios increased discharges in ENP by 20 – 50 %, resulting in similar relative increases in total fish density and density of Least Killifish and Bluefin Killifish. These findings show that discharge is a critical variable for predicting responses in fish density and that further flow restoration is expected to benefit fish production.

Open Data, Diatoms, and Decision-Making: A Pilot Application of a National Diatom MMI Using NEON Diatom Data

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Freshwater practitioners rely on biological monitoring to assess stream condition, yet the broader adoption of diatom-based assessment remains limited by challenges of accessibility, standardization, and scalability. Diatoms are used as bioindicators because they integrate cumulative ecosystem responses to environmental change over time. However, applying diatom-based indices at broad spatial scales often requires specialized taxonomic expertise, substantial analytical capacity, and access to compatible data infrastructures, constraining their implementation beyond research contexts. Recent syntheses of biomonitoring practice highlight the need for open, interoperable data and transferable assessment frameworks that translate biological information into decision-relevant metrics (Yates et al., 2025). The National Ecological Observatory Network (NEON) provides long-term, open-access, taxonomically standardized diatom data that directly respond to these challenges. The National Diatom Multi-Metric Index (MMI) developed by Carlisle et al. (2022) offers a standardized framework for converting diatom community data into ecologically interpretable measures of stream condition that also respond to these challenges to support practitioners. In this pilot study, we evaluate whether NEON diatom data can be operationalized within the Carlisle et al. national MMI framework to assess stream condition across the conterminous United States. Using this approach, we generate a national map of diatom-based stream condition across NEON aquatic sites. This work assesses whether integrating an existing national diatom MMI with an open-access ecological data infrastructure can reduce barriers to implementation and support more scalable applications of diatom-based biomonitoring. This pilot study represents the first stage of a longer-term effort toward refining diatom-based MMIs at smaller regional scales using NEON data to increase MMI performance and, in future phases, integrating social data alongside biological assessments. Incorporating the social context has the potential to enhance interpretability and relevance for identifying disparities in monitoring coverage and restoration prioritization. Ultimately, this work contributes to ongoing efforts to develop accessible biomonitoring tools that support ecologically and equitably informed freshwater decision-making.

1. Carlisle, D. M., Spaulding, S. A., Tyree, M. A., Schulte, N. O., Lee, S. S., Mitchell, R. M., & Pollard, A. A. (2022). A web-based tool for assessing the condition of benthic diatom assemblages in streams and rivers of the conterminous United States. *Ecological Indicators*, 135, 108513. <https://doi.org/10.1016/j.ecolind.2021.108513>.
2. NEON (National Ecological Observatory Network). Periphyton, seston, and phytoplankton collection (DP1.20166.001), RELEASE-2026. <https://doi.org/10.48443/758r-c486>. Dataset accessed from <https://data.neonscience.org/data-products/DP1.20166.001/RELEASE-2026>.
3. Yates, A. G., Brua, R. B., Culp, J. M., Aguiar, F. C., Ajayan, A. P., Aspin, T., Bundschuh, M., Calderón, M. R., Csabai, Z., Dallas, H., Datry, T., Silva, K. D., Dzavi, J., England, J., Erős, T., Gebler, D., Goedkoop, W., González-Ferreras, A. M., Hamilton, D. P., ... Wanderi, E. W. (2025). Charting a course for freshwater biomonitoring: The grand challenges identified by the global scientific community. *Ecological Indicators*, 176, 113646. <https://doi.org/10.1016/j.ecolind.2025.113646>.

Biological assessment of a grazing impacted first-order stream in central California by undergraduate students

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The conservation and management of freshwater ecosystems are of the utmost importance to the well-being and long-term interests of life on earth. An important step toward protecting our rivers and streams is to expand monitoring efforts. Tracking the

abundance and diversity of benthic macroinvertebrate (BMI) species offers an effective way to assess how well a stream's ecosystem is functioning and its ability to provide essential ecosystem goods and services. In September 2025, the freshwater ecology class of California State University Monterey Bay conducted a bioassessment of a 150 meter reach of Agua Mala Creek, near the confluence of Rana Creek, to determine the ecological health of the stream. Agua Mala Creek is a first-order stream with a length of 1.91 km. It is situated within Rana Creek Preserve in the upper Carmel Valley in central California. This bioassessment followed the California SWAMP Standard Operating Procedures (SOP) for wadeable streams. The California Stream Condition Index (CSCI) was used to quantify stream conditions in California based on biological and physical data. The CSCI 0.47 score indicates that the stream is very likely altered. The MMI 0.63 score suggests that the stream's ecological structure and function are dominated by species that tolerate degraded conditions, while the O/E 0.31 metric indicates that the number of species present is lower than what would be expected in a healthy stream system. In general, the poor CSCI score suggests that Agua Mala Creek might be more biologically productive were not for decades of disturbed landscape spanning a range of physical habitat degradation, primarily associated with livestock grazing, the possibility of over-pumping groundwater from the numerous wells on-site, and some altered channel geomorphology: including erosion and sediment pollution. Removing cattle from the stream corridor is recommended to reduce bank erosion, sediment input, and nutrient loading, allowing riparian vegetation and stream habitat to recover. In addition, continued collaboration with the Esselen Tribe—on whose traditional homeland Rana Creek Ranch is located—should be prioritized.

INVURTS & NEON: Macroscale vs local effects on stream macroinvertebrate community composition

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The INVURTS project (Investigations with NEON: variability of macroinvertebrates versus urban and rural temperature dynamics) is looking at similarities and differences among stream macroinvertebrate communities at the continental scale and how those are related to stream temperature variability and thermal regime. The National Ecological Observatory Network (NEON) provides a wealth of data from less impacted sites throughout the US. Here, we focus on macroinvertebrate community similarity to investigate the effects of ecoregion (NEON Domain) and of watershed urbanization on similarities among communities. Using multivariate methods, we compared data from less impacted sites (NEON sites) and from sites with more watershed development (preliminary data from INVURTS sites). We hypothesized that communities within ecoregions (NEON domains) will generally be more similar to each other, but that more urban sites lose this ecoregion effect and become more similar to each other regardless of ecoregion due to the homogenizing effects of urbanization.

In general, the expected pattern was observed. Among the less impacted sites, some ecoregions, including the Northeast (NEON Domain 1), Mid Atlantic (2), and Appalachians & Cumberland Plateau (7), had very similar macroinvertebrate communities, whereas others, including the Southern Plains (11), Prairie Peninsula (6), and Ozarks Complex (8) had more distinctive assemblages. In our preliminary INVURTS dataset, the more developed INVURTS sites happened to be in the Mid Atlantic, and they were similar in community composition to the less impacted NEON sites in the Eastern USA (1, 2, and 7). However, when comparing within the Eastern USA region only, the more impacted Mid-Atlantic INVURTS sites did seem to be more similar to each other than to other sites in the Eastern USA, suggesting homogenization from urbanization could be occurring. These results somewhat support our hypotheses, but more data, especially from more developed sites in other ecoregions across the USA, will be illuminating. Future investigations will look more directly at the role of temperature variability in shaping the ecoregion and urbanization patterns.

Carbon in the Capillaries: DOC Lability from Ice-Wedges to Rivers in Arctic Alaska

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Permafrost regions store large quantities of organic carbon that have been stored, frozen for millennia. Rapid warming across the Arctic is destabilizing these reservoirs through permafrost thaw. Melting runoff increases hydrologic connectivity between terrestrial pools and aquatic systems. In ice-wedge tundra, thaw-driven subsidence reorganizes surface flowpaths, which creates capillary networks that transport dissolved organic carbon (DOC) from soils into streams and rivers. How reactive the DOC exported from these thaw features is, and how its lability changes as water moves downstream, remain poorly understood. This study investigated how DOC concentration and composition change when flowing from degrading ice-wedge troughs through capillary networks and into the mainstem of the Jago River on Alaska's North Slope. Surface water and ice-wedge meltwater

were collected from a range of landscape positions during the summer thaw season. Laboratory incubation experiments were used to quantify microbial DOC consumption, while the optical absorbance of dissolved organic matter was analyzed as a proxy for DOC composition along the hydrologic network. We hypothesized that ice-wedge troughs export more biologically labile DOC than downstream sites because the DOC originates from ancient frozen organic carbon sources. As water moves from thaw features through capillary networks to the Jago River mainstem, microbial processing and mixing with downstream DOC sources are expected to reduce overall lability, thereby lowering biological consumption rates. Incubation experiments revealed absorbance coefficient spectra consistently higher CDOM absorbance in ice-wedge trough and capillary network sites relative to mainstem river samples, with systematic decreases in absorbance downstream. These downstream shifts in CDOM optical properties indicated changes in dissolved organic matter composition along the flowpath. These results support that small permafrost thaw features affect how dissolved organic carbon is processed as water moves through Arctic headwaters.

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Climate Change and Decomposition: How Will Temperature and Precipitation Affect Decomposition Rates in Rivers Under Future Climate Scenarios?

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Decomposition plays a key role in carbon cycling, as it determines how long carbon remains stored in recalcitrant organic matter, before it is converted into more labile forms like CO₂. Changes in temperature and precipitation due to climate change have the potential to alter ecosystem processes such as decomposition. Predicting the effects of climate change on decomposition rates in freshwater streams could be a critical step toward elucidating how organic matter availability will change under future climate scenarios. This study used the global CELLDEX decomposition data set and an existing boosted regression tree (BRT) model to predict stream decomposition rate at watershed scales under two climate scenarios: shared socioeconomic pathway 126 (SSP126) and shared socioeconomic pathway 585 (SSP585). Three climate variables extracted from Earth System Models (ESMs) predicting conditions in 2100 were used as inputs in the BRT model: mean annual air temperature, temperature range and mean annual precipitation. We compared predicted changes in decomposition rates in equatorial Lake Victoria and temperate Lake Superior watersheds, as well as across smaller watersheds in five different biomes from polar to equatorial. A sensitivity analysis was conducted to determine which climate variable(s) had the greatest impact on decomposition rate in each watershed. There was a general trend of increased decomposition rates under both shared socioeconomic pathways in all but one of the watersheds studied. However, the sensitivity analysis showed that the changes in decomposition rate and the degree of impact of each climate variable were regionally specific. The combined change in temperature and temperature range explained most of the variation in decomposition rates. However, precipitation change was the most important climate variable in the boreal Agashashok River watershed. Additionally, temperature change was the most important in the Lake Superior watershed, possibly due to maritime influence from the lake causing sustained periods of high temperatures. The results of this study highlight the sensitivity of a key ecosystem process to climate change and provide biome-specific guidance on important climate drivers.

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Permafrost thaw causes hydrologic change in interior Alaskan upland headwater streams.

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Climate change impacts high-latitude ecosystems, accelerating permafrost thaw, altering hydrology, and affecting carbon cycling. In interior Alaska, we examined the influence of permafrost thaw on dissolved organic carbon (DOC) flux and temporal variation in carbon age across upland headwater streams in the Caribou-Poker Creeks Research Watershed (CPCRW), a boreal forest watershed underlain by discontinuous permafrost. Radiocarbon (¹⁴C) analysis, excitation-emission matrices (EEMs), and end-member mixing model analyses were used to understand DOC sources and transport in subcatchments of the CPCRW. Samples were collected across discharge and permafrost extent gradients. Initial $\Delta^{14}\text{C}$ -DOC results indicate that DOC near headwaters is significantly older than DOC downstream, with $\Delta^{14}\text{C}$ -DOC ranging 4,000 to <6,000 years among headwaters, and 2,000 to >4,000 years further downstream. Fluorescence spectroscopy, was used to identify fluorophores associated with terrestrial or microbial sources and track temporal and spatial variations in DOC composition in response to environmental factors such as wildfire, hydrologic change, and permafrost thaw. DOC flux was calculated as the product of stream discharge and DOC concentration to evaluate overall trends in carbon export. To test if carbon fluxes have changed, the measured concentrations of DOC, cation, and anion tracers were compared to values predicted by Bayesian end-member mixing models. To assess long-term hydrological pattern shifts, 20 years of water chemistry data from the Bonanza Creek Long-Term Ecological Research site will be analyzed. An initial linear mixed-effects model on DOC data from the summers of 2001-2023 showed an average yearly decrease in DOC of -8.3 μM ($p < 0.001$).

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Carbon cycling in a novel floodplain under wet-dry conditions

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Wetlands play a critical role in carbon cycling, but the dynamics of carbon storage and release in engineered wetlands that experience wet and dry phases remain poorly understood. I examined carbon biogeochemical dynamics in stream-wetland complexes created through a novel floodplain reconnection method called 'Stage 0' or 'dynamic alluvial valley restoration' in western Pennsylvania. Three Stage 0 restored and three traditionally restored sites were compared to explore how wet-dry cycles

influence soil moisture and soil carbon. At each of the six sites, I collected soil cores along two cross-sectional transects representing the hydrological gradient at the site and quantified terrestrial leaf litter inputs from traps set out in fall 2024. Leaf litter was sorted, identified, and weighed to determine the dominant contributors to leaf litter entering the systems. Percent soil moisture was measured gravimetrically to determine patterns of water storage in the channel and riparian zone compared to drier upland sites. A laboratory experiment was conducted with soil samples from different moisture zones to quantify the amount of carbon, phosphorus, and nitrogen exported during a simulated flood pulse. Soils were prepped by homogenizing dried soils and sieving 150 g of sample through 6 sizes of sieves. Soil particle size distribution was recorded for each soil. HOBO data loggers in place at the sites were used to collect continuous hydrological measurements and models of the hydrologic returns of the sites were generated to improve the resolution of the model for the headwater sites sampled. STIC sensors were deployed during the initial sampling dates in fall 2024. Comparing data acquired from the deployment of STIC sensors in summer and fall 2025 to the hydrological data for the sites in the same year provides more insight into the hydrological connectivity of Stage 0 sites compared to traditionally restored sites. By expanding soil characteristics and hydrology, I hope to better identify the patterns of soil moisture and soil nutrient leachates. This work provides valuable information on how the hydrology of a reconnected floodplain system affects carbon cycling, and may assist in managing the ecosystem services, including carbon sequestration, of future wetland restoration efforts.

A nitrogen river continuum model

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The River Continuum Concept (RCC; Vannote et al., 1980) is a historically and ecologically important framework for predicting community structure, energetic flow, and geomorphic characteristics of lotic ecosystems as they increase in size. While the RCC is helpful in making predictions of stream community ecology, source of energy, and the balance between primary production and respiration, it does not encompass the change in nitrogen (N) balances along the continuum. Here we asked the question: how does the stoichiometry of total dissolved nitrogen (TDN) and dissolved organic matter (DOM) change along Stahler stream orders gradients and across environmental gradients? We used stream chemistry data from five watersheds in different biomes that each included a range of 0 to 4th order streams to assess the elemental composition of TDN and the change in DOM C:N ratios (as DOC: DON ratios) along a river continuum. Preliminary results from the northern temperate ecosystem site indicate that while DON dominated the composition of TDN, relative abundance of NH_4^+ was greater than that of NO_3^- in 1st and 2nd order streams, while the opposite was true of 3rd and 4th order streams. These data suggest changes in nitrogen cycle reaction rates with increasing stream order. Additionally, DOC:DON ratios significantly decreased as stream order increased pointing to changes in how the nutrient fraction of DOM is used along the RCC. Understanding how the availability and processing of nitrogen changes along a stream order gradient can better inform energy flow and biogeochemical cycling in the context of the RCC.

1. Robin L. Vannote, G. Wayne Minshall, Kenneth W. Cummins, James R. Sedell, and Colbert E. Cushing. 1980. The River Continuum Concept. *Canadian Journal of Fisheries and Aquatic Sciences*. 37(1): 130-137. <https://doi.org/10.1139/f80-017>

Concentration-Discharge Hystereses Reveal the Dominant Hydrological and Biological Controls on Nitrate Concentration Dynamics along the Upper Clark Fork River, Montana, USA

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The Upper Clark Fork River (UCFR) in western Montana is part of the largest Superfund project in the United States and has undergone extensive restoration following decades of mining-related contamination. Monitoring associated with ongoing floodplain restoration activities, including contaminated sediment removal and floodplain reconnection, provides a unique opportunity to examine long-term nitrate dynamics in a heavily managed river system. This study examines how the coupling of seasonal ecosystem uptake with a snowmelt-driven hydrologic regime influences temporal and longitudinal patterns of nitrate concentrations in the UCFR. We analyzed nitrate as NO_3^- -N from 13 sites spanning ~200 km of the UCFR, sampled biweekly to monthly in a growing dataset from 2017 to 2024. We hypothesized that the primary controls on seasonal nitrate patterns are flushing or dilution during the snowmelt season and uptake of nutrients by algae during the growing season, and that spatial differences in nitrate loading from groundwater produce distinct concentration-discharge (C-Q) relationships along the river. We present consistent evidence from the hystereses of C-Q curves at most monitoring sites that nitrate concentrations rapidly decrease, likely due to dilution at peak flows; remain low during decreasing and warm low-flow conditions during the algal growing season with increased biological demand; and increase during cooling low flows due to higher concentrations in groundwater inputs and decreasing ecosystem uptake. The spread of these hystereses appears to increase in upstream segments of the river, where groundwater nitrate loading is stronger, and then decrease downstream where groundwater nitrate loading is weaker. Comparisons of two UCFR tributaries representing contrasting land-use influences provide some evidence that sites with very low nitrate loading from groundwater may exhibit little to no hysteresis and either chemostasis or a modest flushing

effect of higher nitrate concentrations at high flows. Together, these results demonstrate how hydrology, nitrate loading, and biological uptake interact to control nitrate dynamics in a recovering river system with a snowmelt-dominated hydrologic regime.

1. Beegum, S., Malakar, A., Ray, C., & Snow, D. D. (2023). Importance of snowmelt on soil nitrate leaching to groundwater: A model study. *Journal of Contaminant Hydrology*, 255, Article 104163. <https://doi.org/10.1016/j.jconhyd.2023.104163>.
2. Fazekas, H. M., Royer, T. V., & Bernhardt, E. S. (2020). Dissolved organic carbon and nitrate concentration–discharge behavior across scales: Land use, excursions, and misclassification. *Water Resources Research*, 56.
3. Ohte, N., Sebestyen, S. D., & Kendall, C. (2004). Tracing sources of nitrate in snowmelt runoff using a high-resolution isotopic technique. *Journal of Geophysical Research*, 109(G4).
4. Pellerin, B. A., Saraceno, J., Shanley, J. B., Sebestyen, S. D., Aiken, G. R., Wollheim, W. M., & Bishop, K. (2011). Taking the pulse of snowmelt: In situ sensors reveal seasonal, event, and diurnal patterns of nitrate and dissolved organic matter variability in an upland forest stream. *Journal of Geophysical Research*, 116(G3).
5. Piatek, K. B., Mitchell, M. J., Silva, S. R., Kendall, C., & McHale, P. J. (2005). Sources of nitrate in snowmelt discharge: Evidence from water chemistry and stable isotopes. *Water, Air, & Soil Pollution*, 165, 13–35. <https://doi.org/10.1007/s11270-005-4641-8>.
6. Valett, H. M., D'Andrilli, J., DeGrandpre, M. D., Payn, R. A., & Peipoch, M. (2017). LTREB: Collaborative research – River ecosystem responses to floodplain restoration (NSF Grant No. DEB-1655197). University of Montana.
7. Valett, H. M., Peipoch, M., & Poole, G. C. (2022). Nutrient processing domains: Spatial and temporal patterns of material retention in running waters. *Freshwater Science*, 41(2), 195–214. <https://doi.org/10.1086/719991>.

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Nitrate budget for the Nyack Floodplain reach of the Middle Fork Flathead River

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The rates and mechanisms through which rivers transport nutrients reflect the health and productivity of an ecosystem. Understanding nitrogen processing and removal is essential in the context of anthropogenic inputs. Nitrate often infiltrates watersheds through fertilizer runoff or decaying organic matter, and excess concentrations can drive eutrophication and algal blooms. Floodplains in gravel bed rivers are highly productive and have high surface water-groundwater exchange, potentially increasing nitrate processing. Here we developed a nitrate budget for the 15-km Nyack floodplain reach of the Middle Fork Flathead River. While most studies focus on small streams, single-station approaches, or tracer-methods, we sought to quantify nitrate dynamics at a larger spatial scale through a two-station, high-frequency monitoring approach. We established primary monitoring sites upstream and downstream of the Nyack along the main channel of the Middle Fork and in a groundwater fed spring stream. Additionally, we sampled four intervening tributaries to account for external nitrate inputs contributing to downstream variation. Nitrate concentrations were measured using submersible ultraviolet nitrate analyzers (SUNAs) deployed for 8d alongside deployed alongside dissolved oxygen loggers, while discharge was quantified through use of an acoustic doppler current profiler. To quantify the budget, we calculated average nitrate mass flux at each site and calculated contribution based on relative location to the floodplain. Results indicated minimal net change in nitrate flux between upstream and downstream sites, with measured downstream flux exceeding expected values by only 7.9% (0.219 mol/s measured vs. 0.203 mol/s expected), suggesting the Nyack floodplain was neither a large net source nor sink of nitrate during our sampling period. No net change does not mean nitrate is unreactive; future research should estimate the gross fluxes in NO₃⁻, specifically as they relate to hyporheic and groundwater interactions within the Nyack.

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Determining InStream Transport and Natural Complexity of eDNA: Using model eDNA methods to determine the drivers of the fate of genetic material in streams

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Environmental DNA (eDNA) analysis is widely used as a low-impact, low-cost method for detecting sensitive species in aquatic systems. The application of eDNA techniques has been vital in a variety of biomonitoring, particularly with efforts to detect the presence of endangered or invasive species. While the fate of eDNA within lentic systems can be correlated with species abundance, there is greater difficulty interpreting eDNA “hits” in lotic environments. Moreover, successful mesocosm studies on eDNA decay and transport have brought to light the need to repeat these experiments in natural systems. Therefore, we describe the “DISTANCE” project, which aims to understand how abiotic factors, including pH, turbidity, water chemistry, and temperature, impact how far eDNA is transported in a natural stream. Using a method like constant-rate nutrient additions, we plan to conduct monthly eDNA injections of Atlantic Herring (*Clupea harengus*) material at the Mayfield Creek aquatic National Ecological Observatory (NEON) sites of central Alabama. We will establish rates of eDNA degradation and transport and leverage available NEON data to determine the dominant drivers of eDNA fate. We will replicate our experimental approach at NEON sites across multiple ecoregions of the contiguous U.S. Overall, the DISTANCE project will establish the connections between stream properties and the fate of eDNA, which remains a significant challenge for its effective use and interpretation.

Overlooked Nitrogen Dynamics in Reservoir Sediments: Insights from Phosphorus Flux Studies

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Excessive nutrient inputs from agricultural and urban fertilizer use and inadequate wastewater treatment have led to nutrient accumulation in freshwater systems. Reservoirs slow downstream nutrient transport, retaining nutrients in sediments and biomass and promoting internal cycling of legacy nutrients. Consequently, reservoirs are often hotspots of algal and cyanobacterial growth associated with degraded water quality and harmful algal blooms. While intact sediment core experiments are commonly used to quantify internal phosphorus (P) fluxes, internal fluxes of dissolved nitrogen (N) species are frequently overlooked, despite their potential importance, often due to assumptions that external N inputs dominate eutrophication or the availability of alternative methods to assess N dynamics, such as stable isotope tracer studies.

In this study, we quantified nitrate+nitrite-N (NO_x-N) and ammonium (NH₄-N) fluxes using data from historical sediment P-flux experiments conducted across reservoirs in Northwest Arkansas and Eastern Oklahoma. Although these prior studies focused on dissolved P dynamics, concentrations of NO_x-N and NH₄-N were also measured. Intact sediment cores were collected from reservoirs between 2015 and 2025 and incubated in the dark, at ~20 °C under oxic and anoxic conditions. Changes in dissolved NO_x-N and NH₄-N concentrations over time were used to estimate sediment–water N fluxes. NO_x-N fluxes in oxic cores ranged from -2.2 to 4.1 mg m⁻² h⁻¹ and from -4.5 to 0.8 mg m⁻² h⁻¹ under anoxic conditions. Ammonium fluxes showed distinct redox responses: oxic cores exhibited both positive (1.3 to 2.6 mg m⁻² h⁻¹) and negative (-4.6 to -0.1 mg m⁻² h⁻¹) fluxes, whereas anoxic cores consistently released NH₄-N (0.34 to 3.0 mg m⁻² h⁻¹). Together, these results highlight a critical but underutilized opportunity to quantify internal nitrogen loading from existing sediment core experiments traditionally focused on phosphorus.

Seasonal fluctuations in greenhouse gas emissions and nutrients across the stream-riparian interface.

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In urban watersheds, small order streams face a variety of pollution delivered through surface and subsurface runoff. But the composition of the runoff changes depending on the season; higher instances of nutrient pollution in the summer and increased salt pollution in the winter. Pollution often affects the function and metabolism of a stream by influencing respiration of microbial communities. We look to explore the seasonal variations in the production of greenhouse gases (GHG), including carbon dioxide, methane, and nitrous oxide, as a measure of the system's response to pollution. The study site is Flat Branch Creek in Columbia, Missouri. Flat Branch Creek's watershed is 8.31 square kilometers with 97% of the watershed possessing slow or very slow infiltration rates. Flat Branch Creek experiences high levels of pollution due to heavily developed land use coupled with poor infiltration and high runoff. Flat Branch Creek is currently a sample site for a larger synoptic sampling project. This data has demonstrated that the stream experiences variations in chloride levels of 31 mg/L to 690 mg/L, total dissolved nitrogen levels of 0.13 mg/L to 1.37 mg/L, and total dissolved phosphorus levels of 0.012 mg/L to 0.15 mg/L. Furthermore, during present sampling, conductivity levels have averaged around 1000 µS/cm but experienced a 5-fold increase to 5000 µS/cm after the first snowfall and melt. The high variability in stream chemistry demonstrates the need to understand the response to dynamic pollution within the stream, hyporheic zone, and riparian zone. Across a transect of the stream riparian interface, we placed wells and soil collars for measurement at stream surface, bank, and riparian zone. From October to March, weekly sampling events occur to collect surface and hyporheic water and GHG flux. YSI probes are used to measure the dissolved oxygen, oxidation-reduction potential, and conductivity. In the lab, filtered samples are analyzed for total dissolved nitrogen, total dissolved phosphorus, nitrate, ammonium, and dissolved organic carbon. Through exploring the biogeochemistry of the stream-riparian interface over the course of three seasons, we expect to see the influence of urban pollution on the stream-riparian interface's GHG production.

Distributed embedded system design and performance for dynamic temperature control in tank and mesocosm experiments

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Responses of individuals, populations, and communities to future climate change can be elucidated via experimental manipulation. However, researchers' ability to realistically simulate future temperature regimes has been constrained by existing technology, which currently allows manipulation of daily maximum and minimum temperatures but does not allow for fine-scale thermal changes that mirror diel temperature increases and decreases in aquatic environments. We are presenting both the design and performance of a low-cost temperature control system that was developed to address this problem. The configurable system is able to dynamically control the temperature of aquarium or mesocosm tanks.

This temperature control system uses common hardware interfaced with Raspberry Pi microcomputers that can be configured via software to perform in different experimental contexts. The hardware and software support configurable temperature probe and heater assignments, with the ability to leverage multiple dynamically controlled heaters per tank, as well as multiple temperature probes per tank. The system utilizes off-the-shelf aquarium heaters to flexibly scale to various experimental contexts. The system has been successfully used in several experiments. One of the experimental contexts this system was designed for was for installation in temperature controlled rooms. In this configuration the system supported ongoing dynamic temperature control of 2.5-gallon aquarium tanks. The system has supported complex multifactorial experimental designs of up to 14 temperature combinations that have run for 10+ weeks. The same system components were also installed to support outdoor mesocosm experiments. In this context temperature set-points can be derived from configurable ambient tank offsets. An ambient treatment can be configured with other treatments maintaining a dynamic temperature offset. The mesocosm configuration supported an experiment utilizing 48 190-L tanks.

Creating realistic temperature profiles is necessary for a clear understanding of the potential effects of future climate change in aquatic systems. However, most scientists do not have access to equipment which allows for customizable, flexible temperature scenarios that change on appropriate timescales. Our system is relatively inexpensive and serves as either a customizable blueprint for a similar system, or a framework from which others can expand to test other aspects of future climate change in aquatic systems.

Thresholds in the responses of Pacific chorus frogs (*Pseudacris regilla*) to a gradient of temperatures and a heat shock

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Climate change is altering the temperature regimes that organisms experience. In freshwater ponds, water temperature influences organisms' growth and development rates. While many studies have demonstrated a strong effect of water temperature on amphibian growth rates, hatching timing, and emergence timing, few have investigated the thresholds at which organisms respond to differences in water temperature. In this study, we sought to determine the threshold at which the growth and development of Pacific chorus frogs (*Pseudacris regilla*) respond to increases in water temperature from egg through metamorphosis stage. We hypothesized that (1) the effect of water temperature on *P. regilla* growth and development would plateau at an increase from control tank temperatures of 3°C; and (2) a heat shock would decrease the response of *P. regilla* to increases in average temperatures. We assigned groups of three *P. regilla* eggs to a total of 94 tanks in a temperature-controlled room in Spokane, Washington that had one of seven dynamic temperature treatments representing baseline temperature (BT), BT+1°C, BT+2°C, BT+3°C, BT+4°C, BT+5°C, and BT+6°C. A subset of tanks experienced a heat shock treatment. We measured the date of egg hatching, date of tadpole metamorphosis, mid-experiment stage and mass of tadpoles, final mass of metamorphs, and morphological traits of metamorphs. Preliminary results indicate that eggs in warmer tanks hatched earlier ($p < 0.05$). This advance in egg hatching timing was reduced in tanks that received the heat shock treatment compared with those that did not ($p < 0.05$). For tanks without the heat shock, eggs reached a threshold in tanks with BT+4°C ($p < 0.01$), where tanks warmer than BT+3°C experienced earlier hatch dates. Further data analysis on tadpole development rate, growth rate, and morphological characteristics is ongoing. This work identifies the thresholds at which a common amphibian responds to temperature stress. Further, we highlight the importance of an early-life heat shock to egg development in this amphibian.

A Comparative Study on Climate Change and Geographic Origin's Impacts on *Pseudacris regilla*

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Global anthropogenic climate change and evolutionary history may interact to influence species responses to altered environmental conditions, such as increases in temperature. However, populations within a species may react differently to these changes based on their geography and evolutionary history. *Pseudacris regilla*, commonly known as the Pacific chorus frog (PCF) is widely distributed across Western North America, spanning a wide geographic range. In an effort to test the response of two geographically separated populations to changes in temperature, we compared growth and development rates of PCF tadpoles from western Oregon and eastern Washington near the cities of Corvallis and Spokane, respectively, during the summer of 2025. We used experimental treatment groups from two sequential experiments to opportunistically explore the differences between the two populations' responses to shifts in temperature. We investigated tadpoles' rate of mass and stage change per day under two different temperature conditions, one representing historical average temperatures and one representing warmer future conditions (+3°C above historical average). We used 24 PCF tadpoles from Oregon and 20 PCF from Washington. We found that tadpoles from both locations reached metamorphosis faster in the elevated temperature groups ($p < 0.0001$). Furthermore,

tadpoles from Oregon developed faster compared to Washington tadpoles ($p < 0.0001$). In addition, tadpoles from Oregon had less overall change in mass regardless of temperature treatment ($p = 0.0002$). Analysis is still ongoing, but these data suggest potential population-level differences in response to environmental shifts expected with climate change.

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Biodiversity within Kalamazoo River mnomen beds

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Mnomen (wild rice) is an essential source of food for Indigenous people in the Great Lakes region, including the Nottawaseppi Huron Band of the Potawatomi (NHBP). A collaboration between the NHBP and Albion College allows us to fuse Indigenous knowledge traditions and Western scientific knowledge to study the relationship between aquatic macroinvertebrates and mnomen. We use birch bark collectors, known as *wigwasmkok*, deployed in mnomen beds to collect samples of invertebrates, which are then identified utilizing dichotomous keys. We present results from two studies here. 1) Along the Kalamazoo River in southern Michigan are many mnomen beds of varying sizes. Our overall goal was to understand the effects of abiotic factors of the mnomen beds (bed size, depth, and water velocity) on the level of macroinvertebrate biodiversity found within the beds. We hypothesized that larger mnomen beds would be home to a more diverse community than smaller beds. Samples were collected in June and July of 2023 from four sites with different-sized mnomen beds to examine the effects of bed size and water depth on community composition, species diversity, and abundance. 2) Additional samples were collected in June 2025 to test if other variables, such as water velocity changes caused by mnomen influenced macroinvertebrate species diversity across the rice beds, with the hypothesis that higher species diversity would be found at the upstream edge and middle of rice beds in response to immediate changes in water velocity. *Wigwasmkok* were deployed in both real and artificial rice beds constructed from wood and cotton curling ribbon to assess the role of rice-bed structure in community diversity and test the artificial beds as models. From these two studies, it was concluded that mnomen bed size does not strongly influence macroinvertebrate species diversity and richness, and there was no significant difference in species diversity or abundance between or within real and artificial beds. Determining the impacts of abiotic factors on biodiversity provides further insights into community relationships within the river, which are essential to conservation efforts for mnomen and the broader Kalamazoo River ecosystem.

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Changes in biological metrics in small streams in the Puget Sound lowlands

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The Stormwater Action Monitoring (SAM) project is an ongoing long-term study investigating how well stormwater management protects stream health through physical, biological, and water quality sampling. Beginning in 2020, 26 to 33 (depending on year) unique streams across a range of percent total impervious area (TIA%) within their basin are surveyed each summer. As part of the biological sampling component of this project, benthic macroinvertebrates and periphyton samples are collected from multiple locations at each site, then composited and identified for that site. These data are used to generate five biological metrics. The Benthic Index of Biological Integrity (BIBI) is an overall score assessing the impacts of human influence on macroinvertebrate communities. The Hilsenhoff Biotic Tolerance Index (HBTI) assesses macroinvertebrate sensitivity to pollution based on species diversity and abundance. The Fine Sediment Sensitivity Index (FSSI) is a metric used to determine the sensitivity of a macroinvertebrate community represented in the sample to fine sediment accumulation. The Metal Tolerance Index (MTI) measures the impact of metal pollution on macroinvertebrate communities. The final metric, the Trophic Diatom Index (TDI), is used as an indicator of nutrient pollution based on diatom assemblages. These five metrics are then used to categorize streams as good, fair, or poor relative to regional reference conditions. The patterns in these metrics were similar from 2020 to 2024; as TIA% increased, streams tended to rank in worse condition, reflecting a lower macroinvertebrate diversity and an overall higher community tolerance to pollution. TDI values also showed impacts from nutrients related to TIA% within the basin. Across years, median scores for BIBI ranged from 32 to 69, median HBTI ranged from 4.7 to 4.95, median FSSI ranged from 30 to 55, median MTI ranged from 1.76 to 2.0, and TDI ranged from 52 to 65. There were also differences over time within the different TIA% categories with the highest decreases in scores in the most urban sites. These data are crucial for evaluating stream sensitivities to human influence and will help prioritize certain streams for future restoration efforts.

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Effects of Fish on Macroinvertebrate Community Structure in Non-Perennial Streams

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Stream fish influence macroinvertebrate prey both directly (via consumption) and indirectly through non-lethal effects. Non-lethal effects include life-history responses, such as faster development rates, and behavioral modifications that reduce exposure to predation. Fish predation commonly alters macroinvertebrate communities by reducing vulnerable prey while favoring taxa with behavioral or life-history traits that confer resistance to predation. Such non-consumptive effects of predators have been widely studied in perennial streams but are largely unexplored in non-perennial streams. We examined whole macroinvertebrate assemblages in two small, physically similar non-perennial streams in the Ozark Highlands ecoregion (central USA), one fishless and the other periodically colonized by fish during floods. We asked if any observed differences in community structure were predictable according to our understanding from perennial systems. We quantitatively sampled benthic macroinvertebrates from both streams at two points (Feb and April) during a ~6-month flow season, in two years (2021, 2025). We also collected from two

substrate types: exposed bedrock and loose gravel. Between-stream differences strongly outweighed temporal differences within streams and revealed consistently greater diversity in the fishless stream but greater densities in the fish stream. Several mayfly taxa, including ameleids and baetids, contributed to the greater densities in the fish stream, which also showed a greater relative abundance of taxa with traits conferring resistance to predation, such as increased mobility and fast seasonal development. In contrast, non-insects (e.g. amphipods and isopods) and longer-lived insects were more common in the fishless stream. Macroinvertebrate densities were lower on bedrock than loose gravel substrate, and this pattern was amplified in the fish stream. Overall, results suggest that macroinvertebrates in these non-perennial streams respond to fish similarly to expectations from perennial streams, although some of the variation between streams could also be attributed to minor abiotic differences. Future work to continue addressing this question could increase the sample size of fish and fishless streams and include experimental manipulations of fish presence and abundance.

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Effects of elevated and extreme temperature events on freshwater zooplankton assemblages.

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Freshwater systems harbor an abundance of biodiversity and are increasingly threatened by climate change, particularly through elevated temperatures. Central to freshwater food webs, zooplankton act as important intermediaries among micro- and macro-foodwebs, transferring nutrients from phytoplankton to higher trophic levels. Previous work found that larval salamanders respond to elevated temperatures in part by altering the size of zooplankton on which they feed. However, there is less known about how climate change-driven increases in water temperature affect freshwater zooplankton assemblages. This lack of knowledge is in part due to the time-consuming nature of collecting and analyzing zooplankton community data. To combat this challenge, we developed an affordable sample scanning technique, and used this along with an open-source imaging technology (PlanktoScope) to classify zooplankton communities before and after experiencing heat treatments. We used a 7 x 2 gradient-factorial experimental design, with 7 temperature treatments (Baseline, +1°C, +2°C, +3°C, +4°C, +5°C, and +6°C) and an extreme heat event (present/absent). We hypothesized that with increasing temperature, the abundance of smaller-bodied zooplankton would increase until a thermal threshold is reached, with cladocerans increasing at the greatest rate. Additionally, we hypothesized that both total Shannon diversity and species richness would decrease in tanks with heatwave treatments, with the magnitude of decrease proportional to the amount of temperature stress. Preliminary counts of zooplankton indicate higher total abundance in communities at the end of the experiment compared to the beginning. Low rates of mortality were also observed throughout the experiment across all treatments, suggesting resilience among zooplankton assemblages. We will create multiple regression models to analyze the relationship between temperature and subsequent shifts in zooplankton assemblage composition. Further data collection and analysis of body size will help us predict how communities of zooplankton, an important group of organisms in freshwater systems, will change under future climate conditions.

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Using Size Spectra to Analyze the Effects of Sampling Bias on Stream Macroinvertebrates

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The individual size distribution (ISD), also known as size spectra, describes the relationship between body mass and abundance, independent of species identity. ISDs are widely used to characterize community structure and infer energy flow within aquatic food webs. However, accurately estimating ISDs is challenging because common macroinvertebrate sampling methods introduce size-dependent biases. These biases can distort our understanding of individual sizes and lead to inaccurate estimates of community-level density and biomass, ultimately affecting interpretations of ecosystem structure and function. Power law models have been developed to estimate undersampled portions of the size spectrum, yet few studies explicitly apply these models to correct biased field datasets. In this study, benthic macroinvertebrate data from Brule Creek, SD is used to estimate a power law model and extrapolate undersampled sizes, with the goal of generating more accurate estimates of community biomass. Preliminary analyses indicate substantial undersampling, sometimes exceeding 50% of the data. We will examine how power-law corrections change community biomass compared to the original data. Differences in ISD slopes and biomass estimates will be used to evaluate how missing size classes influences overall community patterns.

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Baseline Survey of a Headwater Stream Before Weir Installation

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In 2021, 43 wetland pools were constructed at Oakwoods Nature Preserve (Hancock County, Ohio) with funding from the H2Ohio program to capture and sequester nutrient runoff before entering the Blanchard River (Maumee River/Lake Erie Watershed). Post-construction hydrologic analyses, however, indicated that the wetland pools do not intercept intermediate-sized flood events originating from a headwater stream draining the preserve. To improve floodwater capture during these spate events, the installation of a small weir has been proposed. The purpose of this study is to characterize the stream benthic community prior

to weir installation. Thus, in fall 2025, macroinvertebrates, periphyton, fish, and physicochemical data were collected from riffle habitats upstream and downstream of the proposed weir location (10 total samples). These surveys will be repeated in spring 2026 to strengthen baseline characterization of benthic community structure and assess potential post-installation impacts in the future.

Agriculture in, Algae Out? Land Use Controls on Phytoplankton Species Diversity and Productivity

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Freshwater ecosystems disproportionately contribute to global species biodiversity. These systems, which include lakes, rivers, and creeks, are sensitive to environmental and climatic changes, leading to dramatic shifts in local species assemblages. Global freshwater biodiversity is declining due to land use changes and pollution, which affect habitat integrity, water chemistry, and species composition. Understanding how human-driven changes in surface water chemistry affect freshwater macroinvertebrate biodiversity and algal communities is critical for guiding management and restoration efforts. Sulfate and nitrate in freshwater ecosystems can significantly affect ecological processes, biogeochemical cycling, and overall water quality. Elevated sulfate concentrations may be toxic to freshwater flora and fauna, while excess nitrate inputs can stimulate rapid phytoplankton growth. The concentrations of these chemical compounds are often influenced by land use in the surrounding watershed and are subsequently transported to streams as runoff. Our project focuses on McDiffett Creek in Kansas, which is surrounded primarily by agricultural land, and the Blue River in Oklahoma, which is largely used for recreation and protected under two conservation programs. McDiffett Creek has experienced greater ecological disturbance than Blue River. Using data from the National Science Foundation's (NSF) National Ecological Observatory Network (NEON), we analyzed sulfate and nitrate concentrations as indicators of agricultural runoff. Nitrate and sulfate data were collected bi-monthly from 2017 to 2023. Taxonomic and biomass data for local phytoplankton communities and macroinvertebrate taxonomic data were analyzed for the same time period. Our data indicate no significant difference in nitrate concentrations between sites, while sulfate concentrations were four times higher in the agricultural basin. This suggests that agricultural landscapes may increase sulfate levels within nearby waterways. Despite having lower macroinvertebrate species richness, Shannon diversity was higher at the agricultural stream. Conversely, the agricultural site had twelve times higher algal biomass than the non-agricultural stream, but lower species diversity. These findings highlight how land use can influence organismal communities in complex ways, affecting both species richness and biomass.

Distribution and diversity of freshwater snails in Ichawaynochaway Creek, southwest Georgia

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The southeastern US is a freshwater biodiversity hotspot in North America, particularly among invertebrates and mollusks. Freshwater gastropods play important structural and functional roles in rivers, but there are limited studies about their natural history and abundance. We aimed to describe the distribution of freshwater snails in a tributary of the Lower Flint River watershed in southwest Georgia. We surveyed ten shoals in the lower reaches of Ichawaynochaway Creek in late summer 2025 to determine the species composition, abundance, and habitat associations of freshwater snails. At each shoal, we conducted 15 five-minute snorkel visual surveys using a transect-quadrat method across a 25-m reach. For the dominant species, *Elimia boykiniana*, we measured the individual snail shell short and long axis dimensions, axes of the operculum, and wet mass. Additionally, we characterized substrate, velocity, and depth at each quadrat, and water physical chemistry (dissolved oxygen, temperature, pH) at each shoal. We identified three snail taxa: *E. boykiniana* (n = 1804), Family *Hydrobiidae* (n = 327) and *Viviparus goodrichi* (n = 1), but focused on *E. boykiniana* as it was the most abundant taxa. *E. boykiniana* shell length (2.0-18.6mm), wet mass (0.0061-1.8g), and mean density per shoal (0.5-135.8 snails/m²) varied across shoals. *E. boykiniana* density decreased as water deepened across the shoals, but the density-velocity relationship was site-specific. For substrate preference, snail density was greater on bedrock relative to cobble, sand, and wood habitats. Overall, our results indicated that *E. boykiniana* density is influenced by habitat characteristics within this specific creek. These findings offer an insight into the freshwater gastropod community in Ichawaynochaway Creek to inform future conservation and monitoring efforts in the southeast USA.

A cryptic concern: *In situ* surveillance of *Cryptosporidium serpentis* in wet-prairie dependent snakes of Killdeer Plains Wildlife Area

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Cryptosporidium serpentis is a protozoan parasite specific to and pathogenic in squamate hosts. Knowledge of this pathogen in wild populations is limited but studies from captive snakes indicate the parasite causes severe gastroenteritis, long-term immune impairment, and probable death. *C. serpentis* was discovered in quarantined captive breeding colonies of Ohio endangered Plains Gartersnakes (*Thamnophis radix*), a species dependent on Ohio's scarce wet-prairie habitat. It was hypothesized that the pathogen entered these colonies via breeding stock from Killdeer Plains Wildlife Area (KPWA; Wyandot and Marion Counties, Ohio). To assess prevalence of *C. serpentis* in wild snakes of KPWA we collected cloacal swabs and opportunistically collected fecal samples from all snake species during the annual Ohio Division of Natural Resources' (ODNR) snake survey in 2024 (May and June). Following DNA extraction, we used quantitative PCR (qPCR) targeting the 18S rRNA gene region of *Cryptosporidium* spp. together with a *C. serpentis* specific FAM-labeled probe to examine parasite loads across samples. *C. serpentis* was detected in 10.4% of snakes (N=209 total samples) and in 3/7 snake species sampled, including two imperiled species (*Opheodrys vernalis* and *Clonophis kirtlandii*). In 2025, we expanded our survey to sample snakes from April to October (389 samples) to capture any apparent changes in prevalence over time since this is an intermittently shed parasite. While our pathogen surveillance results thus far may illicit concern, it is important to next assess whether infected wild snakes develop the associated disease state, cryptosporidiosis, as often seen in animals housed in captivity. Furthermore, our results encourage consideration of potential environmental reservoirs for the pathogen's infectious stage (oocyst) since wet-prairie conditions are ideal for this parasite's persistence outside of the host. Potential parasite mitigation strategies including land management techniques such as prescribed fire are being considered.

Response of benthic organisms to patterns of alternating channel stability along the Gasconade River, Missouri (USA)

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Climate change is altering flow regime in rivers worldwide. In central North America, key changes include increasing frequency and magnitude of floods. More frequent bankfull floods lead to physical readjustments to stream channels and increased mobilization of streambed sediments. We have observed that ongoing channel instability is variably distributed along the longitudinal gradient of the Gasconade, a 6th order, gravel bedded river in Missouri, USA. For this study, we asked how various streambed-associated organisms have responded to these alternating degrees of physical instability along the Gasconade. Along an approximately 55-km stretch of the river, we selected three sections, each with a paired stable and unstable sample reach. During baseflow in July 2025, we collected data from ~100-m lengths on the downstream ends of glides, just upstream of riffle crests. In each glide, we quantified the spatial distribution of and area occupied by a common emergent macrophyte (water willow, *Dianthera americana*), and we surveyed mussels along ten 1-m wide transects. We also semi-quantitatively collected full invertebrate assemblages from benthic gravels, root mats, and large woody debris (LWD). Both water willow and mussels showed more clumped distribution in stable than unstable reaches, although densities of both taxa were similar between the two reach types. Preliminary data on invertebrate assemblages suggest that benthic and LWD densities were similar between reach types. However, root mats appear to have greater invertebrate densities in unstable than stable reaches, suggesting an interactive effect of reach type with habitat type, and that root mats may function as a refugium from physical disturbance. In addition, invertebrate diversity preliminarily appears to have been greater in unstable than stable reaches, potentially due to increased flow-related habitat heterogeneity in the unstable reaches. Given the potential role of physical and flow-related habitat heterogeneity in this system, a planned next step is to expand the scope of sampling within each reach to include riffles and runs. Our results can help inform watershed management decisions, such as determining ecological benefits of stream bank stabilization practices.

The effects of sedimentation on Odonata predator and prey sensing

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Sedimentation and turbidity in rivers are known to have adverse effects on aquatic life, including decreasing primary production and total abundance of macroinvertebrates. There also is evidence of chemical contaminants in aquatic systems interrupting macroinvertebrate chemosensory, specifically in detecting predators. However, it is unclear whether sedimentation and turbidity interrupt macroinvertebrate chemosensory detection and subsequent response to predators. On the Pea River in southeast Alabama, the Elba Hydroelectric dam is scheduled to be removed in Spring of 2026. Through dam removal, an increase in suspended solids is expected in the already sediment-laden and turbid river. A common damselfly genus in the Pea River is *Calopteryx* (Calopterygidae: Odonata). To determine the effects of increased turbidity on damselfly response to predators and prey, *Calopteryx* naiads were placed in individual containers of water and raised to eighth instar where they were then used for behavioral experimentation. After reaching eighth instar, individual odonates were placed in 37 Laquariums with either turbid

water or clear water. Turbidity was maintained using a mixture of kaolinite clay and artificial freshwater. Predator cues were then applied to the systems by pipetting 5 mL of water from a bluegill (*Lepomis macrochirus*) holding tank into each experimental aquarium, and multiple odonate movement responses were recorded via an ethogram. Similarly, to determine the effects of turbidity on Calopteryx prey sensing, individuals were placed again in 37 L aquariums with either turbid water or clear water. Five Hydropsychidae (Trichoptera) were added to each aquarium, and the proportion of successful feeding attempts was recorded. The data were analyzed using ANOVA. Anticipated insights of this study include a determination of the effects of suspended solids on odonate response to prey and predator cues as well as improved odonate rearing and holding techniques for future experimentation.

Exploring stream chemistry in post fire watersheds using a large monitoring dataset

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Wildfires have increased in extent and severity across the western landscape, heightening concerns for stream and watershed resiliency given the potential effects on measures of stream health, such as water chemistry. Previous research suggests that fluxes in chemical constituents are largely dependent on time since fire, fire severity and watershed characteristics such as vegetation type. Much of the existing research was designed to assess stream chemistry changes immediately post fire on a localized scale. In this project we present a novel approach to assess affects of fire on stream chemistry by using the Bureau of Land Management's (BLM) Lotic Assessment Inventory and Monitoring (AIM) program, a large-scale dataset. Lotic AIM is a standardized protocol designed to monitor stream conditions on BLM managed lands and, along with other methods, collects several water quality indicators. Because Lotic AIM sampling occurs across years and ecoregions depending predominantly on random sampling designs, not specifically to assess post fire conditions, we anticipated uncertainty in detecting fire related changes. Here, we paired Lotic AIM water quality indicators, total nitrogen; total phosphorous; specific conductivity; and pH, with wildfire perimeter and severity data obtained through the Monitoring Trends in Burn Severity (MTBS) program in the watershed delineated upstream of the sampling points (excluding fires that occurred more than 10 years before sampling and that burned less than 10% of the watershed). We found median total nitrogen values were similar with 186.5 $\mu\text{g L}^{-1}$ and 185.2 $\mu\text{g L}^{-1}$ in burned and unburned watersheds, respectively, and median total phosphorous values were 58.7 $\mu\text{g L}^{-1}$ to 42.0 $\mu\text{g L}^{-1}$ in burned and unburned watersheds. We additionally noted a distinct bimodal distribution for total nitrogen and total phosphorous in watersheds that burned greater than 75%, where a high mode exceeded the values of the collective median of all watersheds with qualifying fire and of unburned watersheds. We will explore potential relationships contributing to this observed bimodality as well as how these indicators differ with variations in burn severity, precipitation variation post fire, and between watersheds with differences in dominant vegetation.

The Effects of Hydrological Fluctuation and Macrophyte Species on Detrital Food Web Dynamics in Experimental Ponds

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Allochthonous leaf litter falls into ponds, where they provide microhabitats and energy sources for microbes and invertebrates. After microbial colonization, benthic macroinvertebrates colonize and begin to further process leaf litter, supplying the upper levels of the food web with energy via trophic transfer. Hydrological fluctuations are expected to drastically alter pond community assembly and decrease energy assimilation across the food web, with severity likely to increase as climate zones shift, but the interaction of leaf litter with lentic macrophytes has not been studied. The purpose of this study was to illuminate how interactions between drought and macrophytes influence the detrital food web. Specifically, we aimed to (1) quantify the differences in leaf litter decomposition, (2) macroinvertebrates community structure, and (3) carbon and nitrogen flux from leaf litter to aquatic macroinvertebrates in experimental ponds with known macrophyte communities and hydrologic fluctuations at the Lewisville Aquatic Environment Research Facility (LAERF) in North Texas. Sycamore leaf packs labelled with ^{13}C and ^{15}N were deployed in ponds to measure leaf decomposition and trace macroinvertebrate nutrient assimilation in a lentic system. After a six-week establishment period, three ponds underwent a two-week dry-down period (emulating a drought), while three remained filled (emulating ambient, non-drought conditions). After the 2-week dry-down phase, drought ponds were flooded to simulate hydrologic fluctuation. Leaf packs were added to cages with focal macrophytes: forty-nine leaf packs were added in each pond, with three cages without plants in each pond serving as controls to elucidate ambient macroinvertebrate communities in the ponds. The experiment started in October and was designed to emulate natural dry-down conditions and leaf-drop phenology typical of ponds in the central plains region. After seven weeks post-rewetting, packs were harvested, flash frozen, and transported for storage until further processing. Results from this study will inform how fluctuating hydrological regimes interact with macrophyte communities to influence leaf litter decomposition, macroinvertebrate biodiversity, and carbon and nitrogen fluxes up the detrital food web. Understanding how nutrient subsidies are impacted by drought in ponded systems is critical for understanding how ecosystems are linked and how these linkages are expected to change in the Anthropocene.

Defining Blackwater Systems through the Examination of Physicochemical and Biological Parameters

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Blackwater systems are traditionally defined as water bodies with a high concentration of tannins, low pH, dark color, and low levels of dissolved oxygen (DO); however, many freshwater systems considered “blackwater” do not fit this set of criteria. The lack of an operational definition for such a system is a pertinent issue in areas like the eastern United States, where blackwater systems dominate the coastal landscape but are poorly researched and defined.

The aim of this study is to select several methods to determine a set of physicochemical conditions and biological assemblages that most consistently distinguish blackwater systems from non-blackwater systems across the coastal plains of Georgia. By evaluating how physicochemical parameters traditionally used in observing blackwater streams may correlate with biological parameters such as species abundance, life history adaptations, and community composition, which are typically not included in blackwater monitoring efforts, we hope to broaden the definition of blackwater into a framework more definitive and comprehensive to better inform the classification and management of such systems by natural resource managers.

Through the development of this framework, aquatic organisms that specialize in or are endemic to blackwater systems, such as fish and macroinvertebrates, will serve as additional parameters to classify a blackwater system. The unique life history traits, community composition, and abundance of a species in a given system may be indicative of a biological assemblage unique to blackwater streams. Thus, designing an approach to collecting biological assemblage data in conjunction with traditional physicochemical traits of blackwater systems may prove vital to further informing management of these species and the conservation of their habitats.

1. Flotemersch J E., Blocksom K A., Herlihy A T., Kaufmann P R., Mitchell R M., and Peck D V. Distribution and Characteristics of Blackwater Rivers and Streams of the Contiguous United States. *Water Resources Research*. 2024; 60(2).
2. Colvin S. A. R., Helms B. S., DeVries D. R., and Feminella J W. Environmental and Fish Assemblage Contrasts in Blackwater and Clearwater Streams. *American Fisheries Society*. 2020; <https://academic.oup.com/tafs/article/149/3/335/7819427>

Bridging a knowledge gap between the social and the ecological in riverscape restoration on the Willamette River

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Billions of dollars are invested annually in the Pacific Northwest on riverscape restoration – millions in the Willamette River Basin alone. These projects are issued primarily to mitigate ecological degradation for Endangered Species Act listed salmonids. Salmonids are a culturally and economically important species. Currently, restoration sites are chosen based on land ownership and efficient use of funding without considering the social values of the surrounding community. Correlations between human demographics and restoration site selections in the Willamette River Basin remain under-explored. My study utilized ArcGIS Pro to map and quantify current riverscape restoration worksites across the Willamette River Basin. By integrating spatial data with US Census demographic and socioeconomic layers, I assessed emerging patterns of project distribution relative to human community composition. Preliminary spatial analysis reveals an absence of riverscape restoration projects within urban boundaries, areas which typically house higher proportions of Black and Brown citizens. By quantifying these patterns, my research provides a critical social lens for valuing riverscape restoration for salmonids. The findings will aim to recommend management strategies that integrate a socio-ecological framework ensuring restoration efforts benefit salmonids without neglecting the importance of the riverscape to their human neighbors.

1. Dunham, J.B., Angermeier, P.L., Crausbay, S.D., Cravens, A.E., Gosnell, H., McEvoy, J., Moritz, M.A., Raheem, N., and Sanford, T. 2018. Rivers are social–ecological systems: Time to integrate human dimensions into riverscape ecology and management. *WIREs Water* 5(4): e1291. doi:10.1002/wat2.1291.
2. King, B., and Fonner, R. 2024. Exploring restoration efforts from a social lens: statistical models reveal relationships between salmon habitat restoration efforts and ecological and social characteristics of the Puget Sound basin, USA. *E&S* 29(4): art14. doi:10.5751/ES-15272-290414.

Stream ecosystem metabolism in a dystrophic Scottish highland tributary network.

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Short-term fall open system metabolism estimates were conducted in four stream locations in the Dubh Lochan watershed in late October/early November 2025. Methods involved placement of miniDOT dissolved oxygen data loggers in the thalweg of each stream for a minimum of 72 hours. Light was recorded (lux) using submerged data loggers (Hobo Pendant). Reaeration was analyzed empirically using the energy dissipation model. All sites were strongly heterotrophic (mean net daily metabolism and community respiration ranged from -2.4 to -3.8 and -1.3 to -3.7 g O₂ m⁻² 24hr⁻¹, respectively) and gross primary production was negligible. In-stream metabolic activity is almost entirely driven by heterotrophic respiration (e.g., microbial mineralization of organic carbon). This carbon is likely derived from allochthonous input of OM in the form of leaf litter and dissolved organic carbon. The Dubh Lochan outflow and Dubh Lochan downstream sites both were significantly more heterotrophic ($p < 0.05$, ANOVA) relative to inflowing tributaries, likely due to the high concentration of dissolved organic carbon and other refractory dissolved compounds present therein and derived mainly from the dystrophic Dubh Lochan. This outflow consists of highly tannic and "stained" water. Free-water dissolved oxygen measurements taken at 1 and 8 meters below the surface of Dubh Lochan indicated strong heterotrophic conditions. The results of this short-term project indicate that the major tributary feeding Dubh Lochan is heterotrophic, and the accumulation and processing of OM in the Lochan likely drives higher levels of heterotrophy (e.g., CR was greater in the stream receiving Dubh Lochan outflow). Implications and outstanding questions include the following: What is the relative importance of legacy carbon derived from extensive peat formations in the headwaters vs. leaf-litter inputs for aquatic food webs? And how might these carbon sources fluctuate in the context of a changing climate? Site-specific research like this is critical, given that small, dystrophic lakes are common in boreal and subarctic bioregions, but are understudied.

Endocrine and Gonadal Responses of a Tropical Catfish to Anthropogenic Environmental Stressors in Two Nigerian Lagoons

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Environmental estrogens and other endocrine-disrupting chemicals (EDCs) impact the reproductive health of fish in many aquatic ecosystems. Gonadal alterations and levels of sex hormones and vitellogenin, a family of glycolipoproteins that serve as precursors to egg yolk protein, are effective biomarkers of endocrine disruption. Although these biomarkers have been widely applied, their use in identifying EDC impacts in developing countries are lacking. Such is the case in Nigeria. This study investigated the endocrine and gonadal responses of a commercially important catfish, *Chrysichthys nigrodigitatus*, in Lagos and Epe Lagoons.

A total of 195 adult fish (Lagos: $n = 105$; Epe: $n = 90$) were collected monthly between November 2020 and May 2021. Blood plasma was extracted for analysis of vitellogenin protein and two sex hormones (testosterone and 17 β -estradiol) using ELISA. Gonadal tissues were examined histologically. Fish muscle samples and sediment, also collected monthly, were analyzed for heavy metals (Hg, As, Cr, Cd) and harmful phthalates (DEHP, DBP, DEP) using atomic absorption spectroscopy. Physicochemical parameters, including pH, DO, Temp and EC, of both lagoons, were measured using a multiparameter probe.

Intersex conditions were observed in 3.81% of the *C. nigrodigitatus* from Lagos Lagoon (males: 50%; females: 50%) and 3.33% from Epe Lagoon (males: 33.3%; females: 66.7%). Gonadal alterations, including empty seminiferous tubules and distorted ovaries, were recorded in male and female fish, respectively, from both lagoons. Significantly higher testosterone levels (ng/ μ L) were observed in females (3.00 ± 0.71) relative to males (1.94 ± 0.48) from Epe Lagoon, but not from Lagos Lagoon. 17 β -estradiol levels (ng/ μ L) were higher in females than in males in both lagoons. Vitellogenin (ng/ μ L) in male fish was statistically higher than those in female fish in both the Lagos (0.28 ± 0.08 versus 0.23 ± 0.02) and Epe (0.31 ± 0.03 versus 0.28 ± 0.05) lagoons.

Multivariate analyses revealed sex-specific patterns of metal uptake and positive relationships between sediment pH and metal and phthalate concentrations in both locations. These results indicate that *C. nigrodigitatus* populations in Lagos and Epe Lagoons experience endocrine disruption linked to exposure to environmental estrogens and other EDCs.

Effects of glyphosate on growth and photophysiology of the cyanobacterium *Microcystis aeruginosa*

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Glyphosate is an active ingredient in increasingly used herbicides. Runoff and aerosolized glyphosate can enter river and lake systems leading to selective pressure on different species, including harmful algal bloom (HAB)-forming plankton. We examined the effects of glyphosate on the growth and photophysiology of the cyanobacterium *Microcystis aeruginosa* (UTEX-LB-2386). We grew *M. aeruginosa* in the presence of environmentally relevant concentrations of glyphosate ranging from (0 - 10 μ g/L), and measured growth using spectrophotometry. Photophysiology of cells in exponential growth was analyzed using PAM-fluorometry. Assessments included in-vivo chlorophyll (F0), the maximum quantum yield of Photosystem II (QYmax, a proxy for photosynthesis efficiency and cell stress), and relative electron transport rates across PSII (a proxy for photosynthesis). Growth rates ranged from (0.05 to 0.07), and QYmax values ranged from (0.45 to 0.55). However, no measurable dose-response relationship was observed across all glyphosate concentrations tested. The lack of inhibition in *M. aeruginosa* growth or photophysiology at environmentally relevant levels of glyphosate suggest that these glyphosate levels could potentially provide a selective advantage to this bloom-forming species in contaminated freshwater ecosystems. Furthermore, given that the glyphosate levels tested were significantly below EPA thresholds for safe drinking water, our work also highlights the need to integrate robust contaminant monitoring into HAB-monitoring strategies - especially in watersheds with high agricultural activity.

Individual and Combined Effects of Fluoxetine and Microplastics on *Physa acuta*

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Fluoxetine and microplastics are increasingly detected in freshwater ecosystems, yet their combined effects on aquatic invertebrates remain poorly understood. This study evaluates the individual and combined effects of the antidepressant fluoxetine (FLX) and microplastics (MP) on predatory avoidance behaviour, movement, reproduction, and egestion in the freshwater snail *Physa acuta*. Snails were exposed to four treatments: control, fluoxetine (350 ng/L), microplastics (10 µg/L), and a combined fluoxetine and microplastics treatment, with 17 replicates per treatment.

Predatory avoidance behaviour was measured as the proportion of snails using refuge on or above a tile and by escaping their container. Movement was quantified as the number of grid crossings on a 1 cm² grid. Reproduction was assessed by counting egg masses and eggs per egg mass, and egestion was measured by weighing the dry waste mass.

We hypothesize that fluoxetine will alter behavior and reproductive output due to its interaction with conserved serotonergic pathways, while microplastics will impair digestion and reduce energy allocation to growth and reproduction through physical and physiological stress. Additionally, we predict that combined exposure will produce stronger negative effects than either stressor alone due to potential contaminant interactions. This study contributes to understanding how multiple emerging contaminants may interact to influence freshwater invertebrate fitness and ecosystem processes.

Comparing Long-Term Educational Pond Studies at a Wildlife Sanctuary and Working Farm with Prior Intensive Inventory Results

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With worldwide concern about amphibian declines in the 1980s, Massachusetts conservationists surveyed amphibian populations; strengthened protection for breeding areas, including temporary ponds commonly known as vernal pools; and inventoried aquatic communities in breeding pools statewide. Over the past 35 years, teacher-naturalists at Massachusetts Audubon's Drumlin Farm Sanctuary have introduced children and adults to macroinvertebrates and amphibians of temporary and permanent ponds through regular sampling programs, and some educators began monthly surveys. Recently there has been interest in reviewing the long-term sampling data to identify changes in pond communities, assess effects of climate change on pond hydrology, and guide ecological management practices.

We compared long-term data from surveys by sanctuary staff, school groups, and volunteers from 2009 through 2025 ("Sanctuary data") with results of comprehensive inventories by staff scientists in the 1990s ("Inventory data"). The data sets differ in sampling details (time-limited dipping with a kitchen strainer on a 2-m section of bank vs standardized D-net sweeps at multiple random within-pool locations), level of taxonomic identification (field identification, mostly to Order or Family, vs. best achievable taxonomic level in the laboratory), and frequency (Sanctuary sampling occurs almost weekly to monthly, year-round; Inventory sampling was in spring over several years, with limited additional sampling).

There was considerable similarity in Order and Sub-Order results from the two data sets for specific ponds. The limited Sanctuary taxonomic resolution prevents confirmation of amphibian breeding in some ponds and years. The Inventory found high diversity in Dytiscidae and other Coleoptera; this is not reflected in the Sanctuary records limited to "beetle" or "diving beetle." Similar limitations exist for other taxa.

We recommend some modifications to the data form, including: Add "amphibian egg masses" and species names of amphibians (e.g., "wood frog") to "salamander" and "frog;" provide three size classes for Dytiscidae and Hydrophilidae and add Haliplidae to increase the detail on Coleoptera; and more. The additional detail may challenge some Sanctuary samplers, but we are confident that many will be eager to make the distinction, enhancing the usefulness of the data and increasing the educational value for schoolchildren and adults becoming aware of biological diversity.

Kern River in Crisis: Rapid dewatering of the river drastically impacted fish communities

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Excessive water diversions have substantially altered natural river flows throughout riverine systems within California (USA), often leading to drastic shifts in the availability of suitable habitat to support freshwater wildlife. The Lower Kern River (LKR) historically functioned as a perennial-like river system; however, currently ecologically functions as an ephemeral river due to a series of weirs and canals that often divert 100% of flows away from the river. We present findings on how an unexpected, rapid dewatering event impacted fish community dynamics in the LKR. A planned visual fish transect survey was conducted late August 2024 along 2.4 km of the LKR to identify fish taxa and their abundances. Rapid dewatering of the LKR began downstream of our site concurrent to the timing of our survey, with the dewatering expanding throughout our site in less than seven days; therefore, we conducted a second survey one week after the first survey to compare changes in the fish community before and after the dewatering event. Photographs and videos taken were analyzed using ImageJ software to confirm fish identification, measure total lengths, and to determine life stages. River water was also sampled for fish eDNA during the first survey to confirm and expand field taxa identifications. The fish community before dewatering was dominated (>80%) by mosquitofish (*Gambusia affinis*) and largemouth bass (*Micropterus salmoides*). Rapid dewatering of the LKR resulted in

a mass fish death event (>2,000 fish), which demonstrated that the fish community was instead dominated by mosquitofish (*Gambusia affinis*), yellow bullhead (*Ameiurus natalis*), and bluegill sunfish (*Lepomis macrochirus*). Juvenile fish were the most frequent life stage among taxa. Fish eDNA results added the Sacramento sucker (*Catostomus occidentalis*) to the taxa list. One taxon of interest was the endangered juvenile rainbow trout (*Oncorhynchus mykiss*), which has been observed in the LKR by scientists and citizens, but was not detected using eDNA. Freshwater ecosystems provide critical habitat for fish communities, which are affected by water availability and quality; therefore, it is important to find alternative pathways that support anthropogenic freshwater needs without sacrificing fish populations.

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Investigating the Osmoregulatory and Physiological Responses of an Aquatic Salamander (*Amphiuma means*) to Acute Salinity Stress: A Comparison of Coastal and Mainland Populations

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Coastal freshwater wetlands (CFWs) are increasingly vulnerable to habitat degradation driven by land loss, hydrological modification, and salinity intrusion. These systems support diverse faunal assemblages whose biodiversity is threatened as salinity exposure intensifies. Amphibians are generally intolerant of saltwater, with elevated salinity negatively affecting their survival, development, and overall physiological condition. Because of this sensitivity, amphibians have strong potential to serve as indicators of saltwater intrusion in coastal wetlands. Two-toed Amphiuma (*Amphiuma means*) are large, fully aquatic salamanders endemic to the southeastern Coastal Plain. Although cryptic, they function as important mid-trophic consumers, feeding on crayfish, invertebrates, and small vertebrates while serving as prey for larger predators. *A. means* populations occur on three of Georgia's barrier islands, where they are spatially isolated and face heightened risk of salinization through land-use change, storm-surge inundation, and atmospheric salt deposition. Increased exposure to high-salinity conditions may lead to differential fitness between coastal and mainland populations. This study evaluated potential differences in the physiological response of *A. means* from barrier-island and mainland populations when exposed to acute salinity stress. Individuals from both groups were subjected to elevated salinity treatments within an intermittent flow-through respirometry system to quantify cutaneous respiration rates and blood plasma osmolality. **Preliminary analyses indicate that cutaneous respiration and plasma osmolality exhibit similar response patterns across both 5-ppt and 10-ppt exposures, suggesting that *A. means* may possess a capacity to regulate respiratory and osmotic function under acute salinity stress.** This work provides novel insight into the physiological tolerance of *A. means* and enhances our ability to anticipate how coastal amphibian populations may respond to future environmental change in imperiled freshwater ecosystems.

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Using eDNA for Population Monitoring of American Eel (*Anguilla rostrata*) in the Susquehanna River Basin

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There are growing concerns regarding the safe passage of aquatic organisms over large hydroelectric dams on major rivers like the Susquehanna River, as well as lesser barriers such as culverts. It is unknown if efforts to mediate this issue (e.g., increased stocking and fish passage facilities) are significantly effective at restoring populations of *Anguilla rostrata*. *A. rostrata* (American eel) faces population decline due to impediment of anadromous migration from the North Atlantic. They are major hosts of important native freshwater mussels such as *Elliptio complanata* (Eastern elliptio mussel) and are key food web members. Eels are cryptic organisms, which poses a challenge for traditional electrofishing sampling methods; environmental DNA (eDNA) is a popular method for detection of cryptic species. In the fall of 2025, we began an eDNA sampling initiative in the Susquehanna basin, targeting sites upstream and downstream of all major barriers on the main stem of the river and its tributaries. Samples were taken using the Smith-Root eDNA Sampler Backpack. We filtered ~1L (± 0.10) of water through 0.45 μ m filters. Filters were stored in ethanol until they were processed via DNA extraction and qPCR. DNA was extracted using the Qiagen DNeasy Blood & Tissue Kit, and qPCR procedures were completed using a recently developed PCR marker for American eels referred to as AME1. The Bio-Rad CFX Opus 96 Real-Time PCR System was used for 15 μ L qPCR reactions, and the CFX Maestro computer program was used to analyze PCR results. Water quality measurements were taken at each site to better our understanding on potential predictors of eel movement. Current findings indicate presence of eels downstream of major hydroelectric dams on tributaries feeding into the West Branch of the main river, but no presence around low head dams on the main river. More sampling will be completed in eastern and southern Pennsylvania, around the mouth in northern Maryland, as well as the headwaters in upstate New York. Understanding where eels are present as well as how they are moving will guide future stocking and conservation efforts, especially in relation to fish passage systems.

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Modeling Multiple-Stressor Effects on Stream Fish Functional Diversity and Redundancy in the Tennessee River Valley

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The Tennessee River Valley is simultaneously one of the most species-rich and most imperiled drainage basins in North America for freshwater fishes. As extinction rates of North American freshwater fishes are projected to increase in the coming decades

due to ongoing human impacts and climate change, more research aimed at improving the efficiency, economy, and effectiveness of conservation efforts in the Tennessee River system is needed. Trait-based approaches offer a powerful framework for addressing this need by shifting focus from taxonomic composition alone to the structural and functional organization of assemblages. By quantifying the diversity of ecological, morphological, and life-history traits in stream fish assemblages, trait-based approaches allow for the assessment of functional diversity, an important component of biodiversity which may have more direct implications for ecosystem function and resilience. In this study, we will examine how multiple environmental stressors influence both the functional diversity and functional redundancy of stream fish assemblages across the Tennessee River Valley using long-term fish monitoring data provided by the Tennessee Valley Authority. We will use random forest models, a machine-learning approach capable of modeling the complex, non-linear relationships between landscape- and site-level stressors (e.g., land use/land cover, habitat) and fish assemblage responses. Similar methodologies have been used to characterize fish assemblage responses to multiple stressors in other river systems, but to our knowledge, large-scale analyses of this nature have yet been applied to the Tennessee River Valley. Our analysis will address three conservation-relevant questions: i) what is the current status and spatial distribution of fish functional diversity in Tennessee Valley streams, ii) how do environmental stressors affect the functional diversity and redundancy of stream fish assemblages, and iii) how might projected future stressors alter functional diversity and redundancy. By leveraging machine-learning models within a multiple-stressor, trait-based framework, our study aims to identify key streams, traits, and stressors of management concern and provide actionable insights for regional freshwater conservation and management.

Confirmation of Species Identity and Age Estimation of Pygmy Whitefish from Sullivan Lake WA.

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Native to Washington, Pygmy Whitefish (*Prosopium coulterii*) are distributed across nine lakes statewide. Seven of these lakes are east of the Cascade Range, including Sullivan Lake. The population status of Pygmy Whitefish in Sullivan Lake is unknown. This study provides background data on the population status of Pygmy Whitefish in Sullivan Lake. Information collected from sixty-seven samples includes species confirmation, age, and condition factor. Species determination was done using DNA barcoding. DNA was extracted using QIAGEN DNeasy Blood and Tissue extraction kits, then sequenced by GENEWIZ Laboratories. We confirmed species identity by uploading sequences to the NCBI BLAST database. Ageing was done using back-calculations of length, length-frequency histograms, and counting the number of annuli on scales and otoliths. Scale samples were taken from beneath the dorsal fin and above the lateral line. Analysis of length-frequency histograms revealed expected age classes between zero and five years. Condition factor (K) displayed strong seasonal variation, with mean K increasing from 0.268 (SD = 0.067) for fish sampled in June to 0.696 (SD = 0.077) for fish sampled in September. Collectively, these results establish a baseline for population status and condition metrics of pygmy Whitefish in Sullivan Lake. This baseline is crucial for informing fisheries management decisions, including the evaluation of potential Bull Trout introductions into Sullivan Lake.

Weak complexity-stability relationships in tropical and temperate stream food webs.

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Interactions among species shape the organization of life, influencing patterns of distribution, richness, behavior, and the partitioning of energy within ecosystems. Food webs are a central framework for studying these interactions, describing energy flow among organisms and the mechanisms governing community dynamics. A long-standing debate in network ecology concerns the relationship between complexity and stability, defined as a system's ability to maintain or recover its original state following perturbations (Kratina et al., 2012). Temperature is a key driver of food web structure because it modulates metabolic rates, population dynamics, and species interactions (Brown et al., 2004); however, its effects on food web stability remain poorly understood.

This study investigates the relationship between complexity and stability in aquatic macroinvertebrate food webs, evaluating how temperature influences interaction structure and network dynamics. We hypothesized that temperate food webs were expected to exhibit lower species richness, larger-bodied organisms, fewer interactions, and higher energy flux per interaction, leading to increased stability through greater total biomass. In contrast, tropical food webs were predicted to show higher species richness, smaller organisms, and a larger number of interactions, with stability sustained by numerous weak, low-energy-flux links.

We conducted monthly sampling over one year in tropical streams (Intervales State Park, Brazil) and temperate ones (Stony Creek Watershed, USA). Macroinvertebrates were collected, identified, and their biomass estimated. Hypothetical food webs were reconstructed using a model based on functional feeding groups and predator-prey allometric relationships. For each food web, we quantified various complexity metrics and estimated stability.

Preliminary results indicated that temperate food webs exhibited higher species richness, larger average body size, and greater complexity across metrics, and were closer to stability than tropical ones. No direct relationship between complexity metrics and stability was detected in tropical networks. In contrast, temperate networks showed a positive association between average

population body size and stability, suggesting that food webs become more stable as insects approach emergence and as community structure develops throughout the growing season. These findings enhance our understanding of the mechanisms regulating food web structure and stability and provide insights into how temperature may shape ecosystem persistence under ongoing climate change.

1. Brown, J. H. Gillooly, J. F. Andrew, P. Allen, Geoffrey B. West. Toward a metabolic theory of ecology. *Ecology*, v. 85(7), pp. 1771-1789, 2004
2. Kratina, P., LeCraw, R. M., Ingram, T., Anholt, B. R. Stability and persistence of food webs with omnivory: Is there a general pattern? *Ecosphere*, v. 3(6), pp. 1-18, 2012.

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Dynamics of flow intermittency affect cross-directional trophic subsidies of ephemeral streams

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Intermittent rivers and ephemeral streams (IRES) comprise more than 50% of global river networks. Despite their prevalence, IRES remain understudied, particularly with respect to effects of periodic flow cessation and dry conditions on aquatic-terrestrial food web linkages. While theoretical models of large perennial rivers emphasize the importance of terrestrial subsidies to aquatic consumers, comparable frameworks are rarely applied to IRES, in which hydrologic variability may result in dynamic cross-directional subsidies.

We assess the relative importance of aquatic and terrestrial resources in IRES using stable isotope analysis of carbon, nitrogen, and hydrogen. Focusing on five drought-prone streams in north Texas that span a gradient of flow intermittency, we test the hypothesis that aquatic resources are important subsidies for terrestrial consumers during periods of drying, while reliance on terrestrial inputs increases during early rewetting phases when in-stream production is reduced. Aquatic and terrestrial basal resources (C_3 and C_4 plants, periphyton, and macrophytes) and consumers (benthic macroinvertebrates, fish, and terrestrial invertebrates) were sampled across multiple seasons. Isotopic mixing models estimate the proportional contribution of aquatic and terrestrial sources to consumer diets, and mixed-effects models evaluate how water availability and quality mediate these subsidy pathways. By testing how aquatic-terrestrial linkages vary across hydrologic regimes, this study aims to clarify the role of dynamic flow intermittency in structuring the food webs of drought-prone stream systems.

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Butterflies and turtle tears: lachryphagy is common and widespread in tropical South America

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Among the many examples of freshwater-terrestrial resource subsidies that have been documented, one of the more unique is the phenomenon of butterflies drinking the tears of basking river turtles and caimans. 'Lachryphagy' by butterflies was observed in the field on the yellow-spotted river turtle (*Podocnemis unifilis*) on the Napo and Tiputini Rivers of Ecuador. Here, we explore this phenomenon using photographs and ancillary data drawn from the citizen-science project iNaturalist to answer questions such as: How many butterfly taxa engage in lachryphagy? How widespread is this phenomenon? Are there butterfly preferences for turtles across sexes, sizes, and basking substrates? We found that butterfly lachryphagy on *P. unifilis* is common; of the 1,387 research-grade photos of *P. unifilis* uploaded to iNaturalist, 9% (126) contained evidence of this phenomenon. Butterfly-*P. unifilis* lachryphagy is also widespread and spans 2,700 km longitudinally and 1,500 km latitudinally. We observed 14 butterfly genera showing evidence of lachryphagy on *P. unifilis*; *Dryas iulia* was the most commonly observed species, constituting almost half of the butterflies observed. By far the greatest number of *P. unifilis* photos were of turtles as they basked on protruding large wood (87%), a finding reflective of the importance of this habitat for basking by *P. unifilis*, and by extension, for butterfly lachryphagy. While the exact resources provided to butterflies through lachryphagy remain to be determined, candidate substances include: minerals (e.g., sodium), proteins, and trace elements. These results matter because butterflies and other insects that engage in lachryphagy are pollinators of terrestrial plants, including those in the upper rainforest canopy, and butterfly larvae can be important herbivores. While some of these results reflect the biases of the photographers, they nevertheless constitute steps towards understanding a unique type of resource subsidy, one involving a freshwater vertebrate of conservation concern; *P. unifilis* is IUCN red-listed as 'vulnerable'. These results highlight the myriad and complex connections that exist between tropical freshwaters and seemingly unconnected ecosystems such as the upper canopy of rainforests.

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Do freshwater mussel beds influence the nutritional quality of aquatic subsidies?

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Terrestrial and riparian environments are intricately linked through the exchange of resources; however, these subsidies are most often measured in terms of biomass quantity, with comparatively less attention given to nutritional quality. The quality of these trophic linkages is expected to vary depending on the community composition of the organisms. In this study, we examined whether the presence of freshwater mussel beds influences the nutritional quality of aquatic insect subsidies, and whether such variation is reflected in the condition and physiological metrics of riparian predators. We sampled web-building riparian spiders

(family *Tetragnathidae*) and emergent insects from sites featuring and lacking mussel beds, as well as three different creeks within Kisatchie National Forest, Louisiana. To characterize subsidy quality and predator responses, we conducted fatty acid analyses on both insects and spiders, measured spider body condition, and assessed immune function. In parallel, we quantified the emergent insect community composition, biomass, and fatty acid content to characterize nutrient availability and quality. Together, these data will be used to evaluate links between aquatic community structure, subsidy quality, and terrestrial predator health. This work aims to advance understanding of how benthic community composition may influence the nutritional pathways connecting aquatic and riparian food webs. Preliminary analyses indicate no detectable differences in spider body condition between mussel and non-mussel sites.

Pathways of pigments: Investigating carotenoid dynamics in lake food webs

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Carotenoids are diet-derived pigments that play important roles in antioxidant defense, coloration, immune function, and reproductive success in aquatic organisms. In lake ecosystems, carotenoids are synthesized by phytoplankton and are transferred through zooplankton to higher trophic levels, making them useful indicators of trophic linkages and energy flow. Copepod zooplankton can store carotenoids within lipid “bubbles;” understanding such patterns of carotenoid storage across taxa and developmental stages would allow us to identify pathways of carotenoid trophic transfer. This study examines the distribution of carotenoids in pelagic food webs of small Adirondack lakes by quantifying carotenoid content in copepod zooplankton and brook trout (*Salvelinus fontinalis*). Zooplankton were collected from four lakes using vertical plankton tows (80- μ m mesh) at each lake’s deepest point during spring and summer seasons. Preserved samples were transferred to a plankton maze and a 1-mL subsample was examined under compound microscopy (40x) for identification of order (i.e., calanoid or cyclopoid copepod) and life stage (i.e., nauplius or adult). Carotenoid content was estimated visually based on the proportion of orange–red pigment stored within the body and categorized into four pigment-intensity quartiles ranging from minimal to extensive coverage. Preliminary analyses of spring samples reveal clear taxonomic patterns in carotenoid storage. Calanoid copepods were more frequently classified in the upper two pigment-intensity quartiles (approximately 50–100% body coverage), whereas cyclopoids were predominantly observed in the lower quartiles (<50% coverage). Adult copepods exhibited higher carotenoid intensity than nauplii across both orders, with high pigment-intensity categories (50–100% body coverage) observed exclusively in adults. These results are consistent with previous findings that direct phytoplankton feeding by calanoids is associated with greater carotenoid accumulation. To evaluate carotenoid transfer across trophic levels, age-2 brook trout (>12 in) were collected from the same lakes and are being analyzed for carotenoid content using high-performance liquid chromatography. Together, these data characterize carotenoid distributions across multiple trophic levels within Adirondack lake ecosystems and help to establish methods for using carotenoids as food-web tracers and indicators of nutritional condition in freshwater fishes.

1. Vilgrain, L., F. Maps, S. Basedow, E. Trudnowska, M.-A. Madoui, B. Niehoff, and S.-D. Ayata. 2023. Copepods' true colors: Astaxanthin pigmentation as an indicator of fitness. *Ecosphere* 14(6): e4489.
2. Nakano, T., and G. Wiegertjes. 2020. Properties of carotenoids in fish fitness: A review. *Marine Drugs* 18(11): 568.

Environmental DNA (eDNA) as a tool for rare freshwater mussel detection and restoration planning

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Freshwater mussels are one of the most imperiled faunal groups in North America, with over 70% of freshwater mussel species considered to be endangered, threatened, or of special concern. Causes for imperilment include habitat fragmentation, habitat degradation, and climate change. Traditional survey methods, such as snorkeling or using view buckets in transects or quadrats, are commonly used to measure freshwater mussel species presence and abundance; however, these methods are time-intensive and spatially limited, and it can be difficult to detect rare mussel species due to their patchy distribution and burrowing behavior. To overcome these difficulties, environmental DNA (eDNA) surveys, which use non-invasive water samples to assess species presence, have become a desirable tool for freshwater mussel conservation. In this study, we evaluated eDNA as a method for detecting brook floater (*Alasmidonta varicosa*), a state-listed endangered freshwater mussel species in Massachusetts, where only four viable populations remain. First, we sampled eDNA at sites with known populations of brook floater. At each site, water samples were collected upstream, immediately downstream, and 100 m downstream of the known population. In addition, habitat (e.g. wetted width) and water quality (e.g., pH, specific conductivity, dissolved oxygen, total suspended solids) were collected at each location. In the lab, water samples were filtered, DNA was extracted, and then amplified using qPCR. Once methods were verified at sites with known populations of brook floater, we sampled for brook floater eDNA at sites where the species has not been recently recorded to 1) identify the existence of brook floater and/or 2) determine potential for the reintroduction of stocked

propagated mussels. Results will be used by managers throughout the brook floater range (Georgia to Nova Scotia) to inform expanded use of eDNA for species surveys and restoration efforts.

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Effects of varying preservation methods on freshwater mussel fatty acid profiles

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Freshwater mussels (Bivalvia, Unionida) play a crucial role in freshwater environments as benthic ecosystem engineers, providing essential ecosystem services. Their sedentary, filter-feeding life history also makes them valuable indicators of environmental change. Despite their benefits, mussels are among the most imperiled taxonomic groups globally, making investigations into poorly understood mussel necessities like diet imperative for conservation efforts. Natural history collections represent an underutilized resource for addressing long-term ecological questions. Molecular analyses of archived mussel collections offer a unique opportunity to compare past and present dietary patterns, providing valuable insights into long-term ecological change and historical environmental conditions. However, the effects of tissue preservation method on dietary tracers such as fatty acids (FA) remain unknown. In this study, we assess how common preservation methods affect FA profiles in two freshwater mussel species. Specifically, we identify and quantify baseline FA signatures, evaluate FA differences among preservation techniques, and explore approaches for developing preservation-specific FA correction factors. Preliminary results show FA composition and retention time differences between three preservation methods relative to the recommended freeze-dry baseline. The oven-dry method deviated the least from the baseline, ethanol preservation deviated further from the baseline, and formalin preservation deviated the most from the baseline. Results also suggest differences in FA profile composition between the two species at one site. This work provides the needed baseline information to improve the reliability of FA-based dietary analyses in preserved mussel specimens. By addressing methodological uncertainty associated with archived tissues, this study enhances the potential for historical collections to be used in reconstructing mussel diets and ecological change through time.

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Non-lethal methods to create length-mass regressions for freshwater mussels: calculate biomass of two native species in western North America from shell length

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Native freshwater mussels can be pivotal members of aquatic ecosystems, but their numbers are declining and many basic information gaps exist. Biomass is a critical indicator of a species' status, and can be calculated from a relationship between an individual's length and mass to more easily estimate the population biomass of a species. Lethal methods are typically used to create length-mass regressions, but sacrificing individuals of a rare, declining species is undesirable. We used a non-lethal method to create length-mass regressions for the California Floater (*Anodonta californiensis/nuttalliana*) and Western Pearlshell (*Margaritifera falcata*) based on shell and length and wet mass from individuals in the Bear River, Wyoming, USA. We related shell length to the dry mass of shells and soft tissue. Modeled biomass did not differ from measured biomass for Western Pearlshell or California Floater (glm; $p > 0.33$). Biomass is a critical measure to assess populations and our method allows the development of length-mass equations using non-lethal methods for rare mussels of management concern.

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Assessing native, freshwater mussel distribution in the Great Lakes National Forests

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The lack of basic information on native, freshwater mussel (Bivalvia: Unionidae) populations such as distribution, population status (e.g., abundance, size distribution), and required habitat are recognized as major barriers to implementing a Great Lakes basin-wide conservation strategy for unionids, including on National Forest lands. National Forests in the Great Lakes region represent about 5% of the land-base but little is known about unionids in these areas. This knowledge gap reflects a linked set of multifaceted challenges. First, there is a lack of clear, standardized methodology for National Forest biologists, staff, and partners to use for conducting unionid surveys on National Forests. Existing survey protocols for unionids are highly variable and at present, little guidance exists for deciding which protocol to use. Second, many existing protocols are designed for single basins or rivers. National Forests in the Great Lakes basin encompass >3.7 million acres, >7,500 km of rivers, and thousands of road-stream crossings. As such, there is a need to develop site selection guidelines that efficiently target a wide diversity of streams. The survey approach needs to cover large areas robustly but efficiently, while also providing opportunity to survey specific areas to monitor the effects of management activities in targeted areas. Third and finally, there is a need to identify how to best archive and share unionid survey data both within USDA Forest Service and to external partners and the public. Therefore, our project aim is to develop a standardized survey protocol for the National Forests in the Great Lakes basin to assess unionid assemblages. This standardization will increase knowledge-sharing among partners and agencies in ways that support monitoring, protecting, and restoring unionids, particularly within the Great Lakes Basin. Additionally, we are working in

collaboration with federal, academic, and state agencies to coordinate data sharing and ensure our protocol results can be collated, curated, and archived into a searchable database and map(s). We will be pilot testing our protocol this summer (2026) on four national forests in the Great Lakes Basin. Therefore, we welcome both feedback and input at this stage in our protocol development.

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Age and growth in five species of freshwater mussels in Bogue Chitto River, Louisiana

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Freshwater mussels (Order: Unionoida) are a diverse and imperiled group of organisms that make up an important part of global biodiversity, yet they remain understudied. Life history traits such as maximum age and growth vary interspecifically due to trade-offs in energetic investment toward reproduction and growth. Understanding life history trait variation of mussels can aid conservation efforts by indicating how vulnerable populations may be to certain disturbances. Louisiana's coastal plains rivers and subtropical climate offer unique habitats in which to characterize variation in mussel life history traits, and many of the watersheds in the state are data deficient. Our goal is to compare age and growth rates for five different species spanning four phylogenetic tribes and two life history strategies from one site in the Bogue Chitto River, Louisiana. We hypothesize that species within the tribes Amblemini, Pleurobemini would be slow growing and older, whereas species within the tribes Quadrulini and Lampsilini would be faster growing and younger. We also hypothesize equilibrium strategists will exhibit slower growth than periodic species. To date, we have identified and counted annual growth rings (annuli) from radial thin sections cut from 10 individual shells for each of two species (*Quadrula nobilis* (Quadrulini) and *Plectomerus dombeyanus* (Amblemini)) collected in the fall of 2025. Individual lengths ranged from 75-100 mm for *P. dombeyanus* and 51-80 mm *Q. nobilis*. Age estimates based on counts of annuli ranged from 7-12 years for *P. dombeyanus* and 10-16 for *Q. nobilis*. Age estimates for *P. dombeyanus* are consistent with previous estimates for the species, whereas our age estimates for *Q. nobilis* represent the first attempt at aging this species. These results improve our understanding of mussel life history trait variation in coastal plains habitats and fill data gaps for species lacking age or growth data in Louisiana. Future work will model growth rates and maximum age for these species in addition to *Fusconaia cerina* (Pleurobemini), *Lampsilis straminea* (Lampsilini), and *Pustulosa pustulosa* (Quadrulini) and include individuals of all species from eight coastal plains watersheds across the gulf coast, including comparisons of current age and growth patterns to museum-housed historical shells.

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Exploring historical trends in Wyoming freshwater mussel distribution through an archaeological collection

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Are all of Wyoming's extant freshwater mussel species native to the state's watersheds, or were they introduced by human activity? Have Wyoming's mussel populations fluctuated in abundance throughout history, or are declines a modern issue? These and similar questions are difficult to answer in the absence of historically recorded data. Freshwater mussels are often found at archaeological sites and frequently help answer human-focused questions, but few are examined in an attempt to answer mussel-focused questions such as above. Freshwater bivalves are a diverse group that perform many ecosystem services, and over 70% of North America's 300 freshwater mussel species are of conservation concern. Little is known about the historic distribution or population size of any of the 7 extant species of freshwater mussels in Wyoming. We reviewed and sorted thousands of shells housed at the Frislon Institute at the University of Wyoming to find identifiable specimens. Shells with intact diagnostic features were identified, and growth rates, age structures, and distributions were compared to modern populations. We have thus far found 132 identifiable shells. The species we have identified were found at sites near the same watersheds where those species exist today. The presence of Western pearlshell (*Margaritifera falcata*) in Sweetwater and Sublette counties suggests that this species may be native to the Green River, which is a long-standing uncertainty. We have also discovered a species not known to Wyoming. Information from molluscs in archeology collections helps frame current distributions and population trends into a longer-term context to inform future management plans.

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Trends in sediment and nutrient concentrations at multiple temporal scales in a southwest Ohio watershed.

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Over decadal time scales, water quality in streams often changes due to shifts in land use or climate. Studies that examine shifts in water quality typically quantify changes at annual time scales. Furthermore, some studies report annual findings based on calendar years, while others report water years. For seasonal comparisons, some studies utilize meteorological seasons while others utilize astronomical seasons. Stresses applied on water can have variable effects depending on season (e.g. summer algal blooms) and water quality improvements in one season and a decline in another can be masked if data are only examined at annual scales. In this study, we will be evaluating water quality at various temporal scales to test if annual trends mask significant seasonal changes. This research analyzes 30 years of data from 3 adjacent streams within the Acton Lake watershed in southwest Ohio. We investigate patterns in nitrate (NO₃), ammonium (NH₄), soluble reactive phosphorus (SRP), and

suspended solids (SS) concentrations. We also examine if trends in water quality are consistent between temporal grouping options. For example, for annual comparisons, we compare calendar year vs. water year trends and for seasonal comparisons, we compare meteorological vs. astronomical seasonal trends. This improved understanding can be used to judge where efforts are best spent towards improving water quality and can help to inform future watershed management approaches.

Hyporheic resilience: Invertebrate communities minimally affected by variable channel stability along a gravel-bed river.

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The major ongoing effect of climate change in central North American rivers is increasing frequency and magnitude of floods, driving increased streambed sediment mobilization, lateral migration, and bank erosion. The hyporheic zone is a physical habitat within these shifting sediments, but little is known of how hyporheic invertebrates might respond to increasing habitat instability. Given previously observed patterns of alternating stable and unstable channels along the Gasconade River (Missouri, USA), we asked if invertebrate communities in hyporheic gravels varied between stable and unstable locations. We chose 3 sites along a ~55-km length of the Gasconade, each with paired stable and unstable reaches. During baseflow in July 2025, we collected 4 Bou-Rouch style samples within each ~100-m reach, all located at the downstream end of glides. All samples were collected from ~45 cm hyporheic depth at points of shallow surface-water depth (~0-0.3 m) and with minimal evidence of upwelling or downwelling. We also measured dissolved oxygen and temperature of hyporheic water prior to invertebrate sample collection. Invertebrate communities were diverse, with 29 unique taxa overall (25 from stable and 22 from unstable reaches), with a blend of benthic- and groundwater-associated taxa indicating the ecotonal nature of functional hyporheic zones. Samples also included an amphipod (*Allocrangonyx hubrichti*), previously known only from true groundwater. Unstable reaches tended to have greater invertebrate densities (individuals L⁻¹), but overall community structure varied just as much within individual reaches as among reaches, suggesting high heterogeneity at fine spatial scales and that broader-scale patterns of channel stability have not fundamentally affected hyporheic communities. Furthermore, local-scale variables such as dissolved oxygen and water temperature did not help explain differences in community structure among samples. These results imply resilience of hyporheic communities to the increasing movement of gravel beds associated with climate change. Future studies directly following bed-moving floods will increase understanding of how hyporheic communities respond to variably shifting channels. Hyporheic invertebrates are a major source of secondary production in gravel-bed rivers, so it is important to understand their response to ongoing changes in disturbance regime.

How does fire stewardship affect carbon cycling in wetlands?

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Wetlands influence global carbon (C) cycling through high rates of C sequestration and methanogenesis. The net greenhouse gas impact of wetlands is still being debated. Wetlands are highly valuable to Indigenous peoples, who hold cultural connections to these ecosystems. Indigenous peoples have long understood that water and fire are interconnected, practicing fire stewardship in wetlands for ecological and cultural objectives. Fire suppression led to an accumulation of flammable undergrowth, and together with climate change, creates conditions that favor larger, more destructive wildfires. My doctoral research focuses on human-fire regimes in wetland ecosystems through Indigenous fire stewardship (also known as *cultural fire*) and Western science-informed fire stewardship (also known as *prescribed fire*). Alterations in wetland fire regimes can influence greenhouse gas accumulation; therefore, research on fire stewardship in wetlands is essential.

My experiment compares three treatments: cultural fire, prescribed fire, and no fire (control) in a wetland in northeastern Kansas. I will measure C in wetland soils, vegetation, water, and gas emissions before and after prescribed and cultural fire, and in the control plot. I will monitor water depth in both treatment plots and the control, and measure temperature during the two fire treatments. Soil cores will be used to collect data on physical soil characteristics and to determine LOI, which provides an estimate of soil organic matter. I will perform elemental analysis to obtain an accurate estimate of the C stored in the soil. I will determine above-ground plant biomass from vegetation quadrat samples. Elemental analysis will also be performed to quantify the amount of C stored in the plants. Gas samples collected from static chambers will be analyzed for gaseous C (carbon dioxide (CO₂) and methane (CH₄)) and other greenhouse gases. I will monitor water table depth in minipeizometers and collect water samples for water quality testing, including parameters for organic C. Studying C in various forms before and after the fire treatments and in the control plot will clarify the effects of fire on C cycling in wetlands, highlighting wetland fire stewardship as a prospect for climate change adaptation.

1. Eriksen, C., & Hankins, D. L. (2015). Colonisation and fire: Gendered dimensions of indigenous fire knowledge retention and revival. In A. Coles, L. Gray, & J. Momsen (Eds.), *The Routledge Handbook of Gender and Development* (pp.129-137). New York, United States: Routledge.
2. Lake, F. K., & Christianson, A. C. (2019). Indigenous fire stewardship. In S. Manzello (Eds.), *Encyclopedia of Wildfires and Wildland-Urban Interface (WUI) Fires* (pp. 1-9). Springer. https://doi.org/10.1007/978-3-319-51727-8_225-1
3. Medvedeff, C.A., Inglett, K. S., Kobziar, L. N., & Inglett, P. W. (2013). Impacts of fire on microbial carbon cycling in subtropical wetlands. *Fire Ecology*, 9(1), 21-37. DOI: 10.4996/fireecology.0901021
4. Mitsch, W. J., & Gosselink, J. G. (2015). *Wetlands* (5th ed.). Wiley.
5. Temmink, R. J. M., Lamers, L. P. M., Angelini, C., Bouma, T. J., Fritz, C., van de Koppel, J., Lexmond, R., Rietkerk, M., Silliman, B. R., Joosten, H., & van der Heide, T. (2022). Recovering wetland biogeomorphic feedbacks to restore the world's biotic carbon hotspots. *Science*, 376, eabn1479. DOI: 10.1126/science.abn1479

Limnological monitoring of Lake Roosevelt for tribal fisheries resilience

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The Lake Roosevelt Fisheries Evaluation Program was established to monitor and evaluate how hydropower operations affect water quality and fishery resources of Lake Roosevelt in northeastern Washington. Prior to the construction of the Grand Coulee Dam (1933–1941), the upper Columbia River supported salmon runs that were both culturally and nutritionally essential to Indigenous communities. The dam's construction blocked salmon migration and inundated tribal lands, creating the reservoir now known as Lake Roosevelt. Operational impacts from Grand Coulee Dam alter flow regimes, water temperature, and turbidity, influencing ecosystem structure and function - from primary productivity to fish migration, survival, and reproduction. Working under Spokane Tribal Fisheries, the Limnology Team contributes to this tribal-led management effort by monitoring the lake's baseline food-web dynamics and the abiotic conditions that shape habitat suitability and overall water quality. Our mission is to preserve, protect, and enhance the long-term sustainability of tribal fisheries for present and future generations who remain closely connected to this culturally significant waterbody. Here, we present the limnology component of this long-term monitoring program, highlighting multi-decadal data collection efforts and outlining directions for future research and adaptive management.

Shapley Solutions for Sustainability: Contextualizing Equitable Cost-Sharing in Rural Water Infrastructure Financing in Nepal

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This paper presents a novel application of Shapley value decomposition to determine equitable cost-sharing plans for the operation and maintenance (O&M) of rural drinking water systems in Nepal. Rooted in cooperative game theory, the Shapley framework provides a fairness-based metric to divide financial responsibility among important stakeholders, such as households and the government, based on marginal contributions by each to the system. Utilizing a nationally representative survey of 9,600 households and targeted fieldwork in Rupandehi (Terai), Kaski (Hills), and Chitwan (Inner Terai), the analysis incorporates K-means clustering to identify regional cost-sharing profiles that represent institutional, ecological, and economic variation.

As per the analysis the ideal national contribution was determined to be 66.69% from the government and 33.31% from households. Significant geographic heterogeneity however, emerged which shows household contributions should be lowest to 26.42% in the Terai plains, and rises to 45.78% in the Inner Terai. Meanwhile the cost burden rises to 53.15% in hill regions. These discrepancies are a result of variations in household engagement, service delivery reach, and sunk costs associated with terrain. The results show that equity in infrastructure financing needs to be context-responsive, challenging one-size-fits-all co-finance formulas.

The study makes a methodological contribution to public infrastructure financing models by establishing mathematical equity principles under duress of real world governance complexity of real-world governance. Study also provides a proper policy prescription in creating sustainable co-finance plans under federalism. Philosophically study helps to address Fraser's Crisis of Care by revaluing maintenance labor through a cooperative and equitable cost-sharing lens.

Assessing the Impacts of Microplastic Pollution on *Lasthenia fremontii* in California Vernal Pool Ecosystems

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Ephemeral pools, found worldwide and once abundant along the Pacific coast of the United States, are biodiversity hotspots that support threatened and endangered species and are highly susceptible to anthropogenic agents of disturbance and pollution. Although previously abundant in the California floristic region, over 90% of vernal pools have been lost in the state since European colonization, due to habitat destruction as a result of agriculture and other forms of land development. The remaining pools exist within a highly fragmented landscape and are often located in proximity to roads, highways, and landfills. Microplastic pollution has been observed entering terrestrial ecosystems from a variety of sources and has accumulated over decades of discardment and slow degradation. Due to their seasonal pattern of inundation and recession via evapotranspiration, vernal pools may be hotspots for microplastic accumulation and fragmentation, but to date, our understanding of vernal pools and microplastic accumulation remains limited. This study examines the occurrence of microplastics within vernal pools located in the Central Valley of California and their subsequent effects on *Lasthenia fremontii*, an annual herb endemic to these pools. Soil and water samples were collected from five vernal pool complexes in the Central Valley of California during the dry and wet seasons in Fall 2025 and Spring 2026. Additionally, in Spring 2026, we designed and implemented a full factorial greenhouse experiment investigating the effects of four microplastic treatments (concentrations of <2 mm polyethylene fragments- 0, 0.5%, 1.0%, 2.0%), and two hydrologic treatments (well-watered and drought) on the growth and reproductive fitness of *Lasthenia fremontii*. Vernal pools play a unique and essential role in supporting threatened and endangered taxa. Understanding the microplastic load within these systems and investigating their effects on an endemic plant will improve our understanding of environmental health and security, while highlighting the role terrestrial ecosystems, such as vernal pools, play in the plastisphere.

Effects of drying frequency and duration on ecosystem metabolism in non-perennial streams

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More than half of the river miles in the United States are characterized by intermittent flow conditions, and this proportion is expected to increase as a result of climate change and anthropogenic activities (e.g., groundwater pumping). Despite the ubiquity of non-perennial streams, the effect of stream drying on ecosystem function is not well understood. Our goal is to understand how the frequency and duration of stream drying affects ecosystem metabolism. Ecosystem metabolism, which represents the dual processes of gross primary production (GPP) and ecosystem respiration (ER), is a key metric of stream ecosystem function that integrates nutrient and carbon cycling. Using USGS, NEON, and field-collected data, we estimated monthly ecosystem metabolism from 37 USGS gages, 2 NEON sites, and 6 additional sites classified as non-perennial. Linear mixed models revealed that temperature and precipitation were the dominant controls on both GPP and ER, while drying metrics showed more nuanced effects. Specifically, the duration of flow decline before stream drying negatively influenced GPP. Low-flow event frequency positively influenced ER, consistent with microbial respiration pulses following rewetting events. Notably, the duration of individual drying events showed minimal effects on either metabolic process, suggesting that drying time may be more important than absolute dry period length. These findings indicate that increasing frequency of drying under future climate scenarios may alter stream carbon cycling through repeated drying-rewetting cycles rather than through extended drought duration alone.

Ten Years of Timber Harvest Effects on Stream Intermittency in Southwest Washington

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Streams classified as non-fish-bearing perennial (Np) streams in Washington include dry channels that may occur downstream of the highest point of perennial flow. During the low-flow summer months these dry sections can disconnect areas of surface flow from other flowing reaches. To understand the spatial and temporal shifts of the dry reaches in Np streams, we used data collected as part of a larger study by Ehinger and colleagues (2021). This wetted extent survey measured the length of every dry reach in 11 non-fish-bearing watersheds, located in southwest Washington, for 10 years (2013-2020, 2024-2025). In 2014 and 2015, a clearcut timber harvest treatment was implemented at 8 of the 11 basins (3 were left as unharvested references). We used a generalized linear mixed model to test treatment effects on the proportion of the stream network that retained surface flow. We saw an increase in the Proportion of Wetted Channel (PWC) in the first two years after harvest and a return to baseline conditions by year three which continued through year six. A ten-year follow-up survey was conducted in 2024 and 2025. This resulted in substantial declines in the PWC at the treatment sites, with little change in the surface flow patterns of the reference basins.

1. Ehinger, W., Bretherton, W., Estrella, S., Stewart, G., Schuett-Hames, D., & Nelson, S. (2021). Effectiveness of Forest Practices Buffer Prescriptions on Perennial Non-fish-bearing Streams on Marine Sedimentary Lithologies in Western Washington. Washington State Forest Practices Adaptive Management Program, Washington Department of Natural Resources, Olympia, WA.

Adult emergence in aquatic insects: contrasting intermittent and perennial mountain streams at the HJ Andrews Experimental Forest

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The emergence of aquatic insects influences the diversity and ecology of aquatic and terrestrial systems. With an increasingly variable climate worldwide, streams may be shifting between perennial and intermittent states. How the community composition will respond to or be impacted by these shifts is uncertain. In this study, we studied the environmental variables influencing the composition and structure of adult aquatic insect communities among adjacent streams with different flow conditions. For this purpose, during the summer of 2022 at the HJ Andrews Experimental Forest, we collected aquatic insects using emergence traps each three days two months, then identified them in the laboratory to the lowest taxonomic level and analyzed the structure and composition of these groups. At the time of collection, we also measured environmental variables, including water temperature, channel width and depth, canopy cover, and substrate. Across 9 intermittent and eight perennial streams, Ephemeroptera and Plecoptera showed higher abundances in intermittent sites, while Trichoptera maintained consistent abundances under differing conditions of water flow effect. Our preliminary results showed that the wetted width, temperature, and altitudinal range are good predictors of insect community composition for these intermittent and perennial tributaries of Lookout Creek (OR). Also, some genera, such as *Yoraperla* and *Wormaldia*, may respond to desiccation. Regarding the flow of aquatic-derived biomass into adjacent terrestrial ecosystems, peaks in emergence during the first sampling dates (late June- late July) correspond to periods

of elevated cross-ecosystem energy transfer, providing key subsidies that support riparian predators such as spiders, beetles, and birds, as well as other terrestrial consumers.

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Precipitation Effects on fDOM C-Q Relationships of Intermittent Streams in Kansas

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Intermittent streams play several important roles for both the organisms who inhabit them and the humans who rely on them for drinking water. As precipitation patterns shift in intensity and variability, the question arises of how variable streamflow patterns will impact downstream water quality. Fluorescent dissolved organic matter (fDOM), a proxy for dissolved organic carbon (DOC), varies with changing streamflow, but the constraints on these concentration-discharge (C-Q) relationships in intermittent streams are less known. To understand how different wetting and drying patterns influence the solute export from these systems, we examined fDOM C-Q relationships across three intermittent watersheds from 2022-2024 to (1) identify the C-Q behaviors during high and low flow events and (2) assess the drivers linked to these behaviors. We monitored fDOM using EXO water sensors deployed at the outlets of three streams: Kings Creek (KNZ), Shane's Creek (SHN), and Youngmeyer Ranch (YMR). In our preliminary findings, we examined the percent contribution of carbon (C) export during high flow events for each year and site. Preliminary results show that C export during the top 10% of peak flows was high in 2022 at KNZ and YMR (82.4% and 66.7%, respectively), low in 2023 for KNZ and SHN but moderately the same for YMR (22.8%, 49.1%, and 63.4%, respectively), and high for all sites in 2024 (KNZ = 55.9%, YMR = 75.9%, and SHN = 99.2%). We plan to examine the frequency and duration of dry antecedent conditions and identify site- and year-specific C-Q behaviors using total C export alongside stream connectivity and conductivity. Although annual precipitation rates varied differently each year, other factors—such as total annual discharge—could influence export rates. Thus, predict that KNZ and SHN will shift between transport during years of high precipitation to source-limited during drier years, while YMR remains transport-limited across all conditions. Our analysis will reveal differences in fDOM C-Q behavior across intermittent stream networks in the Great Plains and identify key drivers of these C-Q behaviors. This work lays the foundation for better predicting how DOC loading will change under future climate and streamflow variability.

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National horizon scan of terrestrial and aquatic invertebrates assessing the social media pathway of the pet trade

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The pathways of importing organisms provide the opportunity for a significant number of potentially invasive species being transported to novel environments and ecosystems, providing an avenue for establishment and harm where introduced and established. The pet trade is an important component of this organism-in-trade pathway. Invertebrates have been a widely traded sector of the pet trade, with the number of traded individuals increasing coinciding with an increase in internet and social media usage within the trade. Social media and internet-based stores allow for more people to engage with the pet trade as social media reduces the need for personal connections, provides anonymity, eases purchase process, and decreases prices. This research focuses on the attitudes of buyers and sellers moving freshwater and terrestrial invertebrates within the pet trade and the differing perceptions of the media consumed, specifically determining the incentives for people to post and interact with this media. We will analyze account holders, individual posts, and comments on 5 different platforms to reveal intentions of the account holder when making posts and the reaction from the public to these posts. We additionally will survey account holders regarding their practices within the trade and their attitudes regarding the movement of invasive species within the trade. Results of this study will help to create a list of species being moved through this pathway, and the risk of these species potentially being introduced as an invasive species. This work will inform public perception of terrestrial and aquatic invertebrates, risk of import of species, and potential risk of becoming invasive to best direct regulations and introduction prevention methods.

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Ecological traits mediate contrasting effects of stream network connectivity on the success of invasive fish

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Biological invasions in freshwater communities can result in great economic and ecological harm, including the loss of native species and ecosystem function; prediction of the course of invasions is important in targeting efforts to prevent or mitigate such damage. The aim of this research is to combine ideas from metacommunity theory with invasion ecology to develop a predictive model of the success of invasive fishes in stream systems. Principles of metacommunity theory and invasion ecology suggest that invasive species might be disproportionately successful in either well-connected or poorly-connected sites within stream networks; the former because of the ease of dispersal for invaders into well-connected sites, the latter due to poorly-connected sites having smaller, less diverse native communities which may be more vulnerable to invasion. This research will investigate how the balance between these two contrary effects for a given invader is related to the traits of that species and of the watershed

it has invaded. Using fish community assessment data from thousands of streams in Minnesota, Iowa, Wisconsin, Illinois, and North Carolina, we will analyze how the effect of connectivity (measured as network centrality) on invasion success relates to selected species traits, using a generalized linear mixed model. We hypothesize that ecological traits related to the ability to disperse to many new sites (e.g. fecundity) and the ability to establish a foothold despite biotic resistance (e.g. habitat generalism) may mediate the effect of connectivity on invasion success. Our model will provide valuable insights into what types of invasive species are more likely to establish populations in different areas of the network.

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Ecological and Hydrological Thresholds Differentiating Native and Non-native Fish Occurrence in Great Plains Streams

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Stream fish assemblages are shaped by natural hydrologic variability, but flow alteration and environmental change can push systems beyond ecological thresholds that trigger rapid shifts in community structure. In stream systems characterized by naturally variable flow regimes, small changes in hydrologic or environmental conditions may result in disproportionate losses of native species and facilitate the establishment of non-native fishes. Understanding where these thresholds occur and how assemblages respond is essential for anticipating community change under ongoing hydrologic modification. We used fish occurrence data from Great Plains streams and integrated these records with hydrologic and environmental variables. Species were classified as native or non-native and assigned to trophic groups. We examined the influence of hydrologic and environmental gradients on assemblage and species level threshold responses using gradient forest. We expect to observe nonlinear shifts in assemblage composition along environmental gradients. Native species occurrence and site-level persistence are anticipated to remain relatively stable across portions of environmental gradients but then decline more abruptly once particular hydrologic or environmental thresholds are exceeded, rather than exhibiting simple linear declines. We also anticipate that the specific position and magnitude of these threshold responses may vary among sites or hydrologic contexts, which would highlight the value of identifying specific benchmarks for management. These results aim to inform fisheries conservation and management strategies focused on maintaining native biodiversity and ecosystem function across hydrologically diverse Great Plains stream ecosystems.

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Combining bioenergetic and demographic modeling to estimate consumption of native organisms by invasive Sacramento Pikeminnow in the Eel River, California

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Around 1980, Sacramento Pikeminnow were introduced to northern California's Eel River Basin and spread rapidly, further straining a system where large declines in native fish abundance had already been observed over recent decades. Invasive Sacramento Pikeminnow are highly piscivorous and have been identified as an impediment to the recovery of threatened Eel River Coho Salmon, Chinook Salmon, and steelhead. Though funding and resources have been and will continue to be dedicated to removing these invaders, there has to date been no attempt at quantifying the amount of native fish they consume. Without this information, there can also be no corresponding estimates of the potential reduction in predation of native fish as a result of pikeminnow removal. In this project, I will first seek to develop a bioenergetics model to estimate the amount of Chinook Salmon, Coho Salmon, and steelhead consumed by Sacramento Pikeminnow. This will provide ecosystem managers with the ability to set objective, measurable management goals directly relating to the fish they are trying to recover. Second, I will combine this model with an already-developed demographics model to compare and optimize suppression regimes. This will provide managers with the optimal size selectivity, suppression gear types, and suppression timing and spatial regimes to design evidence-based suppression programs for effective pikeminnow management.

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Impact of Polystyrene Microplastics on the Bioaccumulation of Silver Nanoparticles in *Daphnia magna*

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Microplastics (MPs) are a significant environmental concern due to their persistence and potential interactions with other contaminants. The interaction of MPs with AgNP, for example, could pose additional threats to organisms. Here, we assessed the influence of polystyrene microplastics (PS-MPs) on the bioaccumulation processes (uptake, elimination, and dietary

assimilation) of silver (Ag) in the filter-feeding zooplankton *Daphnia magna*, using environmentally relevant concentrations of Ag. Results show that the presence of PS-MPs increased the influx rate (i.e., waterborne uptake) of Ag into *D. magna* by eightfold compared to that from AgNP alone. In addition, the presence of PS-MPs altered Ag elimination kinetics in *Daphnia magna*, resulting in faster and more complete depuration, with 98% eliminated after 5 days compared to 66% in their absence. The presence of PS-MPs reduced the algal food-ingestion rate in *D. magna*, an effect that was independent of Ag concentration. Modeling Ag concentrations at steady state using a biodynamic model showed that, when PS-MP are present, the primary pathway for Ag bioaccumulation in *D. magna* shifts, making waterborne exposure more dominant than dietborne exposure. These findings provide critical insights for assessing the ecological risks of metals to zooplankton. They also highlight that for filter-feeding organisms such as *D. magna*, particles such as PS-MPs can act as Trojan horses for contaminants like AgNP. Additionally, these results underscore the complexity of Ag bioaccumulation when microplastics and nanoparticles interact.

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A survey of gastropods across the state of Missouri using a wide-spread volunteer participatory science network

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Freshwater mollusks are widely used as bioindicators in the development of water quality criteria, including those for ammonia. EPA's most recent freshwater ammonia criteria guidelines incorporate prosobranch gastropods. Freshwater mussels are commonly surveyed in Missouri, however, comparable data for gastropods are limited. In 2020, a volunteer-based statewide survey of freshwater snails was initiated to generate baseline occurrence data to support the development of ammonia water quality criteria for Missouri. Specimens were collected by volunteers participating in a statewide participatory science program and submitted for identification and data analysis. We will give an overview of the data collected over the last 5 years and discuss the future of this project. This project documents the presence of multiple freshwater snail taxa across diverse habitat types and geographic regions, represents the first coordinated volunteer-based effort to survey freshwater snails in Missouri, and contributes to a growing statewide dataset on gastropod occurrences.

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Macroinvertebrate assemblages associated with blackwater streams and wetlands in Alabama

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Wetlands provide 40% of global ecosystem services despite covering just 1.5% of Earth's surface. However, wetlands are disproportionately threatened by human development, pollution, and climate change. Alabama is home to a variety of wetland types, each harboring unique flora and fauna, including blackwater wetlands. These wetlands are typically associated with blackwater streams that originate below the fall line, with increased tannins, organic matter, and acidity. The Alabama Department of Environmental Management (ADEM) is responsible for monitoring surface waters, including blackwater wetlands, yet full incorporation of wetland monitoring has been difficult. To assist in these efforts, we are establishing baseline reference conditions for aquatic macroinvertebrates in blackwater systems in southwest Alabama.

Seventeen stream/wetland systems were selected based on stream pH, color, and conductivity in southwest Alabama to establish a tannic gradient (13 sites along the gradient and four sites as alluvial outgroup). Macroinvertebrates were sampled during summer 2025 in the streams of each system in all available habitat types using d-nets according to ADEM protocols. Physicochemical were collected during stream sampling this previous summer (temperature, pH, conductivity, dissolved oxygen, and color) along with habitat conditions (discharge, flow and depth, organic matter content, and sediment profiles). Macroinvertebrates are currently being processed, and wetland sampling will commence in spring 2026.

Relationships between abiotic conditions and biotic communities will be visualized using nonmetric multidimensional scaling (NMDS), and differences between blackwater and alluvial systems will be quantified using permutational analysis of variance (PERMANOVA). Linear mixed-effect models will be used to describe macroinvertebrate metrics (e.g., diversity, richness, etc.) along the tannic gradient. From these results, taxonomic and functional baseline criteria will be established for improved, targeted monitoring and assessment through ADEM as well as for groundwork for future studies in these unique systems.

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Hydropsychidae density and diversity in relation to substrate embeddedness and flow in the Spokane River watershed

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Hydropsychidae (net-spinning caddisflies) are common riffle macroinvertebrates whose larval attachment and feeding depend on stable, coarse substrate and flowing water. Fine sediment deposition can increase substrate embeddedness, reducing interstitial habitat quality, while reduced current velocity can limit food delivery and net-spinning efficiency. Hydropsychidae are among the most abundant invertebrates in the Spokane River. They can have significant ecological effects through their filtering of seston and through creating microhabitats that favor bacterial production and nutrient cycling. They are also widely used as sentinels of pollutant bioavailability. Although they are abundant in the Spokane River, their distribution in the Spokane River watershed is not documented. This study tests relationships between Hydropsychidae diversity and density and substrate embeddedness and current velocity.

We sampled benthic macroinvertebrates from sites in the Spokane River watershed during the summers of 2024 and 2025. At each site, four standardized kicknet subsamples were collected from riffles and composited into a single sample. Flow velocity was measured just above the substrate surface using a Hach 950 flow meter for each subsample. Particle sizes were characterized using the Wentworth scale and substrate embeddedness was also estimated for each subsample. We predict that *Hydropsyche* density will be highest in coarse, low-embeddedness substrates under moderate to high flow conditions, and lowest in embedded substrates with reduced current velocity.

Sample processing is still underway. The genera *Hydropsyche* and *Cheumatopsyche* appear to be widespread within the watershed. Preliminary results indicate that *Hydropsyche* were present at 9 of the 12 sampled sites within the watershed. Among sites where *Hydropsyche* occurred, their proportional abundance relative to the total macroinvertebrate community ranged from <1% - 40%, indicating substantial variability in dominance across sites.

Our results should clarify whether *Hydropsychidae* can serve as a practical indicator of fine-sediment and flow-related habitat degradation in the Spokane River watershed.

Cicadas in the Stream: Quantifying Periodical Cicada Drift Densities in Streams

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Every 13 and 17 years, billions of periodical cicadas (*Magicicada* spp.; hereafter cicadas) emerge from subterranean burrows, delivering a substantial pulse of biomass and nutrients that alters ecosystem structure and function across landscapes. Despite the magnitude of these events, critical gaps remain in our understanding of how freshwater ecosystems are affected by cicada emergences, particularly related to the magnitude and composition of cicada-derived subsidies entering streams. Our objectives were to provide a foundation for future research by: 1) estimating the density and biomass of cicadas drifting in streams, and 2) characterizing the taxonomic composition and sex ratios of cicadas entering streams. We counted individuals from three cicada species groups (*Magicicada* -*decim*, *M. -cassini*, and *M. -decula*). During the dual emergence of Broods XIX and XIII in 2024 across Arkansas, Missouri, Illinois, and Iowa, and later Brood XIV in 2025 across Tennessee, Kentucky, and Ohio, we deployed drift nets (26 x 45 cm) for 30–60 minutes in different streams for a total of 88 streams. Cicadas were present in the drift at 88% of sites, ranging from 0 to 79 individuals per deployment. Preliminary analyses indicate that drift densities ranged from zero to 45 cicadas-h⁻¹ and biomass inputs from zero to 5.43 g-h⁻¹. These estimates are likely conservative, as they do not account for cicadas consumed by aquatic macroorganisms prior to reaching the drift nets. Concurrent studies of fish foraging behaviors in a subset of these streams suggests that a substantial fraction of drifting cicadas are intercepted by fish and other macroconsumers. Of cicadas collected from the drift, approximately 50% were male, 35% female, and 15% undeterminable. Approximately 40% were identified as *M. -cassini*, 18% as *M. -decula*, and 6% as *M. -decim*, with 36% unidentifiable. By quantifying not only the magnitude of cicada inputs but also their species composition and sex-specific contributions, this study provides the first broad-scale assessment of periodical cicada subsidies entering lotic ecosystems. These datasets establish a critical baseline for evaluating how rare, pulsed terrestrial resource events influence aquatic food webs, nutrient cycling, and consumer dynamics across spatial and temporal scales.

Effects of nutrient enrichment and groundwater calcium on gastropod distribution in the Spokane River watershed

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Gastropods are the second most diverse animal class after insects. In freshwater benthic communities, they are tolerant to a range of flow regimes, sediment types, and water quality. Gastropods are primarily scrapers that graze periphyton and biofilms. These food sources often increase with nutrient enrichment from agricultural inputs (i.e. high levels of nitrogen and phosphorus that can cause algae blooms). While nutrient loading can create more food, it can also create water conditions that restrict shell formation. Snail shells are composed primarily of calcium carbonate (CaCO₃), and their calcification depends on calcium ion availability and stable pH conditions in the water. The Spokane Valley–Rathdrum Prairie Aquifer is characterized by high concentrations of calcium and bicarbonate, creating chemically favorable conditions for CaCO₃ shell formation in reaches influenced by groundwater discharge. In contrast, agricultural land use is associated with increased nitrogen and phosphorus loading and reduced pH, conditions that promote algae growth but can limit available calcium.

This study aims to examine how nutrient enrichment and groundwater-derived calcium interact to influence gastropod density, richness, and diversity in the Spokane River watershed. Benthic macroinvertebrates were collected from four riffle subsamples per site multiple tributaries of the Spokane River with varying levels of agricultural land use and aquifer inputs during the summers of 2024 and 2025. Gastropods are identified to the lowest practical taxonomic resolution, typically genus or species. So far, we have observed four different gastropod families, and six genera within those families. Percent agricultural land use will be used as a proxy for nutrient load, and published information will be used to identify areas of aquifer influence in Hangman Creek, the Little Spokane River, and the Spokane River.

We hypothesize that gastropod quantity will increase with agricultural land use up to a certain point, beyond which environmental stressors will reduce population density. We further predict that gastropod richness and diversity will be highest in areas of groundwater recharge where calcium availability and pH are more stable. Overall, we expect gastropod indicators to peak where ideal nutrient enrichment coincides with groundwater calcium inputs.

1. Larson, Michele, "Gastropods in lower Snake River and three of its tributaries and the effects of competition with the New Zealand mud snails (*Potamopyrgus antipodarum*)" (2013). EWU Masters Thesis Collection. 171. <https://dc.ewu.edu/theses/171>

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Evaluating the utility of functional feeding groups for predicting aquatic insect diet: a case study from Sucker Brook, New York, USA

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Aquatic insect functional feeding groups (FFGs) were defined in the seminal paper on trophic relations of aquatic insects by Cummins (1973). While FFGs are useful for understanding generalized feeding behaviors of insects, they can sometimes misrepresent the true diversity of feeding habits within functional groups. For example, previous work has suggested that the free-living trichopteran, Rhycacophila, which are characterized as predators during the larval stage, have species that may feed as omnivores or herbivores, in addition to those species that feed as predators. Ingested diets can also vary at the individual level, within species, based on site location or season, suggesting opportunistic feeding behavior. Furthermore, ingested items do not always reflect assimilated resources that support aquatic insect production, which means that FFG classifications alone may not provide information about energetic support of insect consumers. Therefore, we asked: (1) do the ingested diets of aquatic insects conform to FFG classifications, and (2) how do ingested diets of aquatic insects compare to estimates of assimilated resources inferred using stable isotope analysis? We collected insect taxa representing 9 families and 14 genera from Sucker Brook, a forested headwater stream in the Adirondack Park (New York, USA). For each individual collected, we conducted gut content analysis (GCA) to compare their ingested diets to assigned FFGs and stable isotope analysis ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) to compare assimilated energy to ingested diets. GCA involved dissection of specimens, filtration of gut contents, and identification and enumeration of food particles (i.e., diatoms, amorphous detritus, cyanobacteria, animal prey, and filamentous algae) using ImageJ. Preliminary results suggest that individual gut contents largely reflect the food habits of their FFG, but some unexpected diet items were observed across all functional groups. Next, we will use stable isotope analysis (SIA) to determine assimilated diet components, which we will compare with GCA results to understand the relationship between ingested diet items and assimilated resources. Our research advances understanding of the cryptic ecologies of aquatic insect taxa and provides new insights into the relationship between ingestion and assimilation of resources in stream food webs.

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Impact of physico-chemical & hydraulic variables and crayfish occurrence on hellgrammite populations

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The relationship of hellgrammites (*Corydalis cornutus*) with physicochemical variables and the co-occurrence of hellgrammites with crayfish, caddisflies, and black flies at the plot level was assessed in summer 2025 as part of a larger three-year study. Sampling was conducted in five central Ohio streams in June and July. Results indicated that Clear Fork River sites had the greatest hellgrammite abundance and the Alum Creek, Big Walnut Creek, and Kokosing River sites exhibited the lowest abundance. Statistical analyses revealed a positive correlation between hellgrammite density and water temperature ($p = 0.031$) and a negative correlation with dissolved oxygen ($p = 0.001$). Hellgrammite occurrence was also negatively correlated with pH ($p = 0.027$) and dissolved oxygen ($p = 0.004$). Log transformed head capsule width showed positive correlations with conductivity ($p = 0.050$) and water velocity ($p = 0.007$), and a negative correlation with crayfish density ($p = 0.037$). Additionally, head capsule width exhibited a negative correlation with crayfish density ($p = 0.037$) and none of the hellgrammite response variables were correlated ($p > 0.05$) with caddisfly and blackfly occurrence. Our results indicated that hellgrammite density and occurrence within riffles of central Ohio rivers are affected by physicochemical variables, and hellgrammite body size is associated with crayfish density. Future analyses will involve investigating the relationships between hellgrammites and nutrient concentrations and the long-term population trends of hellgrammites.

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Tributary Macroinvertebrate Drift Providing Food Resources to the Mainstem Kootenai River

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The Kootenai River is a large, oligotrophic river that relies on tributary resources to sustain the mainstem food web dynamics. Among these critical food resources are macroinvertebrates, particularly for higher-level consumers like fish. We examined the diel abundance of macroinvertebrate drift on three tributaries during three consecutive summers. Expected diel trends would follow cyclic patterns with peaks in drift abundance occurring at night. In this research, only a few samples conformed with what was expected. The remaining samples have peaks at different times of the day, either not following a clear cyclic pattern or having peak drift occur during the day. Site characteristics tend to influence the composition of macroinvertebrates present. The sites with lower water velocity have higher numbers of the mayfly *Paraleptophlebia* sp. and members of the family Corixidae, both macroinvertebrates that prefer slower-moving waters. Although there are differences in composition among site locations, some key taxa remain similar among all samples. The *Baetis* sp. mayfly is the most abundant macroinvertebrate for all but two samples, with an average abundance of 40.4% and a range of 1.7% - 71.2%. Macroinvertebrate drift is a critical component to understanding food resource availability in large oligotrophic rivers and promotes informed conservation strategies.

Woody wetlands as productivity hotspots: contrasting macroinvertebrate biomass and diversity across land use gradients in the Pend Oreille Basin

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Macroinvertebrate diversity, particularly taxa within Ephemeroptera, Plecoptera, and Trichoptera (EPT), serves as a critical indicator of habitat quality and forms the base of the aquatic food web for native fish species, including Westslope Cutthroat Trout (*Oncorhynchus clarkii lewisi*). As these trout are of ecological and cultural significance to the Kalispel Tribe, understanding the land use drivers of their food sources is vital for watershed management.

We sampled macroinvertebrates from tributaries in the Lake Pend Oreille basin to evaluate how land use influences community structure, biomass, and voltinism. Site-specific land use was classified by extracting pixel values from the USGS Annual National Land Cover Database (NLCD) Collection 1.1 using the 'Extract Multi Values to Points' tool in ArcGIS Pro. Sites were categorized as Evergreen Forest (NLCD 42), Woody Wetland (NLCD 90), or Developed (NLCD 21–24). We collected Surber samples ($n = 22$) across these gradients, targeting Evergreen Forest ($n = 17$), Woody Wetland ($n = 3$), and Developed ($n = 2$) reaches.

Our results revealed distinct productivity hierarchies. Woody Wetland sites consistently supported the highest biodiversity (Mean Shannon $H' = 2.21$) and total biomass (>68 mg/sample), driven by large-bodied, univoltine shredders (e.g., *Tipulidae*). Evergreen Forest reference sites exhibited high variability (H' range: 0.56–2.40), where mean diversity ($H' = 1.66$) was frequently dampened by episodic blooms of multivoltine taxa. In contrast, Developed sites maintained high diversity ($H' \sim 1.97$) comparable to reference reaches but supported the lowest standing crop biomass (~ 10 mg/sample), suggesting that urbanization may limit secondary production despite maintaining taxonomic richness.

These findings suggest that woody wetlands function as critical high-production habitats that may subsidize downstream food webs. Furthermore, the high variability in forest reference sites highlights the importance of distinguishing between natural disturbance regimes and anthropogenic degradation when establishing bioassessment criteria. This study provides a refined framework for identifying key habitats that maximize food web stability for native trout populations.

Influence of Limestone Quarry Discharge on Benthic Macroinvertebrate Communities

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Benthic macroinvertebrates are widely used as indicators of stream health because their community structure responds to changes in water quality, habitat stability, and hydrologic conditions. This study examined how differing flow regimes influenced by quarry discharge affect stream physical characteristics and macroinvertebrate communities in three headwater streams in northwestern Ohio. Hog Creek represented a natural flow regime, Little Riley Creek received continuous quarry discharge, and Aurand Run experienced pulsed quarry inputs. At each stream, we sampled riffle habitats using a Surber sampler and recorded geomorphic measurements, including depth, width, and current velocity. We identified macroinvertebrates to family level and used one-way ANOVA with Bonferroni post hoc test to analyze differences in physical habitat and community structure. Depth was the only variable that differed significantly among streams, with quarry-influenced sites being deeper than Hog Creek. Diversity and taxa richness did not differ significantly across sites. These results indicate that although quarry discharge alters some physical stream characteristics, these hydrologic differences did not produce measurable changes in macroinvertebrate diversity or richness at the family level. The findings suggest that benthic macroinvertebrate communities in these small streams are resilient to moderate hydrologic alteration or that finer taxonomic or seasonal sampling is needed to detect quarry-related ecological effects.

Developing new genetic barcodes for endemic Idaho snowflies to improve biodiversity conservation and stream monitoring

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DNA barcoding has become a tool in modern stream ecosystem health studies because barcoding can improve detection of rare species and environmental impacts. Barcoding can be used to identify insects to species in bulk using DNA metabarcodes whereas morphological approaches require identification of each individual under a microscope. However, many species that are important indicators of water quality have yet to be barcoded, including rare and endemic species of conservation concern. Two rare species of stonefly, *Capnia lineata* and *C. zukeli*, appear to be endemic (restricted to) Latah County, Idaho and are challenging to observe because adults are required for identification but emerge only during a brief period in spring. Our objectives are to 1) develop DNA barcodes from available museum and field-collected specimens that can identify adults and larvae, and 2) summarize available records to better understand the factors limiting the ranges of the two species. *C. zukeli* has been historically found in the upper reaches of the Potlatch and Palouse rivers and was last recorded in 1984. Thus only museum specimens of *C. zukeli* are available for barcoding. *C. lineata* is somewhat less rare in tributaries of the Potlatch River. Specimens of *C. lineata* were collected in 2024–2025 and we are barcoding these and historical samples. DNA Barcodes will allow rapid and extensive surveys of larvae in the future to determine the distribution and habitat requirements of both species. The knowledge will aid in managing stream ecosystem health and conservation of Idaho endemic species.

1. Isaak, D.J., S.J. Wenger, E.E. Peterson, et al. 2017. The NorWeST Summer Stream Temperature Model and Scenarios for the Western U.S.: A Crowd-sourced Database and New Geospatial Tools Foster a User Community and Predict Broad Climate Warming of Rivers and Streams. *Water Resources Research* 53 (11): 9181–205. <https://doi.org/10.1002/2017WR020969>.
2. Merritt, R.W. and Cummins, K.W. eds., 2019. An introduction to the aquatic insects of North America, 5th Edition. Kendall Hunt.
3. Nelson, C R and Baumann. 1989. Systematics and Distribution of the Winter Stonefly Genus *Capnia* (Plecoptera: Capniidae) in North America. *The Great Basin Naturalist*. 49: 289–363. <https://doi.org/10.5962/bhl.part.22645>.

The impacts of wildfire and rising temperatures on Spiny Mayfly (Ephemerellidae) diversity and abundance in the Pend Oreille watershed

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The impacts of wildfire and rising temperatures on Spiny Mayfly (Ephemerellidae) diversity and abundance in the Pend Oreille watershed

In the western United States, aquatic macroinvertebrates are increasingly threatened by the dual pressures of rising temperatures and high-intensity wildfires. Spiny mayflies (Ephemerellidae) are critical components of river ecosystems, serving as sensitive bioindicators of water quality and food sources for salmonids. They process detritus and algae, contributing to nutrient cycling and improving water quality through their feeding and burrowing. This study investigates the distribution and diversity of spiny mayflies (Ephemerellidae) in the Pend Oreille watershed.

We hypothesize that wildfire-induced thermal spikes and post-fire sediment runoff would cause a decline in Ephemerellid abundance and diversity. Based on the known sensitivity of Ephemerellidae to temperature and physical habitat changes, we hypothesize that areas impacted by recent forest fires would exhibit lower population densities due to the synergistic effects of increased solar radiation (from canopy loss) and elevated levels of fine sediment runoff. We also hypothesize that water temperature alone would have a negative effect on Ephemerellid abundance and diversity.

We collected 4 benthic invertebrate samples from each of 7 tributaries in 2023 and 13 tributaries in 2024. Some tributaries were sampled in both years for a total of 17 independent sites. Ephemerellid nymphs are identified to the lowest practical taxonomic level, genus or species. Water temperature loggers were used to determine the 7-day maximum temperature for each site (maximum 7-DADMax). Fire records and GIS will be used to determine the proportion of watershed burned within the past 5 years for each site. Ephemerellid diversity in the watershed is high. We have observed 4 genera within the family (*Drunella*, *Ephemerella*, *Matriella*, and *Serratella*), and 3 species within the genus *Drunella*. Number of Ephemerellid taxa per site ranged from 0 to 4. Abundance and diversity of this sensitive and relatively well-characterized family may serve as indicators of the effects of global change within relatively undeveloped watersheds such as the Pend Oreille.

Does elevation outweigh urbanization in shaping BMI communities?

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Big Chico Creek (Northern California, USA) is one of the region's few remaining undammed streams, maintaining a natural flow regime while traversing steep elevational and urbanization gradients. Decades of benthic macroinvertebrate (BMI) sampling show strong community turnover along this gradient, from sensitive specialist taxa at higher elevations to more generalist assemblages downstream. We tested the relative influence of elevation, urbanization, and other environmental predictors on BMI community composition. Ordination analyses show elevation as the dominant driver of community separation among sites. We further evaluate associations between community patterns and additional predictors (e.g., habitat and water-quality variables) to assess how correlated environmental factors may interact with elevation and urbanization to shape BMI assemblages. These findings clarify which gradients most strongly structure communities in Big Chico Creek and inform expectations for how BMI assemblages may respond to environmental change across mixed elevational–urban stream networks.

Disentangling climate vulnerability in lentic and lotic odonates using trait-based and geographic metrics

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Freshwater biodiversity is declining faster than any other ecosystem type. Compared to vertebrates, freshwater insects are often left out of large-scale assessments of vulnerability to climate change. Previous work suggests that lotic species are more vulnerable to climate change than lentic species, possibly due to smaller geographic range sizes and greater habitat specialization. Most metrics of climate change vulnerability are derived from either a trait- or geographic-based approach, however combining these two approaches could help illuminate the relationships between species traits, geographic distributions, and their interaction with the scale of assessment. We used the geographic-based Rarity and Climate Sensitivity Index (RCS) to estimate intrinsic sensitivities for over 500 odonate species in the Nearctic realm at three different spatial extents: major river basins, sub-basins, and catchments. We also included biological traits known to influence odonate sensitivities and extinction risk under climate change and assessed their utility in predicting the geographically-derived RCS scores. RCS scores were more

congruent across spatial scales for lentic species of both dragonflies and damselflies compared to lotic species. Preliminary results suggest limited utility of biological traits in predicting intrinsic climate sensitivities for both lentic and lotic species at the catchment scale but may be useful at larger scales. This discrepancy demonstrates the importance of conducting vulnerability assessments derived from different approaches at multiple scales. Indeed, such an approach can help managers determine if sensitive species within a focal region are also sensitive more broadly, thus helping inform decisions of resource allocation towards species conservation.

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Documenting the presence of Plecoptera in fire-affected mountain streams of the H.J Andrews Experimental Forest.

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Wildfires in the Pacific Northwest have played an increasingly significant role in shaping community composition. To gain a better understanding of how these anomalies and their consequences affect freshwater environments, we sampled emerging adult Plecoptera across several sites within the H.J. Andrews Experimental Forest between June and July of 2024— nearly a year after the Lookout Creek Fire. This taxonomic group can be especially helpful in understanding the impacts of such disturbances, as many species have varied tolerances for habitat change and notable preferences for cool, high-velocity and low-turbidity water. Specimens were identified to species in a laboratory setting, and a literature review was conducted to i) assess the ecology of individuals found in the samples and ii) understand mechanisms that influence macroinvertebrate growth and emergence, such as canopy cover and food availability. We examine how presence and relative abundance of species correlate to burn severity in the forest and relate these differences to limiting environmental conditions of different Plecoptera. Considering the most recent taxonomic inventory for invertebrates in the H.J. Andrews is from about three decades ago, we hope this study can provide new insights for understanding how freshwater communities have evolved over the years and in post-fire conditions especially.

1. DeWalt, R.E., B.C. Kondratieff, and J.B. Sandberg. 2015. Chapter 36: Order Plecoptera. Pages 933-949 in James H. Thorp and D. Christopher Rogers (editors). Thorp and Covich's Freshwater Invertebrates (4th edition). Academic Press.
2. Parsons, G. L., G. Cassis, A. R. Moldenke, J. D. Lattin, N. H. Anderson, J. C. Miller, P. Hammond, and T. D. Schowalter. 1991. Invertebrates of the H.J Andrews Experimental Forest, Western Cascade Range, Oregon. V: An Annotated List of Insects and Other Arthropods. U.S. Department of Agriculture, U.S. Forest Service, Pacific Northwest Research Station.

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New Jersey's Reference Lake Monitoring Network: A Look at Seven Minimally Disturbed Lakes Across the State

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New Jersey's Reference Lake Monitoring Network was created to monitor trends in seven minimally disturbed lakes across the state. These lakes are characterized by the lack of urban and agricultural land use surrounding the lake, minimal human modifications along the shoreline, and other criteria. The lakes are sampled and monitored once a year during the last week of July and the first week of August. The purpose of this network is to gather baseline information on minimally impacted lakes, such as water quality data, ecological integrity, and trophic state, to use as a reference when comparing to other lakes (Franken, 2025). It has been over ten years since the Reference Lake Monitoring Network started. The purpose of this study is to determine if water quality parameters of these lakes have significantly changed since the start of this program. Another objective was to determine if these lakes should still be included in the network based on the assessment of water quality parameters from the Reference Lake Network and other Lake Network projects. The researchers found that the water quality in the lakes has been relatively stable since the program launched ten years ago.

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From open water to wetland: Evaluating changes in shoreline dynamics in Amazonian floodplain lakes

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Amazonian floodplain lakes of South America play an important role in supporting biodiversity and providing essential ecosystem services for riparian human communities. However, Amazonian floodplain lakes can be very dynamic systems, as their location, size, depth, and water conditions are influenced by seasonal changes in water and sediment flows in adjacent rivers, or by other climatic conditions. In recent years, the Amazon has experienced historic droughts, which have likely affected floodplain lakes. In this study, we asked: How has Amazonian floodplain lake area changed over time? We focused on floodplain lakes in the triple border region of Peru, Colombia, and Brazil along the mainstem Amazon during the period 2018-2024. We used high-resolution Planet Basemaps imagery to capture satellite images and ArcGIS Pro to digitize changes in lake area. We also needed to consider the effects of ENSO (El Niño) on Amazonian floodplain dynamics. Our preliminary results indicated variability in floodplain lake area across regions within our study site, with some lakes showing changes in area while others remained stable in size during the same timeframe. These findings are helpful in advancing understanding of hydrological and ecological changes in the Amazon and provide a quantitative baseline for further research on floodplain lakes.

A Disturbance in the Watershed: The Influence of Land Use on Stream Impairment Across New Jersey

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As the most densely populated state in the US with ~1,267 people per square mile, the Garden State relies on strict surface water quality standards to ensure that our freshwater resources are protected against degradation. The NJDEP Bureau of Freshwater and Biological Monitoring's Ambient Macroinvertebrate Network samples streams across the state for macroinvertebrates to determine if they meet the standards set for Aquatic Life Use, and each site is scored using one of three different multi-metric indices that assign a rating of "Excellent", "Good", "Fair", or "Poor". Generally, sites that score "Excellent" or "Good" are considered as attaining surface water standards, and scores of "Fair" or "Poor" are not attaining. To determine if there is a relationship between land uses and the impairment rating of the AMNET sites, we calculated the land use percentages of the contributing watersheds for 730 sites. We then determined the percentage of 'disturbed' land in each watershed by combining the 'Urban' and 'Agriculture' land uses. Sites were grouped into 5 different regions within the state and by the multi-metric index with which they were scored to investigate differences in disturbed land use across the groups. Two-way ANOVAs and subsequent Tukey's tests showed a significant relationship between the percentage of disturbed land use and the impairment ratings when grouped by region ($p < 0.000$), and between percent disturbed land use and impairment rating when grouped by both region and multi-metric index ($p = 0.03$). Additionally, there was a positive relationship between the percentage of disturbed land uses and impairment ratings, with nearly all regions demonstrating an increase in disturbed land use as impairment worsened. The same relationship exists for the multi-metric indices. These relationships can be used by NJDEP to inform management and regulation in areas that fail to attain surface water quality standards, support antidegradation designations, and restoration activities.

Assessing the Impact of Flood Potential and Geographic Features on Water Quality and Pollutant Mitigation in Central Massachusetts Water Bodies

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Abstract:

This study examines how flood potential influences water quality and pollutant levels in various water bodies across Central Massachusetts. By categorizing areas into FEMA-defined flood zones, Red (high risk), Orange (moderate risk), and Green (low risk), we analyze one small stream, a large river, and a pond within each zone. Key water quality parameters, such as fecal bacteria (fecal coliform, *E. coli*), sediment, and excess nutrients (nitrogen and phosphorus), are assessed to understand the variability in pollutant levels across different flood zones. Additionally, the influence of geographic features, including urban versus rural land use, riparian buffer health and size, and surrounding infrastructure is explored to determine their roles in affecting water quality. Particular attention is given to riparian buffer size and health to evaluate their effectiveness in filtering pollutants, utilizing a nutrient pollution reduction model. This approach helps quantify the potential of riparian buffers to mitigate nutrient runoff under different flood risk scenarios. The findings highlight the complex interactions between flood risk, geographic characteristics, and infrastructure, emphasizing their combined impact on water quality. Ultimately, this research provides valuable insights into sustainable watershed management practices in flood-prone regions, aimed at reducing pollution and improving water resilience amid increasing flood hazards.

Assessing land-use and seasonal drivers of volatile and nonvolatile suspended solids in Southwestern Ohio streams

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Total suspended solids (TSS) in streams are influenced by both environmental and anthropogenic drivers. Freshwater biota and watershed health are shaped by both volatile suspended solids (VSS) and nonvolatile suspended solids (NVSS) in streams. Volatile solids represent the organic material in streams that are key for aquatic health. Nonvolatile solids represent the inorganic stream solids (silts/clays) that influence water clarity, sedimentation, and abrasion. This study sets out to investigate how VSS and NVSS vary with land-use cover and seasonality over the course of one water year. Thirteen stream sites in Southwestern Ohio with differing land-use percentages (e.g. urban vs. agricultural), including two directly impacted by wastewater treatment plant discharge, were selected for this study. Biweekly water samples were collected and analyzed for TSS, VSS, NVSS, and VSS% using filtration and loss on ignition methodology. Complementary analysis of carbon-nitrogen (C:N) results were used to quantify the particulate organic carbon (POC) of each sample. We anticipate higher VSS% within agricultural watersheds due to increased canopy cover in riparian areas and higher stream nutrient concentrations leading to increased in-stream production. During cold and wet periods, we anticipate increased NVSS% due to increased runoff and decreased in-stream production. We expect low C:N ratios in agricultural watersheds due to accelerated decomposition and nutrient cycles, and low C:N ratios in urban streams with increased storm-water runoff and reduced vegetation in riparian zones. POC measured by C:N is anticipated to show seasonal variation based on shifts in organic content and in hydrologic changes over the water year. This integrated approach will advance understanding of how watershed land-use and seasonal trends shape the organic and inorganic suspended solids in streams for both ecological assessments and watershed management.

We (mostly) haven't used a p-value in ten years, and so can you

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Freshwater scientists routinely analyze data with linear (e.g., regressions and ANOVAs) or non-linear models, including hierarchical models with random effects and non-normal error structures. A common challenge with these models is how to make an inference on the hypothesis of interest: difference in treatments, slope of a regression, etc. This is typically achieved with a significance test using p-values or confidence intervals. However, significance tests convert rich ecological and quantitative information into a simple dichotomy of "significant" or "non-significant". Such focus on dichotomies often comes at the expense of more ecologically meaningful quantities like effect sizes and uncertainty. We argue that Bayesian data analysis naturally reorients the focus of analyses to effect sizes and to quantifying uncertainty. It does this by placing emphasis on model structure, likelihoods, and parameter estimates, rather than dichotomous significance tests. However, while modern software has made it easier to conduct Bayesian analysis (and frequentist analysis for that matter), there remain deep philosophical differences in these two approaches that are difficult to reconcile just by changing some code. We explore those differences and discuss how they have changed our approach to statistics using examples of studies from aquatic subsidy research, food webs, and ecotoxicology. We also include examples of positive peer-review that contradict commonly held assumptions that Bayesian analyses are harder to publish because reviewers are unfamiliar with it.

The Influence of Land Use on the Biotic Composition and Function of Southeastern Coastal Streams

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Land-use change across watersheds is a major driver of variation in the abiotic and biotic dynamics of stream ecosystems. Converting natural forest cover to agricultural, silvicultural, or urban uses can elevate sediment inputs, alter hydrology, and modify nutrient regimes, ultimately restructuring stream communities and reducing overall ecosystem function. Using standardized leaf-pack assays, we evaluated how dominant watershed land-cover types (Urban, Silviculture/Pine, Natural Forest, and Agriculture) influence detrital decomposition rates and aquatic macroinvertebrate assemblages in streams of southeast Georgia. Because decomposition integrates microbial activity and macroinvertebrate processing, leaf-pack breakdown provides a robust indicator of stream ecosystem function, while the colonizing macroinvertebrates serve as widely used bioindicators of ecological condition. Detrital processing rates differed across land-use categories, with agriculturally dominated watersheds exhibiting the highest decomposition rates relative to forested, plantation, and urban systems. Macroinvertebrate communities also varied with land use: disturbed watersheds (Urban and Agricultural) were characterized by assemblages dominated by tolerant taxa, whereas forested watersheds supported more diverse and functionally complex communities. These patterns highlight watershed land use as a key determinant of both stream ecosystem function and biological integrity. This project represents the southeastern U.S. component of a broader global effort to assess how land-use gradients shape freshwater ecosystem processes and community structure. By integrating our regional findings with those of international collaborators, this work contributes to a growing understanding of how freshwater ecosystems respond to accelerating landscape change and helps build the predictive capacity needed to support resilient watersheds in a rapidly changing world.

Aquatic insect subsidies mediate seasonal bat activity in the Grand Canyon

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Seasonal activity is a fundamental driver of ecological interactions. In dam regulated rivers, flow regimes and water temperatures are often asynchronous with natural seasonal patterns. In this study, we tested whether year-round aquatic insect subsidies in a dam-regulated river system mediate seasonal constraints on bat activity, particularly during the winter months. We quantified seasonal bat activity along the Colorado River corridor in Grand Canyon National Park using two years of passive acoustic monitoring across winter and active seasons. We paired species-level activity collected from April 2023 to November 2024 with functional trait data describing bat flight morphology and foraging ecology, as well as temperature and aquatic insect availability. Bat species clustered into three activity classes: species active year-round, species that were consistently rare throughout the sampling period, and species that were strongly seasonally filtered. Seasonal bat activity and prey availability were both highest at warm temperatures. Prey availability was an important predictor of bat activity year-round ($p = 0.00171$), but was a better predictor in winter ($p = 2e-16$) than the rest of the year ($p = 2.17e-06$). Our findings highlight the importance of aquatic insect availability for structuring consumer activity at higher trophic levels, underscoring desert river corridors as essential subsidies for cross-ecosystem energy transfers. Given ongoing pressures facing bat populations, including the impacts of white-nose

syndrome, these findings could inform management applications for Glen Canyon Dam operations aimed at supporting bat diversity through winter insect availability.

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Hidden Reservoirs: Native Fishes Harbor Greater Genetic Diversity Than Nonnatives in the Lower Colorado River–Grand Canyon Ecosystem

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The Lower Colorado River Basin ecosystem plays host to dozens of nonnative fishes. Within Grand Canyon National Park, nonnative species such as smallmouth bass, rainbow trout, brown trout, flathead catfish and more often co-occur with federally threatened or endangered species, including the native and endemic humpback chub and razorback sucker. Understanding the genetic diversity of fish within the ecosystem is a vital tool for monitoring and conserving these species. We used environmental DNA (eDNA) as a non-invasive method for sampling the 12S mitochondrial region of fish genomes within the Lower Colorado River Basin (LCRB) at 44 unique sites, including the mainstem Colorado River, major tributaries, and tailwaters directly below dams. We isolated eDNA from these water samples and amplified the products via polymerase chain reaction (PCR) and the MiFish-U metabarcoding primer. Amplicons were then sequenced and sorted into operational taxonomic units (OTUs). We found that native fishes (N=4) harbored an average of 16 distinct 12S OTUs (range: 8 to 22), while nonnative fish species (N=26) had an average of 2 distinct 12S OTUs (range: 1 to 6). Mainstem and tributary sites contained three times as many nonnative fish species as native ones, but these nonnative fish have, on average, one-sixth the genetic diversity of native species. Tailwaters directly below dams were dominated by genetically poor nonnative species. The Colorado River is a highly modified system where native fishes, such as humpback chub, have undergone historic genetic bottle necks; the retention of high genetic diversity despite these challenges indicates the persistence of these species within microhabitats. Exploring the mechanisms behind the retention of this high genetic diversity would inform fish conservation efforts.

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Microbiome Diversity Differences Amongst Fish Species in the Colorado River.

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Microbiomes of fish species can provide researchers with an abundance of information on the condition of fishes in regards to their environment. We're interested in the differences in the intestinal microbial diversity of native fish species within the Colorado river. The Colorado river as we know it is an extremely diverse ecosystem with there currently being monitoring of the rare species present and the rapid environmental changes occurring like drought and climate change. Fish that have abnormal gut microbiomes may be consuming new or different food sources or have been subject to environmental contamination. Diets also delineate information about the condition of the habitat and the general health of the fish by providing insights into food sources. When examining the gut contents of species within unique environments like the Grand Canyon, microbiomes provide vital information on their hosts and the function of the ecosystem at large.

We extracted DNA from 56 mainstem and 58 tributary water samples along with fish fecal samples from 20 rainbow trout, 77 humpback chub and 255 flannelmouth suckers from the Colorado River located in the Grand Canyon. Fish were caught using hoop nets and information on their length and weight was also taken with the samples being collected in April 2022-2024. DNA was extracted using Qiagen Powersoil kits and sequenced using the Earth Microbiome Project 16s metabarcoding procedure.

Utilizing the sequenced DNA, we were able to calculate overall microbial richness of the samples and identify patterns of the most common microbial taxa listed. We found that the flannelmouth suckers (FMS, *Catostomus latipinnis*) had a significantly higher microbial richness per sample than the humpback chub (HBC, *Gila cypha*) with FMS having a mean richness of around 120 unique taxa and the HBC averaging at about 90 unique taxa present. We found that Entrobacterales, a common order of gut bacteria found in both species, were significantly more common in humpback chub than flannelmouth suckers. We also described changing trends in microbiomes across sampling years. These differences can assist us in bettering policy and care for the landmark to maintain it for future generations.

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Injury Rates and Condition Factor as Indicators of Health in Native Fish of the Modified Colorado River Ecosystem

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Since its construction in 1966, the Glen Canyon Dam has altered habitat conditions in the Colorado River, affecting the survival of endangered native fish. The river is now characterized by reduced temperature and flow variation and the introduction of nonnative fish species, both of which could contribute to higher risks of infection and injury for native populations. This study's goal was to examine the health of humpback chub (HBC, *Gila cypha*) and flannelmouth sucker (FMS, *Catostomus latipinnis*), to examine differences in overall physical condition and visible injury patterns between the two species.

In the field, data was collected via capture of fish at sites along 280 river miles of the Colorado River in the Grand Canyon. FMS were the most commonly caught fish, with 116 fish captured, while 56 HBC were captured. FMS were caught at nearly all sites along the river, while 56 HBC were primarily caught at miles 61.5 and above 220. Fish had weight and length recorded and were

visually scanned for gross pathology on the skin, eyes, caudal, dorsal, and pectoral fins, and the operculum. Using the collected data, we analyzed condition factors and injury frequencies across the two target species.

Length–weight regressions indicated strong positive relationships between length and weight for both species ($R^2 = 0.980$ for FMS, $R^2 = 0.971$ for HBC), providing a basis for assessing spatial condition trends and identifying individuals with outlier condition factors. While most individuals of both species exhibited no visible injuries, almost all observed injuries were classified as mild. HBC showed higher injury frequencies in the caudal fins and eyes and included occasional moderate to severe injuries, whereas FMS more often had mild injuries on the operculum. Additional analyses expand upon this work by comparing injury severity across river-mile locations, and assessing relationships between injury presence, injury severity, and injury type with condition factor to evaluate how physical condition relates to injury patterns within and between species. Establishing these species-specific baselines will enable the assessment of future environmental changes and their impacts.

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Development of a Web-based Conservation Map and Prioritization Model for the Middle Rio Grande Riparian Bosque, central New Mexico, USA

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Publish consent withheld

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Environmental drivers of bacterial community composition in 46 headwater streams in the upper Delaware River Basin, USA

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Headwater streams are crucial for the health and resilience of entire watersheds; therefore, understanding how human activities disrupt the structure and function of these vulnerable ecosystems is paramount. Bacterial assemblages respond to a myriad of environmental changes and have the potential to indicate subtle anthropogenic impacts to relatively pristine headwater streams. We collected benthic samples from 46 headwater streams throughout the upper Delaware River Basin, USA. Streams were relatively pristine with upstream forest cover >50%. Bacterial community compositions for each stream were characterized following DNA extraction, amplification of the 16S rRNA V4 gene, and sequencing. Nonmetric multi-dimensional scaling was used to visualize between-site differences and potential relationships with environmental variables. We found that variation in bacterial community composition was most strongly related to sodium concentration with canopy cover being secondarily important. These results suggest that road salt associated with winter road deicing and changes in algal dominance of stream biofilms associated with reduced riparian canopy can have measurable effects on the microbial assemblages.

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Arsenic contamination drives homogenizing selection in freshwater lake periphyton

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Bacteria and other microbes are important contributors to arsenic biotransformation processes, which can alter the bioavailability and toxicity of arsenic within a contaminated environment. Multispecies biofilms, known as periphyton, have been identified as a significant site of arsenic bioaccumulation within shallow freshwater lakes impacted by legacy arsenic contamination. We hypothesized that prolonged arsenic exposure results in the formation of distinct prokaryotic communities within the periphyton and other environmental compartments in arsenic-contaminated lakes compared to uncontaminated lakes. We also predicted that the periphyton prokaryotic communities would be distinct from, but partially overlapping with, those found in the surrounding water column and nearby littoral sediment. To test these hypotheses, we determined the taxonomic composition and modeled the assembly processes that yielded the bacterial communities found within three environmental compartments (periphyton, littoral sediment, and water column) of three lakes that had been differentially impacted by legacy arsenic contamination. We identified unique microbiomes within these environmental compartments and observed a clear shift in microbial community composition within high arsenic-contaminated periphyton. Accumulation of arsenic (~400 ppm) in the periphyton correlated with non-random (deterministic) selection for prokaryotic taxa that are more related than expected by chance (homogenizing selection). We also identified key prokaryotic genera within the arsenic-contaminated periphyton that suggest prolonged arsenic contamination may shift iron and methane biogeochemical cycles, which may regulate arsenic accumulation and mobilization. Our results imply that legacy arsenic contamination, by altering bacterial community composition and metabolic potential at the base of the food web, may influence biogeochemical and nutrient cycles at a larger scale within a freshwater lake ecosystem.

Polyphasic identification of filamentous cyanobacteria from anatoxin-a-producing mats in the Shenandoah River

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Cyanobacteria belonging to order Oscillatoriales can form large benthic mats in rivers, some of which produce the neurotoxin anatoxin-a. Mats of potentially toxic cyanobacteria are common, abundant, and diverse in both forks of the Shenandoah River, VA, USA but little is known about their taxonomies and toxicities. In 2023 and 2024, we collected benthic mats growing in the Shenandoah River and conducted integrative molecular, morphological, and toxin characterization of *Microcoleus* strains isolated from the samples. Using isolated filaments from field samples and monoclonal unialgal cultures, whole genome sequencing was successfully completed for three of the cyanobacterial species using either the IonTorrent S5 system or Illumina MiSeq systems to produce short reads which were de novo assembled into contigs. Phylogenetic analyses of the 16S rRNA gene identified two distinct species of *Microcoleus*, one non-toxic and one produced anatoxin-a and dihydroanatoxin-a. According to 16S phylogenetic analysis it is closely related to *Microcoleus anatoxicus* but the species identity is inconclusive. However, the most abundant mat-forming species in Shenandoah River, fit the morphological description of *Microcoleus lacustris* Farlow ex Gomont, but was phylogenetically distant from members of the Family Microcoleaceae. It was placed within the Family Coleofasciculaceae and appears to be a non-toxic undescribed member of the genus *Limnofasciculus*.

Linking Genotype to Phenotype: How Cryptic Diversity Shapes *Prymnesium parvum* Blooms in Inland Waters

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Prymnesium parvum s.l. (*sensu lato*, in the broad sense) is a cryptic species complex of toxic, unicellular haptophyte algae responsible for recurrent fish kills and major ecological disruptions in freshwater and brackish systems worldwide. Although historically considered marine, *P. parvum* s.l. has increasingly expanded into inland freshwater environments. Even though the strains within this species complex are morphologically indistinguishable, recent genomic studies have revealed substantial cryptic diversity. Distinct genetic clades produce different prymnesin toxins (types A, B, or C), indicating that genetic divergence may be associated with functional and ecological variation. Previous studies have documented phenotypic differences among *P. parvum* strains, including variation in growth and toxicity. However, these differences have not been evaluated in a comprehensive, comparative framework across a broad spectrum of strains or between genetically divergent strains co-occurring within the same bloom. In this study, we investigate phenotypic variation among multiple *P. parvum* s.l. strains isolated from a single bloom, with a focus on differences in growth rates. By comparing growth performance across genetically distinct strains under controlled conditions, we assess whether phenotypic variation is structured primarily by clade identity rather than by strain-level variation. This approach provides insight into potential mechanisms that allow genetically distinct strains to coexist within the same bloom and how such differences may scale up to influence bloom dynamics, including bloom formation, persistence, and impact. By linking genotypes to measurable ecological traits, this work highlights the ecological relevance of cryptic diversity in harmful algal blooms and provides a framework for integrating genetic and phenotypic data to improve prediction, monitoring, and management of *P. parvum* s.l. blooms in inland waters.

Investigating the impacts of land use on aquatic biodiversity within an agricultural watershed using environmental DNA

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Biodiversity within riverine systems is crucial for the ecosystem's health and function, yet many of these ecosystems are being disturbed through increased human impacts, threatening the biodiversity within. While the general patterns of decreased biodiversity within riverine systems are well documented, little is known about how biodiversity across multiple taxonomic groups responds to land uses in the surrounding watershed. Environmental DNA (eDNA) can be used to efficiently measure the biodiversity and taxonomic composition of multiple species groups through sampling traces of DNA shed by organisms into the surrounding environment. In this study, we used eDNA metabarcoding to assess alpha and beta diversity across taxonomic groups at sites spanning a gradient of agricultural land use along the Mackinaw River watershed in central Illinois. To assess land use, we created a GIS map with 5-km buffers around each eDNA sampling site and applied a land use raster map to determine agricultural impact. We assessed biodiversity using eDNA metabarcoding with 12S and 18S primers to target fishes and eukaryotes, respectively. Our results revealed that alpha diversity of fishes was significantly higher in slightly and moderately disturbed sites, while eukaryotes were the most diverse in highly disturbed sites. Beta diversity among sites was high for both species groups. These findings demonstrate how land use in the surrounding watershed can impact the biodiversity of aquatic

organisms across different taxonomic groups, providing important insights into potential conservation strategies to sustain the biodiversity of freshwater systems.

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Investigating the impact of glyphosate-containing herbicides on *Aeromonas* growth

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Glyphosate is an organophosphate in commonly used commercial herbicides and may contaminate nearby ecosystems through agricultural runoff. *Aeromonas* is a genus of bacteria ubiquitously found in freshwater ecosystems. It includes species that are known fish and human pathogens. We tested six *Aeromonas* isolates representing three species (*A. salmonicida*, *A. eucrenophila*, and *A. sobria*) to determine if glyphosate-containing commercial herbicides impact their growth at environmentally relevant concentrations. Growth kinetics in the presence of varying levels of glyphosate were assessed in two glyphosate-containing commercial herbicides as well as pure glyphosate. Comparison of exponential growth rates and maximum biomass yields revealed species and strain-specific responses within the tested *Aeromonas* isolates to the presence of environmentally relevant concentrations of glyphosate-containing commercial herbicides. Our findings suggest the potential for altered growth and quantities of bacterial pathogens following glyphosate inputs into freshwater ecosystems.

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Bacterial Isolates from Mill Creek River Pre and Post Fire

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This research study qualitatively assesses the types of microorganisms found at the surface and deep level of Mill Creek River and whether they are potentially pathogenic to humans. Mill Creek River is in Breckenridge Mountain via the Mill Creek trail: located 35–40-mile drive from Bakersfield. The water samples consisted of 70 ml water collected from a stagnant area, and 70 ml water taken from active running water area of Mill Creek River in spring and fall of 2024. The first date of collection was 03/03/2024, and the second date of collection was 11/09/2024. A large fire occurred in the area prior to the second collection date of 11/09/2024, which drastically altered the environment. The water samples were cultured on Tryptic Soy Agar plates before Gram stains; oxidase testing and Micro Gen A and B panels were conducted. The Micro Gen tests assist with quickly identifying gram-negative bacteria within the Enterobacteriaceae. The results from the spring pre-fire water samples indicated three pathogenic bacteria *Salmonella choleraesuis*, *Acinetobacter lwoffii*, and *Moraxella* spp. *Salmonella* causes Salmonellosis, *Acinetobacter* causing blood, lungs, or urinary tract infections, and *Moraxella* can cause systemic infections. The main bacterium documented from 7 distinct cultures from the fall post-fire samples was *Burkholderia pseudomallei*. This is an environmental bacterium known to be a deadly intracellular pathogenic bacteria causing 'melioidosis' in humans. Further PCR genetic testing to confirm these results is underway.

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The combined effect of temperature and light on the growth and photophysiology of the bloom-forming cyanobacterium *Microcystis aeruginosa*

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Cyanobacterial blooms are becoming more common in freshwater ecosystems due to climate-change induced warming and eutrophication. These shifting environmental conditions introduce multiple simultaneous stressors, such as increased temperature and irradiance, which can have a non-linear, synergistic effect on the productivity of bloom-forming cyanobacteria. Here, we examined the combined effects of thermal stress and light on the growth and photophysiology of *Microcystis aeruginosa* (UTEX-LB-2386). This strain was monitored across five temperatures (20–40 °C) and three light regimes in a batch culture setting. Growth was measured via optical density and PAM fluorometry assessed in-vivo chlorophyll (F0), maximum quantum yield of Photosystem II (QYmax, a proxy for photosynthesis efficiency and cell stress), and relative electron transport rates across PSII (a proxy for photosynthesis). A thermal optima (TOPT) for growth was observed at 30°C under low light, but shifted to 25°C as light levels increased. The thermal optima (TOPT) for QYmax and rETR was inconsistent with the optima for growth, and was observed at 25°C under low light, but shifted to 20°C as light levels increased. Our results thus indicate that growth and photophysiology in *M. aeruginosa* is decoupled, and could have implications for bloom formation, carbon allocation, and toxin production by some toxin-producing strains of *M. aeruginosa* in a warming world.

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Predator-prey interactions and body morphology of freshwater species under thermal stressors.

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Freshwater ecosystems are critical for providing numerous ecosystem services, but they are highly susceptible to the effects of climate change. Climate change has increased average temperatures and led to more frequent extreme heat events, which can interact with predator-induced stresses to change freshwater species' interactions, behavior, growth, morphology, and survival rates. We tested the effects of press and pulse heat events in the form of an elevated temperature treatment (press event) and two different types of extreme heat events (pulse events) in tandem with the presence or absence of dragonfly (family *Aeshnidae*) nymph predators on the growth, development, morphology, and behavior of Pacific chorus frog (*Pseudacris regilla*) tadpoles. We used a 2x4 factorial design with two levels of predator treatment (present/absent) and four levels of heat exposure (baseline, baseline +3°C, baseline + short high-intensity heat wave, baseline + long moderate-intensity heat wave). Each treatment was replicated 12 times for a total of 96 tanks. Tadpoles were held in 9-L glass tanks and predator treatments were applied by placing dragonfly nymphs into the tanks in clear plastic cups with perforations. Tadpole and dragonfly size and morphology were assessed at the beginning and end of the experiment. Preliminary findings show that tadpole mass change is influenced by temperature and predator presence—tadpoles grew less with the predator present ($p<0.05$). As the accumulated heat exposure increased, tadpoles developed more quickly overall ($p<0.001$), but this trend was nearly significantly less strong with a predator present ($p=0.067$). Additionally, all treatments with temperature increases led to an increased tadpole development rate ($p<0.01$). Lastly, we found that dragonflies exposed to the short heatwave attacked tadpoles faster than dragonflies exposed to any other treatment ($p<0.01$). In all other treatment groups, dragonfly attack time decreased with increasing tadpole size ($p<0.001$). Understanding predator-prey interactions in different temperature scenarios provides context for community responses under climatic stressors. Our results contribute to the general understanding of how freshwater communities respond to climate change, allowing us to better protect and preserve these important and threatened ecosystems.

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Aquatic ecosystems under pressure: how anthropogenic land use and flow regimes shape stream macroinvertebrate communities and organic matter decomposition

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Freshwater ecosystems are experiencing disproportionate levels of pollution and ecological deterioration in response to climate change and the continuously expanding human population. Anthropogenic activities contribute a variety of stressors, including altered land use, nutrient loading, and hydrologic modification, that influence stream macroinvertebrate community composition and the ecosystem processes they support. Hydrologic regimes are also essential predictors of ecosystem function, shaping benthic macroinvertebrate community structures and organic matter (OM) dynamics. In streams, terrestrial OM represents a predominant energy source and forms the foundation of many aquatic food webs. Prior studies have shown that environmental conditions influence the relative contributions of biological and physical factors driving OM decomposition. To better understand these dynamics, we conducted a 140-day leaf-pack study in 18 streams across the Tennessee River watershed, representing a range of human land use (e.g., urban, agriculture, and forest) and flow intermittency. This study aims to differentiate the contributions of shredding macroinvertebrates, physical abrasion via stream flow, and microbial activity to OM decomposition rates (k). We will use a structural equation model (SEM) to quantify the relative importance of each factor under varying environmental conditions. We hypothesize that in streams with greater human land use, microbial activity will be the dominant driver of decomposition due to nutrient enrichment, increased microbial activity, and reduced macroinvertebrate diversity. Alternatively, we hypothesize that physical abrasion will be the primary driver of decomposition in intermittent streams, where fluctuating flows can mechanically break down leaf material. Lastly, we predict that the drivers of decomposition in perennial, low-impact streams will be more variable and context-dependent, reflecting intact ecological interactions. Understanding the factors that regulate OM decomposition is critical because these processes underpin nutrient cycling, energy flow, and overall ecosystem resilience. Our research will provide insight into the response of freshwater ecosystems to anthropogenic pressures and hydrologic differences, informing conservation and management strategies for maintaining functional aquatic food webs in human-impacted landscapes.

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Assessing the influence of specific conductivity on fish traits in the Chesapeake Bay Watershed

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Anthropogenic activities are increasingly altering freshwater ecosystems, with rising specific conductivity serving as a widespread indicator of environmental stress. Changes in conductivity may be associated with shifts in life-history traits that influence how species grow, reproduce, and persist under altered environmental conditions, shaping a species' ability to adapt. This project aims to explore how ecological traits of freshwater fishes are associated with variation in specific conductivity across the Chesapeake Bay Watershed. We focus on two key life-history traits, maturity age and fecundity, to assess how species' life-history strategies influence their responses to conductivity gradients, with the potential to incorporate additional traits in future analyses. Our primary goals are to (1) investigate species-specific abundance patterns across conductivity levels to identify thresholds at which fish species are no longer detected and (2) determine whether maturity age and fecundity exhibit trait-specific thresholds in response to increasing conductivity. We will analyze long-term water quality and fish assemblage data (1986 -2019) from the Chesapeake Bay compiled by U.S. Geological Survey. To quantify species- and trait-level responses, we will apply Threshold Indicator Taxa ANALysis (TITAN; Baker & King, 2010) along with quadratic-plateau and step models to identify conductivity thresholds beyond which sensitive taxa are unlikely to persist. We hypothesize that fish communities in high

conductivity streams will be dominated by species with shorter lifespans and a greater number of offspring, reflecting earlier maturation and higher fecundity. Conversely, species with longer maturation times and lower fecundity are expected to exhibit stronger negative responses and lower conductivity thresholds compared to species with more opportunistic traits. By linking functional trait variation to environmental thresholds, this project provides a trait-based framework for assessing how freshwater fish communities may respond to environmental change in the context of specific conductivity. These findings may inform stream health indicators and conservation strategies, and can be extended to additional traits, such as spawning season and functional feeding guilds.

1. Baker, Matthew & King, Ryan. (2010). A new method for detecting and interpreting biodiversity and ecological community thresholds. *Methods in Ecology and Evolution*. 1. 25-37. 10.1111/j.2041-210X.2009.00007.x.

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High resolution UV-VIS observations of DOM gradients in the North American Great Lakes

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Dissolved organic matter (DOM) is a major driver of lake biogeochemistry and water quality through its influence on microbial metabolism and light attenuation. Aromatic DOM is delivered primarily via rivers where it reduces light availability and can alter vertical thermal structures through the absorption of solar energy. Further, photodegradation of aromatic DOM converts it into more labile forms which can fuel microbial metabolism. Despite the importance of DOM, plume dynamics and their ecological impacts remain difficult to quantify due to spatiotemporal variation in DOM source inputs and rapid in-situ processing. As a result, accurate descriptions of spatial gradients and the impacts of DOM delivery require high-resolution measurements capable of capturing both plume structure and rapid processing.

To quantify DOM gradients and processing signals across tributary plumes in the Laurentian Great Lakes of North America, we utilized high frequency UV-Vis sensing. Nearshore to offshore transects were sampled in proximity to tributaries in Lake Huron and Lake Michigan using a towed, submersible ultraviolet nitrate (SUNA) sensor to compare plume extent and DOM optical characteristics in both lakes. In a different application, we analyze fixed location SUNA data from buoys in the western basin of Lake Erie during the months preceding harmful algal blooms (HAB) onset between years of different bloom severity. We quantified DOM optical properties using UV absorbance at 254nm (a₂₅₄) to characterize aromatic DOM pools, and the a₂₅₄:a₃₅₀ absorbance ratio to estimate relative DOM lability.

Across transects, tributaries displayed distinct DOM optical signatures and plume structures, with Lake Michigan tributary plumes showing greater average DOM aromaticity than those in Lake Huron. Diel and spatial variability in a₂₅₄ across lakes display patterns consistent with photochemical processing and mixing of distinct water masses. In Lake Erie, a₂₅₄:a₃₅₀ ratios were higher prior to HAB onset in the year with more severe blooms, indicating a greater proportion of more labile DOM and suggesting a possible priming prerequisite in which pre-season DOM quality can enable greater bloom intensity.

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Leaf litter decomposition as a proxy for ecosystem services of riparian forest composition with macroinvertebrate and microbial community assessment

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Riparian forests are major contributors of organic matter to stream ecosystems, and variation in tree species composition can strongly influence detrital processing and the structure of benthic macroinvertebrate communities. Differences in litter quality, such as those between hardwood and conifer species, may alter microbial conditioning, decomposition rates, and the diversity of detritus-associated taxa. Yet the functional consequences of shifts in riparian forest composition remain understudied across many southeastern U.S. landscapes. This project will compare decomposition dynamics and associated microbial, fungal, and macroinvertebrate assemblages in red maple (*Acer rubrum*) and longleaf pine (*Pinus palustris*) leaf packs to evaluate how contrasting litter types shape stream ecosystem function. By deploying standardized leaf packs across multiple sites and sampling over time, we will quantify detrital processing rates, macroinvertebrate diversity and abundance, and the presence of sensitive taxa. We hypothesize that hardwood litter will support faster decomposition and more diverse, functionally richer communities than pine litter. This work will provide foundational insight into how riparian forest composition influences stream ecosystem processes and will inform future assessments of forest management practices and their ecological implications.

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Leaf litter decomposition patterns at a continental scale: leverage the NEON network to understand detrital food webs

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Leaf litter decomposition is a foundational process in stream ecosystems, contributing to carbon and nitrogen availability that supports aquatic food webs. In many headwater streams, terrestrial litter represents the primary energy source, linking riparian vegetation to aquatic consumers. Decomposition rates vary with leaf traits and environmental conditions, and this variation can influence resource availability for macroinvertebrates across stream habitats. Improving our understanding of these patterns will

strengthen predictions of how environmental gradients and changing conditions affect nutrient supply, energy flow, and overall ecosystem functioning in freshwater environments. Quantifying litter breakdown is therefore an important step toward understanding how nutrients enter and move through stream food webs. This study measured leaf litter decomposition of three leaf types to characterize organic matter processing in streams across North America. Leaf packs of varying species were deployed in wadeable streams spanning environmental gradients within the National Ecological Observatory Network. Following in-stream incubation for one month, packs were retrieved and processed to determine percent AFDM loss as an indicator of decomposition. Preliminary results show variation in decomposition rates among litter types and stream environments, highlighting the influence of both local environmental conditions and litter characteristics on processing rates. These patterns suggest that both abiotic and biotic factors regulate how quickly litter is processed and how long nutrients remain available in stream ecosystems. Understanding spatial differences in decomposition is essential for predicting the supply of organic matter to macroinvertebrate communities and the structure and productivity of stream food webs. Since decomposition influences how long leaf litter resources persist in aquatic systems, these processes influence the timing and magnitude of energy transfer to higher trophic levels. These findings represent a first step toward linking litter processing with carbon and nitrogen transfer to aquatic consumers. Ongoing work on this project will integrate decomposition measurements with macroinvertebrate community data and stable isotope tracers to better understand the fate and drivers of carbon and nitrogen in stream food webs across diverse ecosystems and environmental contexts across North America.

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Characterizing seston and fine benthic particulate matter in South Sandy Creek Watershed, Alabama

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Fine organic particles are an important food resource in streams. They also serve as a key vector linking the benthos to the water column and upstream reaches to downstream ecosystems. However, few studies have assessed whether benthic and suspended fine particles are a single pool or if suspended particles are qualitatively different from those on the bed. This study examined temporal and spatial patterns in the composition of seston and benthic fine particles across 20 streams in the South Sandy Creek Watershed, Alabama. The watershed is approximately 95% forested and is dominated by a mixed canopy of Longleaf (*Pinus palustris*) and Loblolly (*Pinus taeda*) Pines, with an understory of Oak and other hardwoods. We collected seston and benthic fine particles monthly and characterized particle abundance, size distribution, and an array of shape and color characteristics using FlowCam 8000 with fluorometric detection. Benthic organic matter was quantitatively collected using a stand-pipe sampler, and partitioned into fine (10-100 μm) and coarse ($>100 \mu\text{m}$) fractions. Ash-free dry mass, percent organic matter, and isotopic signatures were also estimated for both seston and benthic fine particles. Preliminary results from September to December 2025 indicate that seston concentrations were highly variable and lacked a clear longitudinal pattern (0.1 km^2 to 123 km^2). Averaged across sites, September had the highest concentration of particles in suspension (3,014 particles/ml) and concentrations were lowest in November (1,578 particles/ml). The three sampling locations with the largest discharge had the most stable concentrations over time (ranging between 2,000 and 5,000 particles/ml), whereas headwater sites showed extreme variability ($\sim 300 - > 20,000$ particles/ml). We are currently analyzing benthic and suspended particles for elemental content and isotope signatures to better characterize each pool. Preliminary results suggest that fine particle dynamics (e.g. suspension, transport, and deposition) are much more episodic and variable in headwater streams than in downstream reaches that integrate runoff from larger geographic areas. When completed, this research will improve our understanding of the coupling between benthic and suspended fine particles in streams and how they contribute to downstream export of energy and elements in lotic ecosystems.

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Microplastic seasonal variation in coastal shellfish and sediments linked with estuary protection status and watershed anthropogenic development

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Anthropogenic particles and microplastics (APs; $< 5 \text{ mm}$; e.g., cotton and polyester fibers) are pervasive contaminants in aquatic habitats and biota worldwide. Terrestrial-aquatic interactions can move anthropogenic litter into aquatic habitats where litter items may fragment into APs and deposit in sediments or enter food webs. Estuaries are among the most biologically valuable yet widely diminished ecosystems in the world, making these systems an ideal target for AP mitigation efforts. Estuary sediments provide critical habitat and support numerous estuary ecosystem services, including water filtration, nursery and refugia for aquatic biota, and biogeochemical cycling. Shellfish are an important estuary resident that can serve as ecosystem engineers in estuaries and are known to be particularly susceptible to particulate pollution due to their filter-feeding behavior. We sought to understand the importance of watershed anthropogenic development (e.g., percent urban; dominant land-use) and estuary protection status (e.g., state marine conservation area) in connection with the abundance and characteristics of APs in estuarine mussels and sediments. Estuaries were sampled for sediments in autumn 2023 ($n = 15$ sites) and for shellfish in winter, spring, and summer 2024 ($n = 7$ sites/season). Preliminary results from the summer mussel samples indicated that AP contamination was present in mussels across all sites, with fibers contributing approximately 96-99% of all AP particles. There were significantly greater AP concentrations in mussels from unprotected sites compared to marine protected area sites ($t = -3.02$, $df = 18$, $P = 0.004$). Microplastic color patterns were distinct across both site ($\chi^2 = 121.09$, $df = 66$, $P < 0.001$) and watershed land-use ($\chi^2 = 37.53$, $df = 22$, $P = 0.021$). These findings are expected to inform management and policy decisions for crucial coastal environments that will mitigate or prevent harmful environmental impacts from AP pollution.

Microplastics in the Tittabawassee-Saginaw River: an analysis of freshwater river microplastic pollution based in varying landscapes

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Microplastic pollution is an understudied issue that impacts freshwater ecosystems and the Great Lakes ecosystems in particular. This study investigates the abundance and type of microplastics in the Tittabawassee-Saginaw River in Michigan, which drains to the Saginaw Bay of Lake Huron, to assess the impact of the upstream-downstream gradient, wastewater treatment plants (WWTPs), and local land use. Water samples were collected from 12 sites and the particles were identified using a dissecting microscope. The main microplastics shape were clear fibers and no other shapes were found in enough abundance for analysis. Statistical tests revealed no significant relationship between microplastics and the upstream-downstream gradient ($p=0.24$) and there was no significant difference found between samples upstream and downstream of WWTPs ($p=0.88$). Among the land use types only “developed open-space” showed a significant positive correlation with microplastic concentration ($p=0.01$).

Influence of Urbanization on Microplastic Pollution in River Systems: A Comparative Analysis of Brevard (USA) and Karatina (Kenya)

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The term microplastics is rising in society and quickly gaining attention. It has strong effects on communities and ecosystems beyond our current understanding. The presence of microplastics has been extensively studied in marine environments, but there is a lack of clear data in freshwater environments and patterns throughout urbanization. In this study, we aim to understand the presence and impact of microplastics in freshwater systems from two different, yet comparable, urbanized perspectives. Samples from the headwaters of the Davidson River in Brevard, NC, and the Ragati in Karatina, Kenya, have been analyzed. These two collection sites have been chosen based on common urbanization patterns and population densities to ensure fair comparisons. Our analysis process consists of density separation, digestion, and observation under microscopes. We hypothesize that microplastics will be found at lower concentrations near headwater catchments due to less human impact. We expect this concentration gradient to increase further downstream, as streams run through the centers of towns and daily living practices impact nearby ecosystems. We also expect that these two areas may have differing livelihoods and practices, such as waste management, eco-awareness, and socioeconomic status. At the end of this study, we will contextualize what the presence of microplastics means in two separate urbanized settings and how socioeconomic standing impacts data. These findings will add to the scientific community's knowledge on the role of urbanization and economic development on microplastics pollution in freshwater systems, and will further push more research on the matter to the surface.

Mapping Wetland Inundation Regimes using Google's AlphaEarth Embeddings: A Case Study in the Florida Everglades

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The presence and duration of water in wetlands drive core biogeochemical processes that shape ecosystem function and carbon cycling. Yet data on wetland inundation regimes remains limited beyond static inventory maps, especially in data-scarce regions. This gap is compounded by the computational and storage demands of deriving inundation information from dense satellite time series, particularly for small, cryptic wetlands that are ephemerally flooded or masked by vegetation. Emerging Earth observation foundation models, such as Google's AlphaEarth, compress spectral, temporal, and contextual information from petabytes of multi-source observations into generalizable geospatial embeddings that offer a novel approach for monitoring land surface processes. In this study, we use a data-rich wetland site in the Florida Everglades to evaluate how AlphaEarth embeddings could support inundation mapping in both data-rich and data-scarce regions. We first assess embeddings in a supervised setting by training a Random Forest classifier to map annual inundation regimes (e.g., seasonal, permanent, ephemeral) and their trajectories from 2018–2021 at 10 m resolution, using AlphaEarth embeddings and ground-truth data from the Everglades Depth Estimation Network (EDEN). We compare accuracy to the same model trained on annual Sentinel-2 Normalized Difference Water Index (NDWI) composites to quantify the added value of embeddings over raw satellite data. Next, to represent sites lacking ground truth data, we assess embeddings in an unsupervised setting by clustering in embedding space and comparing these clusters to those derived from the Sentinel-2 NDWI composites. In this unsupervised case, EDEN data are used solely to evaluate how well clusters correspond to distinct inundation regimes, not for training. Finally, we benchmark both supervised and unsupervised results against inundation regimes reported by the 30 m Global Surface Water Explorer and the National Wetlands Inventory. Together, these analyses evaluate the potential of AlphaEarth embeddings to advance diverse applications of wetland remote sensing. The resulting maps can help pinpoint biogeochemical hotspots, track regime changes through time, inform the parameterization and fine-tuning of foundational and process-based models, and guide targeted field sampling campaigns.

Hydroelectric Dam Removal and River Restoration in Elba, AL

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Large scale dam removals present opportunities to improve natural habitats, restore connectivity, and improve overall ecosystem integrity. The Elba Hydroelectric Dam Project (EHDP) is a multi-agency effort to restore connectivity to the Pea River in southeast AL through the removal of a damaged and decommissioned dam. This project will alleviate river hazards, restore connectivity and improve habitat for fishes and mussels, including federally threatened/endangered species (Gulf Sturgeon, Southern Kidneyshell). Additionally, this project provides an opportunity to improve environmental connection and social discourse within the Elba community and the broader Wiregrass region of southeast AL. This emphasizes the river as a community resource for conservation, recreation, and education. EHDP focuses on evaluating the effects of dam removal on macroinvertebrate and mussel populations and associated habitat before and after dam removal, while engaging local communities to promote understanding and support for the project. Biological assessments include targeted mussel surveys, macroinvertebrate sampling, and habitat characterization along an ~11km reach above and below the existing Elba Dam pre- and post-removal. Outreach efforts include a combination of public events and an environmental education program (Deep South Student Leadership and Environmental Action Program [LEAP]) involving local schools to implement beneficial, place-based environmental projects to engage and facilitate knowledge transfer to the community. Dam removal is scheduled for Spring 2026 while monitoring and outreach continue. Initial mussel and habitat surveys reveal an effective absence of mollusks and marginal habitat quality in 10 representative reaches along the broader ~11km study reach around the EHDP. Preliminary outreach efforts suggest public interest and support for EHDP, including new school additions to LEAP for spring 2026. By sharing these results, we aim to foster discussion, solicit expertise, and build partnerships to enhance the success of this project and serve as a model for future watershed restoration efforts.

Building a toolbox for monitoring small estuary restorations in Puget Sound

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Salmon species in Puget Sound provide essential ecological and sociocultural value but are increasingly threatened by climate change, habitat alteration, and shoreline development. Small tidal embayments, or barrier estuaries, play a critical role in supporting juvenile salmonids by providing non-natal rearing habitat, particularly for populations that remain within local basins in south Puget Sound. Degradation or loss of these habitats can therefore have disproportionate consequences for salmon recovery. Whiteman Cove, a 29-acre brackish lagoon historically connected to Case Inlet, was isolated in the 1960s by construction of a road and tide gates, resulting in decades of restricted tidal exchange and limited fish passage. Although tidal restoration is increasingly used to address these impacts and improve habitat conditions for species such as juvenile salmon, many restoration projects lack sufficient post-project monitoring to evaluate ecological responses and overall effectiveness. The Whiteman Cove restoration thus represents an opportunity for the Washington Department of Natural Resources Aquatics Division to both reestablish tidal connectivity and to develop and evaluate a suite of monitoring tools, using salmonids, associated fish assemblages and habitat conditions as indicators to assess restoration outcomes and inform future nearshore restoration efforts. This presentation will cover an integrated set of monitoring tools (seining, sediment collection, quadrat sampling, GIS) and the first 1.5 yrs of data collection.

Do constructed wetlands support aquatic top predators?

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Title: Do constructed wetlands support aquatic top predators?

Author: Ethan Isaac, Akira Terui

Abstract

Over the past few decades, significant efforts have been invested in wetland restoration. Many of these restoration projects assumed that rebuilding basal habitat is sufficient to underpin sustainable populations, including those occupying higher trophic levels. However, evidence remains limited on whether constructed wetlands foster populations of top predators. This study aims to evaluate whether constructed wetlands can support populations of aquatic snakes, top predators in wetland ecosystems. We hypothesized that constructed wetlands are less likely to be occupied by aquatic snakes because they provide limited supplies of prey resources, such as fish and amphibians. To test this hypothesis, we collected preliminary data on snake distributions, prey densities (fish and amphibians), and terrestrial habitat structures near the water (plant cover and height). Compared with natural wetlands and streams, prey densities were lower in constructed wetlands, particularly when they are hydrologically isolated from the river network in the landscape. In addition, plant cover and height in constructed wetlands were significantly higher relative to natural wetlands. Snake distributions appeared to follow these differences in prey densities and habitat structure. In total, we detected three occurrences of Northern water snake, *Nerodia sipedon*, and all of these occurrences were recorded in either natural wetlands or streams. However, the limited detections of snake occurrences did not allow us to statistically test the influence of wetland types on their distribution. Our preliminary results imply that constructed wetlands may

not support viable populations of aquatic top predators. However, given the lack of sufficient statistical replicates in our data, these results must be interpreted carefully, and additional data collection would help draw robust conclusions.

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Uncovering the ecological upshots of uncovering urban streams.

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Ecological restoration of urban streams has become common globally. However, many restoration projects lack quantitative measures of success. An engineering-intensive form of restoration involves “daylighting” of buried, culverted reaches, which typically occur in high-density urban areas. The City of Springfield, Missouri (USA) is currently implementing a daylighting project on a headwater stream that has been culverted for nearly a century. We are using macroinvertebrates to evaluate trends in water quality in two stream reaches, one ~3 km and the other ~8 km downstream from the project. The watershed of the upper reach is nearly 100% urbanized, and the lower reach ~85% urbanized. To assess the ecological effects of the daylighting project, we have thus far collected macroinvertebrates and basic abiotic data both before and during the restoration activities. We collected macroinvertebrates semi-quantitatively from three habitats (riffles, pools, and rootmats) and identified them to the highest possible taxonomic resolution. “Before” data indicated relatively good physical habitat structure and high taxa richness overall (75 unique taxa), although a third were Chironomidae, and Plecoptera were absent, indicating poor water quality. A pattern of decreasing specific conductivity (595-533 uS/cm) and increasing taxa richness (43-52) between upstream and downstream sites suggested some improvement of water quality with downstream distance. Macroinvertebrate data are still being processed for the “during” samples, but preliminary results indicate minimal change in the communities between the two time periods. This is likely evidence of resistance and/or resilience of invertebrates already conditioned to the effects of urbanization to any disturbance associated with the ongoing daylighting project. Abiotic characteristics were also comparable between the two time periods. We will continue quantitative monitoring of these sites after the daylighting project is complete, and we hypothesize gradual improvement of water quality over time. Even if ecological effects are minimal, there are other advantages to daylighting urban streams, including a variety of socio-economic benefits.

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Hydrologic response to beaver dam analogs in a semi-arid intermittent stream

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Ephemeral or intermittent streams make up more than 80% of all water courses in the southwestern United States, but the hydrologic response of these systems to restoration remains poorly understood. Within the Southwest, widespread loss of beaver (*Castor canadensis*) and their function as ecosystem engineers has contributed to channel incision, reduced surface to groundwater connectivity, and increased vulnerability of stream systems to aridification. Beaver dam analogs (BDAs) are increasingly utilized to mimic the hydrologic and ecological functions of beavers in systems that are currently unsuitable for recolonization, but the empirical data documenting the hydrologic effects of BDAs in intermittent or ephemeral streams is limited. This study evaluates the hydrologic response of a reach of Glorieta Creek, a semi-arid intermittent stream in Pecos National Historical Park (New Mexico, USA), to the installation of a three-structure BDA complex. The primary objectives were to (1) characterize the natural seasonal variability in groundwater and surface water of the stream, (2) assess changes in water storage and hydraulic connectivity following BDA installation, and (3) evaluate the stability and hydrologic effectiveness of a post-assisted BDA design in a stream with a highly variable flow regime.

This response is quantified through changes in surface water storage, groundwater levels, and surface to groundwater hydraulic gradients using paired treatment and control sites. Monitoring methodologies include pressure transducers installed in shallow monitoring wells and in-stream housings, along with repeated measurements of discharge, channel geometry, and substrate characteristics.

Preliminary results indicated that BDAs increase the seasonal persistence of surface water and raise groundwater levels. Monitoring well data, for example, showed a rise of 18 cm at the treatment site as compared to a 5 cm of rise at the control site over the course of the study. Observations suggest that BDAs alter groundwater-surface water interactions and enhance short-term water storage in systems subject to seasonal drying.

Through its focus on an intermittent stream in a semi-arid landscape, this research addresses a critical knowledge gap in freshwater science and provides insight into the potential role of low-tech, process-based restoration strategies for increasing hydrologic resilience under future climate conditions.

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Vegetation restoration impacts on water temperature, light, and insect emergence patterns along Stroubles creek

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Stream tributaries exhibit habitat heterogeneity from vegetation cover, hydrology, and channel morphology. Human land-uses (i.e. housing development, agriculture, industry) can degrade riparian buffers and impair water quality. Aquatic insect life cycles respond to water-quality changes, particularly higher temperatures that hamper metabolically taxing change from the immature to adult life stage. Mayfly life traits permit higher population densities in cool microhabitats created by riparian vegetation during elevated summer temperatures. However, whether riparian restoration creates equivalent microhabitats for emerging adults remains unknown. Our study compared water quality and insect emergence between upstream unrestored (UR) and downstream restored (R) sections of Stroubles Creek in Blacksburg, Va, a tributary of the New River, to determine how light, temperature, and vegetation affect insect emergence timing and abundance. We predicted that light and water temperature would be similar between sections during fall and winter, but lower in the restored section during late spring and summer when vegetation is denser and air temperatures are warmer.

We divided a 1.5-km section of Stroubles Creek within the StREAM Lab at Virginia Tech into two sections: UR, with sparser and more herbaceous vegetation, and R, with restored riparia of sapling trees planted in 2010. Four replicate cross-sections were established in each section. Light and temperature sensors collected data at 15-min intervals from March 24th to October 31st, 2025. Floating insect emergence traps were deployed at each cross-section during the same dates, with samples collected every 7-9 days. Insects were classified to order, counted, and analyzed for total emergence and proportion of non-dipterans per day. Through June, the UR section had slightly lower temperatures despite 2% lower light levels in the R section. Algal growth or stream morphology variations may explain this discrepancy. The UR section exhibited greater insect emergence (26 versus 18 insects per d), likely due to greater algae and warmer spring temperatures. Analysis of non-dipterans and water conditions from July-October 2025 is ongoing. Understanding how riparian restoration affects microhabitat temperature and dissolved oxygen impacted by land development and climate shifts is critical for implementation of effective conservation practices.

1. Everall, N. C., M. F. Johnson, R. L. Wilby, and C. J. Bennett. 2014. Detecting phenology change in the mayfly *Ephemera Danica*: Responses to spatial and temporal water temperature variations. *Ecological Entomology* 40:95-105
2. Hofmeister, K. L., C. M. Cianfrani, and W. C. Hession. 2015. Complexities in the stream temperature regime of a small mixed-use watershed, Blacksburg, VA. *Ecological Engineering* 78:101–111.
3. Uno, H., and M. E. Power. 2015. Mainstem-tributary linkages by Mayfly Migration help sustain salmonids in a warming river network. *Ecology Letters* 18:1012–1020.

Two years of benthic macroinvertebrate community recovery following low-head dam removal in a semi-arid Rocky Mountain stream

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Removal of obsolete low-head dams is accelerating as an aquatic ecosystem management practice. Dam removal is much less common in the arid and semi-arid West than in the water-rich Midwest, Eastern Seaboard, and Pacific Northwest regions. This paucity of examples makes it important to, when feasible, accumulate case studies of how rivers and aquatic ecosystems in arid and semi-arid regions respond to removal. In 2023, the U.S. Forest Service and collaborators removed the Lake George Diversion Dam on the South Platte River in Colorado. Sediment was excavated from the former impoundment and riparian restoration was undertaken to improve habitat and recreation opportunities. We monitored the benthic macroinvertebrate community as a bioindicator of aquatic ecosystem health using a before-after-control-impact (BACI) design prior to the dam removal and for two subsequent years. Samples were collected and analyzed at impact reaches immediately upstream and below the dam site as well as in an upstream control reach.

Before the dam removal and accompanying restoration, the impact reaches were populated by macroinvertebrate communities characteristic of lentic and degraded habitats and a multi-metric index classified most sites as “Impaired”. This was particularly true in the impoundment (upstream of the dam), which was filled with fine sediment that likely degraded macroinvertebrate habitat and feeding opportunities. Sensitive and intolerant taxa became far more prevalent in impact reaches following the restoration and index values increased by 44% with most sites achieving “Unimpaired” status. Overall density and richness significantly increased at the impact reaches relative to the control reaches and approached reference conditions within two years of the dam removal. Non-metric multidimensional scaling analysis demonstrates significant shifts in the community makeup of both impact reaches towards that of the control reach over time. Our results agree with other recent work from wetter landscapes suggesting that aquatic ecosystems in dammed river reaches can undergo reasonably rapid (two years in our study) recovery from severely degraded conditions to a state more similar to free-flowing reaches. Monitoring will continue through 2026 to assess longer-term trends in recovery following dam removal.

The Mountain Futures YouTube Channel: Sharing Mountain Science With a Global Audience

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Science needs story. Science also needs humanity. Mountain landscapes are critical to the well-being of billions of people, yet they are often remote and difficult to access which makes it hard for people to connect to them. Still, incredible research is

happening in mountains, including in the lakes and rivers they support. Despite the rich and important potential for storytelling, the people doing mountain research often exist in relative anonymity. In 2024, I launched the Mountain Futures YouTube channel with a simple goal: raising the global profile of mountain science and the people studying these critical ecosystems. With no filmmaking experience, it has been a lesson in learning by doing. Just over a year in, we've grown to 600 subscribers from around the world and our videos have been watched more than 90,000 times. In our view, this is just the beginning. For this poster presentation, I will share the ideas behind the channel and the plan for how we will sustain and grow it into the future. I will also invite mountain researchers—particularly those studying mountain lakes, streams, and the cryosphere—to connect with us and perhaps be featured in a future video. If nothing else, we hope this poster will inspire mountain limnologists to think outside of the science communication box and pursue new avenues to share their work.

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Meet the Neighbors: Microcaddisflies (Trichoptera: Hydroptilidae) of the Willamette Basin, Oregon, USA

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Freshwater invertebrates are natural ambassadors that can engage local scientific and non-scientific human audiences. Here, I present a poster about microcaddisflies that I have been using to get people excited about local biodiversity in the Willamette Basin in Oregon, USA.

Microcaddisflies are aquatic insects that are fascinating but can be overlooked due to their small size. Because of this, we probably have an incomplete understanding of which species call the Willamette Basin home. Larval microcaddisflies, like many other caddisflies, build and live in cases, with most genera having portable cases. However, in contrast to other caddisflies, they only build cases in their final larval instar. Adult males in the genus *Hydroptila* have unique, eversible androconia on their heads that produce pheromones for attracting mates, and the females of at least some microcaddisfly species dive and swim to lay their eggs on underwater substrates. Published records indicate there are 33 microcaddisfly species in Oregon, and by consulting literature and the collection records of David E. Ruiter, I could confirm that 14 of these occur in the Willamette Basin. Three additional species not previously reported from Oregon in published literature had records from the Willamette Basin in the Ruiter collection. This brings the confirmed number of microcaddisfly species in the Willamette Basin to 17, but additional species may be found with further efforts. By working together, we can learn more about the local species of these interesting aquatic insects and aim to maintain healthy populations.

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Community Structure and Food Web Dynamics in the Face of Invasion: Using Round Goby (*Neogobius melanostomus*) as a Model System

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Invasive species are found on every continent and cause severe ecological impacts. Many of these impacts are governed by feeding ecology, including predation of native species and competition with native species for food resources. In stream ecosystems in particular, the critical role of aquatic-terrestrial interactions closely governs food web structure, complicating interpretations of invader impacts. The spread of round goby (*Neogobius melanostomus*) in the Lake Erie watershed serves as an ideal model system for investigation of invasion dynamics and river food webs, as invasion is actively progressing in tributaries with diverse riparian plant communities. Our research aims to examine the interactions between community structure and food web dynamics across aquatic-terrestrial linkages. Using DNA metabarcoding, we will first examine changes in round goby diet as invasion progresses. We will additionally examine changes in native fish assemblages, relating community structure to round goby presence. Finally, we will determine whether invasional meltdown occurs across aquatic-terrestrial linkages by examining the impact of Japanese knotweed (*Fallopia japonica*)-dominated riparian communities on round goby diet and broader food web structure through stable isotope analysis and DNA metabarcoding of goby gut contents. Ultimately, this research will expand our understanding of the fundamental drivers and impacts of aquatic invasion through a detailed assessment of aquatic food web structure in the face of invasion.

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From Reaches to Regions: A Decade of NEON Sediment Data Reveals Patterns Across Freshwater Ecosystems

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Sediment biogeochemistry plays a critical role in nutrient cycling, carbon storage, and water quality in freshwater ecosystems, yet long-term, large-scale datasets have historically been scarce. The National Ecological Observatory Network (NEON) provides a unique opportunity to address this gap through its standardized, continental-scale monitoring program. NEON's Sediment chemical and physical properties data product (DP1.20194.001) includes measurements of organic matter, nutrients, and trace elements collected twice annually (spring and fall) from NEON's 34 aquatic sites (24 wadeable streams, 3 non-wadeable rivers, 7 lakes) that span 19 ecoclimatic domains across the United States. This presentation will introduce the NEON sediment dataset and explore approaches for detecting spatiotemporal trends after the first decade of data collection. We will highlight methods for visualizing and analyzing trends in sediment chemistry under conditions of low temporal frequency but high spatial coverage and explore how these data can reveal signals of environmental change, land-use impacts, disturbance events, and ecosystem processes across spatial and temporal scales. By leveraging NEON's open-access, continental-scale monitoring network,

researchers can address fundamental questions about sediment dynamics and their role in freshwater ecosystem function. This work demonstrates how continental-scale monitoring can inform freshwater science and management and provides a foundation for collaborative research using NEON's growing time series.

Characterizing Macrophyte Communities and Abundance of Macroinvertebrates in the Deep Meandering Section of the Kootenai River

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The mainstem of large, channelized, and regulated rivers are often referred to as sterile halfpipes with reduced biological diversity. However, these reaches may contain more complexity and ecological activity than is initially apparent. Macrophyte assemblages can influence hydrology, substrate stability, and organic matter retention, which can influence macroinvertebrate abundance. Yet, in regulated systems, the extent to which macrophyte community structure mediates resource availability remains poorly understood. Understanding habitat characteristics that influence fish and prey distribution across trophic levels is critical for effective management. Macrophytes provide essential nutrients, cover, and substrate for macroinvertebrates and juvenile fish, which, in turn, serve as prey for larger piscivores. This study will quantify macrophyte distribution, diversity, and native versus non-native composition throughout the meander section of the Kootenai River between Bonners Ferry and Porthill, Idaho (~75 river kilometers). This section is characterized by slower flow velocities (0.17 m/s–1) and deeper water (~6.4 m) compared to upstream reaches. Preliminary observations indicate macrophytes accumulate along shallow shorelines and near islands, potentially creating productive foraging areas. Fish have also been found to accumulate near islands, suggesting that island features may create gradients of productivity. As such, sampling will focus on habitats associated with islands, backwater areas, and representative mainstem shorelines, allowing us to capture spatial distribution. Standard rake sampling methodology will be used to collect macrophytes for characterization. Macroinvertebrates associated with macrophytes will be collected and identified to the family level. We will compare macroinvertebrate abundance, diversity, and functional feeding group composition associated with native, non-native, emergent, and submersed macrophytes. We hypothesize that native macrophytes will support more abundance and diversity of macroinvertebrates. Identifying productive refugia will inform management strategies and restoration priorities aimed at enhancing nutrient and prey availability in an oligotrophic river and similar systems. By linking macrophyte community structure and macroinvertebrate diversity, we will highlight localized biodiversity hotspots that may serve as critical food resources. This challenges the notion that large river mainstems are uniformly unproductive.

Investigating spatial patterns of denitrification potential and soil nutrient stocks in a beaver-created ecosystem

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North American beaver (*Castor canadensis*) are ecosystem engineers known to alter physical and chemical processes through hydrologic manipulation, and influence biogeochemical processes and downstream water quality. However, the spatial patterning of biogeochemical processes and nutrient storage within beaver-created ecosystems is not well understood. We aimed to investigate spatial variation of denitrification and soil carbon and nutrient (nitrogen, phosphorus) stocks in a beaver-created ecosystem in northeastern Connecticut, USA. We collected soil cores to a 30 cm depth, divided into 10 cm increments, at random points ($n = 20$) throughout three vegetation zones (open water, emergent wetland, and meadow) in August 2025. Denitrification potential was estimated using denitrification enzyme assays. Preliminary analysis suggest denitrification potential was highly variable throughout the beaver-created ecosystem, ranging from approximately zero to $> 3900 \text{ ng N hr}^{-1} \text{ g soil}^{-1}$. There did not appear to be a clear spatial pattern relating to dam proximity or vegetation zone. However, the highest points appeared to be located adjacent to shrubs and woody vegetation, including alder, a nitrogen-fixing species. Throughout the system, denitrification potential appeared highest and most variable in the top 10 cm of the core, where microbes are most active. Log-transformed denitrification potential was positively correlated to soil moisture in the top 20 cm and negatively correlated to bulk density in the top 10 cm. Soil carbon and nutrient stock processing is ongoing. Our results may allow for a more accurate understanding of how beavers influence nutrient storage and processing behind dams and inform management decisions regarding increased beaver populations in the northeastern United States.

Do beaver ponds serve as sinks or sources of greenhouse gases?

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Aquatic ecosystems can be sources of greenhouse gases. Beaver (*Castor canadensis*) ponds represent a unique and understudied ecological niche with the potential to influence greenhouse gas dynamics. Beavers are well known ecosystem engineers whose dam building transforms flowing streams into ponded wetlands that provide many ecosystem services, including carbon sequestration, while potentially increasing CH₄ and CO₂ emissions. This research aims to identify the role of beaver ponds on CH₄ and CO₂ fluxes through seasonal sampling with static chambers. We also measured dissolved oxygen, temperature, water depth, and nutrient concentrations. Chambers for gas collection, as well as depth sensors, were both deployed at upstream controls and pond locations at three different sites, including a suburban, agricultural, and forested site. In the suburban site, net CO₂ production was higher in the pond location than in the upstream control; the suburban pond emitted CO₂ at a rate of 9.98 mg C/L/m ± 14.27 while the control emitted 7.37 mg C/L/mppm±9.83. In contrast, the pond and control at the agricultural location had very similar fluxes (pond: 2.52 mg C/L/m±0.78, control: 2.42 mg C/L/m±0.5). Based on preliminary results, beaver dam ponds have the potential to emit more greenhouse gases, but patterns vary with land use, suggesting that site characteristics can mediate beaver effects.

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Impact of water temperature and competition on distribution of invasive crayfishes *Procambarus clarkii* and *Faxonius rusticus* in an urban river system

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Procambarus clarkii (red swamp crayfish) is a species of great concern in the Laurentian Great Lakes area due to the recent discovery of established populations in inland and Great Lakes-connected waters. *Faxonius rusticus* (rusty crayfish) is also a non-native crayfish which has spread throughout the Great Lakes and inland waterways in the past 40 years. Populations of both *P. clarkii* and *F. rusticus* are established in the North Shore Channel (NSC) of the Chicago Area Waterway System (CAWS) which connects Lake Michigan to the Mississippi River Basin. Previous work indicates that populations of *P. clarkii* may be temperature limited, particularly in the presence of competitors. Our work combines field and laboratory studies to determine the impact of temperature and competition on *P. clarkii* in the NSC and across the Great Lakes area. First, we leveraged catch-per-unit-effort (CPUE) data from annual crayfish trapping of the entire North Shore Channel since 2018 and water temperature data collected from 2023-2025 using iButton temperature loggers to determine how temperature influences *P. clarkii* distribution and density in our system. Second, we conducted laboratory experiments to test both the thermal tolerance of *P. clarkii* and how temperature affects competitive interactions between *F. rusticus* and *P. clarkii*. Our field study results indicate that the highest densities (average CPUE = 0.36 crayfish/trap) of *P. clarkii* occur where water temperature is higher (mean temp warmest month= 23.7°C, mean temp coldest month= 2.9°C) and *F. rusticus* population is less dense (average CPUE = 0.16). Further, our laboratory experiments indicate that *P. clarkii* is less active in lower water temperatures (13°C) than those in ambient (20°C; p<0.01). Preliminary results also indicate that *F. rusticus* won more competitive interactions with *P. clarkii* in cooler temperatures (5°C) compared to ambient (20°C; n=225, X²=13.15, df = 1 p<0.001). Overall, our studies demonstrate that *P. clarkii* becomes less active in cooler water, which may impact survival and ability to compete with *F. rusticus*. These results are directly applicable to other systems across the Great Lakes where managers are working to control populations of invasive crayfishes.

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Drivers of Crayfish Community Composition in the Altamaha River Basin of Georgia

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Crayfish are widespread, low-trophic consumers that play disproportionate roles in shaping freshwater ecosystems. As ecosystem engineers, they influence bank stability, sediment dynamics, and habitat structure through burrowing, while simultaneously recycling nutrients and supporting higher trophic levels. Despite their ecological importance, many crayfish species remain poorly understood due to cryptic diversity, variable life histories, and limited historical survey effort. In Georgia, the last comprehensive crayfish assessments were conducted in the 1980's, leaving major gaps in our understanding of species distributions, community structure, and conservation status. This project seeks to update and expand baseline knowledge of crayfish assemblages within the Altamaha River Basin (ARB), Georgia's largest drainage and a historically diverse center of crayfish endemism. We are conducting systematic sampling across all 41 HUC10 watersheds in the ARB using a combination of passive trapping, active hand and net collection, and burrow excavation. At each site, we are also quantifying environmental variables, including water quality metrics and habitat structure, to evaluate the abiotic drivers shaping crayfish community composition. Although still in early stages, this work contributes to a broader regional effort to reassess crayfish diversity across Atlantic-draining basins in the southeastern United States. By generating contemporary, basin-wide data on species distributions and habitat associations, this project will provide essential context for understanding how crayfish influence watershed resilience and will inform future management and conservation strategies for native crayfish communities.

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Modeling the influence of environmental factors on invasive *P. clarkii* Catch Per Unit Effort in an urban river system

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Red swamp crayfish (*Procambarus clarkii*) is an invasive species that has recently become established in the Chicago Area Waterway System (CAWS). This species has large ecological impacts and poses a threat of invading the Great Lakes because

it is established in the North Shore Channel (NSC) which flows out of Lake Michigan. We have been intensively trapping *P. clarkii* in the NSC for the last eight years with the goal of reducing population growth and spread of the species. Trapping is focused on the area where *P. clarkii* is at highest density. From early June until late October we maintain roughly 200 traps in this area. Traps are cleared and re-baited twice a week during June – August and once per week thereafter. This has produced a large dataset that gives Catch Per Unit Effort (CPUE; average number of *P. Clarkii* per trap) over several hundred trapping days. CPUE fluctuates both within and among seasons, ranging from near zero to as high as 4.06. Differences can be high even on consecutive trapping days. If we can determine factors that affect CPUE it will be possible to predict days when our efforts could remove more of this invader. The goal of this project is to determine the relationship between CPUE and air temperature (°C), water temperature (°C), lunar illumination (% illuminated), and total precipitation (mm). Results from the 2025 field season show that each of these environmental factors is positively related to CPUE; lunar illumination (linear regression, $R^2 = 0.044$), water temperature ($R^2 = 0.110$), daily mean air temperature ($R^2 = 0.128$) and daily minimum air temperature ($R^2 = 0.150$). Ongoing work is relating these factors to our full eight years of data so that we can better understand the relationship of these environmental factors to CPUE. Results will be used predictively so that we can concentrate our future efforts at times that crayfish are most susceptible to being trapped. This will allow us to further reduce the population density of *P. clarkii* in the NSC and to reduce the chance of the species reaching Lake Michigan.

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Assessing Aquatic Ecosystem Resilience: Bioaccumulation Potential of 6PPD-Q in Rusty Crayfish

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6PPD-Q ((1,3-Dimethylbutyl)-N-phenyl-p phenylenediamine-quinone) is a contaminant of emerging concern found in stormwater runoff. It leaches from tires and tire wear particles and is known to be toxic to a variety of aquatic organisms, including Coho salmon and brook trout. Understanding how aquatic species respond to and potentially resist bioaccumulation of emerging contaminants is critical for assessing ecosystem resilience in the face of increasing urban stormwater inputs. A log KOW of 4.0 suggests potential for 6PPD-Q bioaccumulation, however only a few aquatic species have been assessed. This study evaluates bioaccumulation of 6PPD-Q in Rusty crayfish (*Faxonius rusticus*) and we report findings from a laboratory ecotoxicological experiment and in situ field sampling from three watersheds in the upper Columbia River Basin. The laboratory experimental design exposed crayfish to nine different concentrations of 6PPD-Q (0, 5, 25, 95, 250, 500, 1,000, 5,000, and 10,000 ng/L). Replicate crayfish tissue and water samples were collected at 0, 24, 48, and 96 hr intervals, and 6PPD-Q concentrations in tissues and waters were quantified using a UPLC coupled to a triple-quadrupole mass spectrometer. Field water and tissue samples were collected from 6 sites across a spectrum of rural and urban land uses and quantified using the same methods developed for the laboratory analyses. The findings point to limited bioaccumulation potential of 6PPD-Q in crayfish tissues even under the highest exposure scenarios, though very low concentrations (<1 ng/L) were detected in field-collected samples. These results suggest that crayfish demonstrate resilience to bioaccumulation even under high exposure conditions, potentially through rapid depuration or metabolic transformation. While crayfish may not be ideal biomonitoring organisms for 6PPD-Q, the data provide valuable insight into exposure durations and toxicokinetic dynamics in a common large aquatic macroinvertebrate, informing our understanding of species-specific response mechanisms that contribute to broader ecosystem resilience in watersheds experiencing tire-derived contamination.

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Influence of harmful algal blooms on mercury cycling in New York lakes

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Mercury (Hg) contamination and harmful algal blooms (HABs) are widespread stressors threatening freshwater environments. Despite extensive research on each stressor individually, the influence of HABs on Hg cycling is poorly understood. This poses a critical knowledge gap, as HABs decrease oxygen availability and increase the flux of labile organic matter to lake sediment, which collectively increase favorable conditions for microbial organisms that can convert inorganic Hg to the potent neurotoxin methylmercury (MeHg). Once produced, MeHg is highly bioavailable and can be bioaccumulated through the food chain, eventually reaching humans via fish consumption from contaminated water bodies. Lakes in central New York represent a promising and imperative study area to assess the influence of HABs on Hg cycling. In 2024, more than 500 HABs were reported across the state¹, and over 200 waterbodies were under fish consumption advisories for Hg contamination². As HABs become more common and severe, determining their role in MeHg production becomes essential for predicting and mitigating future exposure risks. This research asks: How do total Hg and MeHg concentrations in lake water and sediments change before, during, and after HABs?

To study how HABs influence Hg cycling and MeHg production, lake water and sediment samples were collected from 6 lakes in central NY—three with a history of HABs and three without—on a biweekly basis from May through August 2025 and weekly during HAB events. At each site, sediment samples were collected from both the shoreline and the deepest part of the lake, while water samples were collected from the shoreline, surface water at maximum depth, and bottom water at maximum depth. Additional in situ measurements were collected at each sampling location (e.g., dissolved oxygen, pH, specific conductivity, temperature). Samples are being analyzed in the laboratory for total Hg, MeHg, and dissolved organic carbon concentrations. Additionally, phytoplankton composition was assessed via microscopy. I hypothesize that concentrations of MeHg in water and sediment will be elevated during and following HABs compared to pre-bloom conditions, as widespread water column anoxia and increased fluxes of labile organic matter would promote anaerobic microbial production of MeHg.

1. (1) Redelmeier, R. Harmful algal blooms found in dozens of New York waterbodies so far this summer. <https://www.wskg.org/regional-news/2024-08-26/harmful-algal-blooms-found-in-dozens-of-new-york-waterbodies-so-far-this-summer> (2024).
2. (2) Waterbodies with Advisories. https://www.health.ny.gov/environmental/outdoors/fish/health_advisories/regional/alphabetical.htm (2024).

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Phytoplankton communities and risk of harmful cyanobacteria blooms in urban stormwater ponds contaminated by road salt

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The salinization of freshwater ecosystems by road salt is a pressing environmental problem in temperate climates. However, despite increasing awareness of this issue, the effects of chloride and sodium pollution (the main constituents of road salt) on phytoplankton communities are poorly understood. While past studies have shown that excessive salt can be toxic and cause osmoregulatory stress for freshwater organisms, little is known about how communities respond in the context of other urban stressors (such as eutrophication and warming). For example, there is potential for salinization to favor toxin-producing cyanobacteria, which can cause serious public health risks. Given these concerns, this study aims to characterize cyanobacteria sampled from a series of 50 stormwater ponds across Brampton, Canada in the summer of 2022. Flow imaging microscopy was used to count individual organisms and estimate the biomass of distinct cyanobacterial genera. The data were then analyzed to assess the relative biomass contributions of toxin producers in relation to increasing chloride concentrations. For example, ponds with higher salinity tended to have higher concentrations of *Cylindrospermopsis* sp., which are associated with the production of the potent toxin cylindrospermopsin. This research contributes to our developing understanding of how ubiquitous urban stressors such as salinization affect ecosystem health and risk of harmful algal blooms.

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Bolstering the mercury biosentinel toolbox: riparian spiders and dragonfly larvae show strong agreement across exposure gradients

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Mercury (Hg), in its toxic methylated form (MeHg), is a potent neurotoxin that impairs waterbodies at a global scale due to its broad distribution through atmospheric deposition, and ultimately bioaccumulates in aquatic and terrestrial food webs. Although Hg contamination of aquatic habitats is widespread, MeHg contamination in food webs varies widely across the landscape. One efficient and inexpensive way to monitor Hg contamination in food webs is to use sentinel organisms: taxa that accumulate contaminants in their tissues without significant adverse effects and the concentration in the tissues of sentinels can be used to monitor the amount of a contaminant that is biologically available. Like fish and dragonfly larvae (Insecta: Anisoptera), riparian spiders (Araneae) are often used as sentinels of Hg contamination, particularly to estimate potential Hg exposure to riparian terrestrial wildlife. However, no studies have established a correspondence between spider Hg concentrations and aquatic-dependent biosentinels. The present study addresses this data gap by assessing the relationship between total Hg (THg) concentrations in riparian spiders and dragonfly larvae, another well-established Hg sentinel, collected from 16 sites with a range of Hg impairment. We also assessed whether there were differences in the relationships between spider and dragonfly Hg concentrations due to spider sex. Preliminary results indicate that site mean THg concentrations of long-jawed orb-weaver spiders (Tetragnathidae: *Tetragnatha* sp.) and dragonfly nymphs were positively correlated ($p < 0.001$; $R^2 = 0.83$, slope = 0.68), indicating that long-jawed orb-weaver spiders can have Hg concentrations that are reflective of Hg impairment in associated aquatic habitats. Spider sex did not have a significant additive effect on Hg concentrations ($p = 0.25$), and there was no significant difference in the slopes of the two sexes, ($p = 0.59$). Our results suggest that long-jawed orb-weaver spider Hg concentrations can reflect the relative Hg levels in other sentinels (dragonfly nymphs), and that both male and female spiders could be used to monitor Hg contamination levels of waterbodies.

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Experimental insights on how selenium speciation drives sediment, biofilm, and macroinvertebrate bioaccumulation

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Publish consent withheld

Employing passive samplers for monitoring urban stream health and informing management of 6PPD-q

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6PPD-quinone (6PPD-Q), a transformation product of the tire rubber anti-degradant 6PPD, has been identified as the cause of Urban Runoff Mortality Syndrome (URMS) in coho salmon. It is highly toxic to coho and other aquatic species, even at low surface water concentrations. While monitoring efforts have expanded rapidly since 2020, characterizing when and where aquatic life are exposed to lethal environmental concentrations remains difficult because 6PPD-Q occurs in episodic, storm-driven pulses, degrades quickly, and has toxic thresholds near analytical detection limits. To address these limitations, we evaluated passive samplers as screening tools for 6PPD-Q in small streams across King County, Washington. Polar Organic Chemical Integrative Samplers (POCIS) and Diffusive Gradients in Thin Films (DGT) were deployed at multiple sites spanning a gradient of urbanization. Samplers provided time-weighted average concentrations, capturing cumulative mass loadings over multi-week deployments. Across basins with 5–53% impervious cover, estimated time-weighted average concentrations ranged from 0.4 to 67 ng/L 6PPD-Q. Results showed elevated loadings consistent with known URMS hotspots, and loadings generally increased with impervious surface cover. Our findings support the use of passive samplers as a sensitive, cost-effective tool for risk assessment, pollution identification, and long-term status and trend monitoring. By identifying hotspots, passive samplers can help prioritize locations for mitigation efforts and inform stormwater infrastructure investment planning.

Temporal and interspecific variation in mercury between three avian species in northwest Greenland

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The Arctic is a sink for globally emitted mercury, which is deposited and methylated into its bioavailable form, methylmercury, in freshwater and marine environments. Methylmercury can then bioaccumulate in the tissues of organisms that feed from aquatic environments, and biomagnify up Arctic food webs. Birds are particularly prone to mercury contamination due to their elevated trophic position within Arctic food webs, and their eggs are a valuable tool to study temporal and interspecific trends of mercury exposure. Eggs provide information on mercury accumulated in the mother from the local environment during the breeding season. Temporal variation in mercury concentrations in eggs is highly variable across the Arctic, with different regions exhibiting distinct trends. However, there have been no studies investigating the long-term temporal variation of mercury in bird eggs from northwest Greenland. The purpose of this study was to assess temporal trends of mercury in the eggs of three avian species that breed in northwest Greenland from 2014 to 2024: Thick-billed Murres, Black-legged Kittiwakes, and Common Eiders. Egg contents were analyzed for total mercury, a proxy for methylmercury in avian studies. We found the highest mercury concentrations in Thick-billed Murre eggs, followed by Black-Legged Kittiwakes and Common Eiders. Concentrations of total mercury in Common Eider increased from 2014 to 2023 and 2024. In the face of multiple stressors threatening Arctic ecosystems, monitoring mercury contamination is key to assessing the resilience of wildlife populations. This study also provides insight into the effectiveness of global mercury-reducing initiatives like the Minamata Convention.

1. Ackerman, J. T., Herzog, M. P., Evers, D. C., Cristol, D. A., Kenow, K. P., Heinz, G. H., Lavoie, R. A., Brasso, R. L., Mallory, M. L., Provencher, J. F., Braune, B. M., Matz, A., Schmutz, J. A., Eagles-Smith, C. A., Savoy, L. J., Meyer, M. W., & Hartman, C. A. (2019). Synthesis of Maternal Transfer of Mercury in Birds: Implications for Altered Toxicity Risk. *Environmental Science & Technology*. <https://doi.org/10.1021/acs.est.9b06119>
2. Ackerman, J. T., Herzog, M. P., & Schwarzbach, S. E. (2013). Methylmercury is the Predominant Form of Mercury in Bird Eggs: A Synthesis. *Environmental Science & Technology*, 47(4), 2052–2060. <https://doi.org/10.1021/es304385y>
3. Braune, B. M., Gaston, A. J., & Mallory, M. L. (2016). Temporal trends of mercury in eggs of five sympatrically breeding seabird species in the Canadian Arctic. *Environmental Pollution*, 214, 124–131. <https://doi.org/10.1016/j.envpol.2016.04.006>
4. Lehnher, I. (2014). Methylmercury biogeochemistry: A review with special reference to Arctic aquatic ecosystems. *Environmental Reviews*, 22(3), 229–243. <https://doi.org/10.1139/er-2013-0059>
5. Dastoor, A., Wilson, S. J., Travnikov, O., Ryjkov, A., Angot, H., Christensen, J. H., Steenhuisen, F., & Muntean, M. (2022). Arctic atmospheric mercury: Sources and changes. *The Science of the Total Environment*, 839, 156213. <https://doi.org/10.1016/j.scitotenv.2022.156213>

Ghost Forest Forensics: What drives variation in soil carbon emissions in coastal forested wetlands?

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Coastal forested wetlands are globally important components of the carbon cycle. These ecosystems, and the carbon they store, are at risk of sea level rise and saltwater intrusion (SWISLR), which can kill tree stands and create ghost forests. Although ghost forests have clear consequences for aboveground carbon storage, how this ecosystem transition affects belowground carbon is less clear. Initial predictions on the effect of SWISLR on carbon emissions from ghost forests suggested methane emissions would decline due to the influx of sulfate in sea water. However, studies to date have found no consistent effect of SWISLR on soil carbon emissions in coastal forested wetlands. We designed a field survey of ghost forest sites to examine how forest mortality and saltwater intrusion impact carbon dioxide and methane emissions in different environmental conditions. We collected soil samples from 17 paired sites in coastal North Carolina, each consisting of one intact forest and one nearby ghost forest. We incubated the soils in anoxic conditions and measured carbon dioxide and methane production over seven days. Further, we explored key soil characteristics like conductivity and organic matter to deduce drivers of carbon emissions in these ecosystems. We found that both carbon dioxide and methane emissions were highly variable, with forest mortality having no consistent effect. Of the 34 individual sites, only six produced notable quantities of methane, distributed amongst live and dead forests, dry and inundated conditions, and across the salinity gradient. The best predictors of carbon dioxide emission rates were soil organic matter and reduced iron, while methane was weakly correlated with sulfide and elevation. Our results suggest that more research is needed to determine how carbon cycling in coastal forested wetlands is affected by forest mortality and SWISLR, and what this could mean for future climate forecasting.

From sediment to invertebrates: tracking metal uptake from 27 years of data on the Clark Fork River

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The Clark Fork River (Montana, USA) is a major tributary of the Columbia River with more than a century of copper mining in the Clark Fork River Basin (basin) that has resulted in metal contributions, impacting both stream ecosystems and public water supplies. Sediment from metal-enriched waste rock is the largest source of metal loads in the Clark Fork River and has been transported more than 121 km downstream from the mines. As a result of mining legacies, the basin contains the largest contiguous complex of Federal Superfund sites in the United States. Remediation efforts in the basin began in 1990 and are currently ongoing. Significant efforts include the creation of tailing ponds to capture metal-rich sediment from travelling downstream and removal of tailings from the basin, resulting in changes to downstream sediment transport. While concentrations of metals in the water column have declined sharply at some sites over the past decades, recent studies have found that once recovered trout populations are again nearly extirpated from portions in the upper river basin. To investigate how remediation influences metal accumulation, the U.S. Geological Survey began collecting sediment and aquatic invertebrate samples across the upper Clark Fork River in 1986. We analyzed this multi-decadal record and found that concentrations of arsenic, iron, and manganese were lower in tissues of predatory macroinvertebrates compared to those of filter feeders. In certain upstream sites, metal concentrations have also been observed to decrease in sediment and insect tissues since restoration began. These findings suggest that metal accumulation might have trophic-specific pathways, even withstanding long term trends of decreasing metal concentrations. Further research aims to quantify the transport of metals from sediment into food webs and how that may be influenced by spatiotemporal patterns of mining-related contaminants across the basin.

The Effects of Toxic Trace Metals on Macroinvertebrate Secondary Production in the Spokane River Watershed

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Toxic trace metal pollution introduced to watersheds through historic mining practices is a significant cause of freshwater impairment worldwide. Toxic trace metals are known to impact macroinvertebrate communities through a diversity of pathways resulting in altered community composition, reduced diversity, and sometimes lower abundance. The translation of these community impacts to secondary production has been much less studied. Macroinvertebrates are crucial contributors to the aquatic food web. Their secondary production determines food availability for aquatic and sometimes riparian predators, as well as potentially transferring contaminants to these consumers. In this study, we are documenting secondary production of the macroinvertebrate community at five sites within the Spokane River watershed with varying levels of toxic trace metals. The Spokane River watershed is directly impacted by historic mining practices that have deposited silver, lead, zinc, and other toxic metals throughout the watershed. A previous study showed that metal concentrations in macroinvertebrate tissue decrease with distance downstream of mining facilities in the Spokane River Watershed. We anticipate that sites with higher toxic trace metal accumulation will have lower secondary production because of reduced macroinvertebrate diversity and lower growth or survival of specific populations. To test this hypothesis, we are estimating annual benthic macroinvertebrate secondary production for the 2025-2026 water year for each site using the cohort production interval method. We are also determining trace metal concentrations in water samples and invertebrate tissue through ICP-OES analysis. In our initial round of sampling (November

2025) and ICP-OES analysis, the Saint Joe River contained diverse mayflies and stoneflies and the lowest zinc concentrations at undetectable levels. The Spokane River sites were dominated by Diptera in the families Chironomidae and Simuliidae with intermediate levels of zinc at 0.023mg/L at the upstream site and 0.019 at the downstream site. The South Fork Coeur D'Alene River was dominated by Ephemeroptera, mostly in the family Ephemerellidae with the highest zinc concentrations at 0.608mg/L. As more macroinvertebrate and metal data is collected secondary production will be estimated for each site, determining if toxic trace metals could potentially have an impact on secondary production in the Spokane River watershed.

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Concentrations and trends of metals in Alaskan transboundary river sediments

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The transboundary watersheds of Canada and the United States contribute valuable economic, social, and ecological benefits to their respective regions. Due to concern for negative potential impacts from Canadian mining operations, Congress directed the U.S. Geological Survey (USGS) in 2018 to begin an ongoing water-quality study to establish baseline conditions in these rivers and monitor changes in water quality and ecosystem health over time. In 2019, USGS Science Centers in Alaska, Washington, Idaho, and Montana began water quality assessments in nine transboundary watersheds. In southeast Alaska, five rivers (from north to south) are included in this monitoring program: the Alsek, Taku, Stikine, and Unuk rivers, as well as the Salmon River, which is located near Hyder, Alaska. USGS stream gages collect continuous measurements of water-quality parameters, including temperature, pH, and dissolved oxygen. In addition to these continuous data, annual sample collection visits are conducted, which comprise discrete water-quality samples, whole body fish, liver and fillet tissue, sediment, algae, and moss from within the bankfull channel. Water and solids samples are sent to national USGS laboratories in Denver, Colorado and Moffett Field, California, for selected metals analysis, including copper (Cu), cadmium (Cd), mercury (Hg), lead (Pb), selenium (Se), and zinc (Zn). Approximately 90 sediment samples have been collected since 2019. Sediment concentrations of metals from samples at each of the five Alaska sites from the last 6 years and their observed trends over time will be presented. In addition, a comparison between sites will be presented and discussed.

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Use of long-term dataset to evaluate how strongly selenium concentrations in fine sediment and fine particulate organic matter correlate to selenium concentrations in benthic macroinvertebrate tissues

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EPA published a national criterion for selenium (Se) in 2016 that prioritizes assessment of fish tissue concentrations in 2016. Guidance was included in this document for two methods of developing alternate Se criteria in streams for which use of the national criterion might not be appropriate based on site-specific conditions. One of these approaches, termed the mechanistic modeling approach, incorporates data from multiple trophic levels, including Se concentrations in water column, particulate matter, macroinvertebrates, and fish. Se accumulation in aquatic environments occurs at the base of the food web from water into particulate matter such as algae, detritus, or sediment, followed by accumulation from particulate matter into macroinvertebrates and macroinvertebrates into fish tissues. EPA noted in the national criterion document that the transfer from Se in the water column into the particulate matter can be highly variable dependent in part on what type of particulate matter is collected. EPA recommends only using sediment measurements if there is also at least one measurement from either algae or detritus based on observations of a lower correlation between Se water concentration and sediments compared to the same analysis between water and algae and detritus. Utilizing an extensive long-term data set of over 20 years that includes Se concentrations in fine sediment, detritus, and macroinvertebrates from a stream sampled in the Pacific Northwest, we investigated the impact of utilizing Se concentrations in fine sediment compared to detritus on the next step in the trophic transfer of Se by determining if the correlation between Se concentrations in particulates and macroinvertebrates followed the same pattern observed between Se concentrations in the water column and particulates, i.e., is the correlation between sediment and macroinvertebrate tissues also weaker compared to the correlation between Se concentrations in detritus and macroinvertebrate tissues? We have encountered sand-bottomed stream systems in which algae and detritus are not readily available in the amounts necessary for Se analysis; this information could be helpful in determining the effect of having only fine sediment samples available for analysis in such streams.

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Insecticides in intermittent streams and their effects on non-target macroinvertebrates

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Intermittent streams, which comprise a large proportion of lotic systems globally, exhibit unique hydrological dynamics that may influence the transport and ecological effects of insecticides. During low-flow periods, isolated pools can accumulate solutes, which are released downstream as high-concentration pulses once flow resumes. We investigated the distribution and ecological effects of imidacloprid, a neonicotinoid insecticide widely used to treat Eastern Hemlock (*Tsuga canadensis*) for Hemlock Woolly Adelgid (*Adelges tsugae*), and its metabolites (imidacloprid-olefin, desnitro-imidacloprid, and imidacloprid-urea) in intermittent

stream systems. While imidacloprid is highly toxic to aquatic macroinvertebrates, prior research has found limited impacts on downstream communities following applications near perennial stream systems, leaving the effects in intermittent systems poorly understood. We investigated these trends in forestry lots in Western New York state using a three-fold approach. First, we tracked continuous hydrological conditions in three stream systems, including two treatment sites and one untreated control, using Stream Temperature, Intermittency, and Conductivity (STIC) sensors. Second, we quantified concentrations of imidacloprid and its metabolites in surface water and sediment using liquid chromatography tandem mass spectrometry (LC-MS/MS), comparing among sites differing in flow regime (intermittent vs. perennial) and location relative to treatment sites (upstream vs. downstream). Finally, we assessed the composition and biodiversity of the aquatic macroinvertebrate community to evaluate the combined effects of insecticide concentration and flow intermittency on non-target aquatic invertebrates. Preliminary results suggest imidacloprid application has little impact on invertebrate community composition in intermittent streams. This may be related to relatively low concentrations in winter surface water samples, though spring water samples and sediment samples are still in progress. This study will provide insight into the sustainability of current integrated pest management practices for Hemlock Woolly Adelgid by evaluating whether existing stream management zone definitions effectively protect aquatic habitats and their non-target macroinvertebrate communities from insecticide accumulation and transport.

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Seasonal emergence patterns of a diverse assemblage of Ephemeroptera, Plecoptera, and Trichoptera in two non-perennial headwater streams

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Streams can provide substantial energy subsidies to terrestrial ecosystems in the form of emergent aquatic insects. Because subsidy fluxes between streams and surrounding ecosystems are asynchronous, terrestrial predators often rely on adult stream insects as prey during periods with low terrestrial arthropod abundance, particularly in winter and early spring. However, most of what is understood about stream-to-terrestrial subsidies is from perennial systems. We quantified seasonal emergence patterns of Ephemeroptera, Plecoptera, and Trichoptera (EPT) from two zero-order non-perennial streams in the Ozark Highlands, a topographically complex karst region of the central USA. Fish colonize one of the streams during floods, and the other stream remains fishless. We collected emergent EPT during six 2-week periods spanning four key non-perennial stream seasons in this system in 2024-2025: rewetting (Nov), full-flow (Feb, April, June), pooling (July), and dry (Sept). Along comparable ~100-m reaches for each stream, we deployed 6 pyramidal emergence traps in the stream channel, as well as 6 Malaise traps placed lateral to the channel to estimate abundance and species richness of emergent EPT undertaking lateral flight. Including both streams, we have thus far documented 14 Plecoptera species, 11 Trichoptera species, and an estimated 8 Ephemeroptera species (mayflies are still being processed). Abundance of emergent EPT increased through the full-flow season, with peak emergence in June. Throughout the study, 64% of total captured EPT abundance across both trap types (~2,300 individuals) were collected by the 12 total Malaise traps. Of these, 487 winter stoneflies representing 4 species (*Allocaenia jeanae*, *A. rickeri*, *A. mohri*, *Zealeuctra warreni*) were captured early in the full-flow season (Feb). Interestingly, the stream with fish produced >2X more total EPT individuals than the fishless stream through the project duration. These numbers were largely driven by mayflies that are likely adapted to withstand fish predation through life-history responses such as increased development rates. However, the fishless stream had greater richness (25 species) than the fish stream (23 species). These findings show that very small non-perennial streams provide abundant and seasonally critical subsidies to neighboring terrestrial ecosystems, stressing the need for conservation management of these heretofore vastly underappreciated systems.

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Influences of fish on ameletid mayfly and nemourid stonefly adult body size in two small, non-perennial streams.

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Emergent aquatic insects can be valuable subsidies to both aquatic and terrestrial food webs. Fish may influence these subsidies through both lethal and non-lethal effects, including selective predation and chemical cues that increase development rate, resulting in smaller adult body size. For example, baetid mayflies and some predatory stoneflies (e.g., perlids and chloroperlids) have shown smaller adult body size in fish presence. However, the effects of fish on non-predatory stoneflies and various other insect taxa remain unknown. We are assessing if some of these understudied taxa emerge at smaller adult size and biomass in the presence of fish in two small, but diverse, non-perennial streams of the Ozark Highlands (USA), one with fish and the other fishless. In both streams, we ran six in-stream pyramidal emergence traps and six Malaise traps placed within 1 meter of the bank simultaneously during the peak emergence season. Insects were collected into propylene glycol to preserve biomass. We have chosen to evaluate two taxa common in both study streams, an ameletid mayfly, which cannot feed in the adult stage, and a nemourid stonefly, which can. For each individual, we measured either mesonotum width for ameletids or head capsule width for nemourids as proxies for body size. We will also directly measure biomass as ash-free dry mass. Preliminary body size measurements indicate that the ameletid emerged at smaller sizes in the fish stream, reflecting previous findings for swimmer mayflies. Conversely, the nemourid showed no size difference between the fish and fishless streams. This pattern may reflect weak fish predation pressure on nemourids, as their small nymphs primarily occupy interstitial spaces within streambed sediments. Biomass measurements are ongoing, but we anticipate that individuals collected in emergence traps will show similar patterns to body size measurements. We also expect ameletids captured in Malaise traps to exhibit a greater reduction in biomass following emergence than nemourids, whose reduction, we predict, will be minimal. Examining life history responses of previously unassessed emergent aquatic insects due to fish presence is key to understanding top-down influences on secondary production and energy subsidies, particularly in unique, understudied, and underappreciated non-perennial streams.

Spatiotemporal variation in ecosystem functioning in a semi-arid intermittent river

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On a global scale, intermittent rivers and ephemeral streams (IRES) constitute the majority of lotic networks by length but are relatively understudied despite serving as critical ecosystems in arid and semiarid regions. IRES are ecosystems where spatiotemporal variation in hydrologic regime can affect multiple functions, including nutrient cycling, decomposition, and ecosystem metabolism. Understanding how hydrological variation affects these functions is crucial for informing water resources management strategies. In this study, we examined the influence of seasonality, watershed position, and hydrologic regime on rates of gross primary production (GPP), ecosystem respiration (ER), and terrestrial leaf litter decomposition in the San Saba River (Texas, USA). This river exhibits substantial spatiotemporal variation in hydrologic regime along its upstream – downstream course: stable and consistent perennial flows, intermittent flows and drying in summer, and seasonally variable but perennial flows. We examined the above ecosystem processes at nine sites that represent these three flow regimes across seasons (Fall, Winter, Spring, and Summer). We hypothesized that light and temperature are principal drivers of GPP and ER, respectively, whereas hydrological characteristics, water temperature, and reach position strongly influence nutrient dynamics and litter decomposition rates. Preliminary analysis indicates the presence of substantial spatial variation in GPP and ER, but that ER and litter decomposition rates were strongly influenced by season. Overall, the results of our study indicate that different facets of ecosystem functioning are affected by diverse environmental drivers at various spatiotemporal scales. Findings from this study have implications for water resources management planning, especially under changing climates.

Where Cattle Meet the River: Predicting Stream Susceptibility to Grazing Disturbance Using Fuzzy Logic

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Across federally managed rangelands in the Intermountain West, streams and riparian areas are limited but frequently used by cattle for drinking and seeking refuge from heat and weather, creating persistent management challenges for agencies tasked with conserving freshwater ecosystems. The Bureau of Land Management (BLM) collaborates with partner organizations to collect standardized stream monitoring data that informs management decisions about grazing and other disturbances on public lands. However, while field-based indicators describe discrete stream monitoring locations, there remains an opportunity to link these data with landscape-wide spatial datasets to generate a network simulation of stream conditions. Modeling with Fuzzy Inference Systems (FIS) is well suited for handling ecological data with vague or qualitative components because it integrates expert knowledge with quantitative predictors to produce interpretable outputs. Here, we apply a FIS to generate a susceptibility index that ranks stream reaches across Nevada by their predicted likelihood of cattle access, which can inform targeted grazing management. We use five spatially derived ecological predictors widely recognized in the literature as influencing cattle distribution across rangeland environments: proximity to upland water sources, cattle-specific heat stress index (HSI), riparian vegetation density, stream-adjacent hillslope, and proximity to upland shade. Preliminary outputs generate continuous susceptibility estimates along stream networks, revealing spatial patterns in cattle–stream interactions. We will test this preliminary model with a field-based disturbance indicator from the BLM's Multiple Indicator Monitoring (MIM) dataset. This collaborative approach between the BLM and its partners highlights how standardized monitoring data and interpretable modeling tools can serve both agency reporting needs and broader scientific applications that generate actionable insights for streams and rivers on public lands.

Investigating dispersal modes of macroinvertebrate occupants of mussel beds

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Freshwater mussels (Unionida) are often found in patchily-distributed, dense communities called beds. Mussel beds frequently harbor larger, more diverse communities of other macroinvertebrates than mussel-poor benthos. While many studies provide explanations for why other macroinvertebrates benefit from occupying mussel beds, few have provided insight into the process of invertebrate community assembly. Furthermore, it is well-established that many macroinvertebrate species are highly mobile, for example, regularly engaging in downstream dispersal behavior called drifting, so occupancy of a mussel bed may be quite brief. In order to investigate the assembly process of mussel-bed occupants, we identified the dispersal modes of organisms collected from mussel beds of varying densities and species composition. During intensive mussel surveys of wadable streams in western New York, we collected macroinvertebrates, characterized substrate, and enumerated and identified all mussels present in randomly selected quadrats. *Ex situ* we identified macroinvertebrates to the lowest practical taxonomic level and used existing literature to assign dispersal capability and modes to all taxa. By doing so we can determine whether diverse bed communities are associated with patterns of dispersal tendencies in the resident taxa. Preliminary data show variation in community composition among quadrats with several taxa showing similar abundance across all samples and select taxa showing high abundance in only certain streams. In the future, we will monitor drifting macroinvertebrates in proximity to mussel beds and in relation to hydrological and physical habitat drivers with the goal of identifying causal relationships between macroinvertebrate community composition and aquatic taxa dispersal modes.

Ichthyological teaching collection curation by undergraduate students

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As part of a taxonomy workshop class I was given the opportunity to work on the SUU fish collection. This collection has had limited curation over the years and was in need of a lot of work. Many specimens were in need of more ethanol with many needing identifications as well as updates to their taxonomic ranking and scientific names. The curation of this collection will aid in its use for teaching demonstrations and in future research and projects. The collection consists of six cabinets full of wet specimen jars. I began by going one cabinet at a time and assessing which jars needed maintenance (ethanol, new jar), and then assessing the identifications on each jar. Various identification books, websites, along with image searches were used to aid with identification. While going through the collection some incorrect identifications were found, along with multiple outdated names and outdated taxonomic arrangements. Corrected identification tags were added to the jars along with the date of the correction. The specimens that are in good shape could be utilized in research relating to various characters of different groups. Further upkeep will be required to keep the collection preserved and as a useful teaching tool. Further maintenance could utilize further undergraduate research and the SUU ichthyology class.

Aquifer to Estuary: Linking Groundwater Source Mixing to Stream and Salmon Responses

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In the Kenai Peninsula Lowlands, Alaska, groundwater discharge comprises ~70% of annual streamflow, playing a key role in modulating flow and temperature in salmon-bearing streams. Before discharging to streams, groundwater flowpaths are often intersected by landscape elements such as peatlands, floodplains, sloughs and oxbows, and lakes. These landscape elements can perform hydrologic functions including lag (e.g., storage), sink (e.g., groundwater recharge), and source (e.g., flow generation), which in turn influence the chemical and biological characteristics of downstream waters. In this region, approximately 40% of mapped surface and subsurface flowpaths intersect peatlands, highlighting their potential importance in conditioning groundwater prior to discharging to streams. The Anchor River is a groundwater-influenced system draining peatland-dominated landscapes and supports one of the largest salmon fisheries in the region. To characterize peatland hydrologic functions, we deployed continuously recording specific conductance loggers in monitoring wells on a flowpath from a hillslope, through a peatland, through another hillslope, through a slough and oxbow, and along a river transect to the estuary, with simultaneous river discharge measurements. We used specific conductance as a conservative tracer to distinguish between precipitation and groundwater sources to flow and to infer relative residence times along the flowpath. Preliminary results from May – October 2025 show distinct spatial and temporal patterns in specific conductance, with measurements distinguishing between shallow, seasonally variable groundwater and deeper, stable groundwater throughout the flowpath, including within the peatland. River measurements reflect a mixture of these shallow and deep groundwater sources. Together, these patterns indicate that peatlands perform multiple hydrologic functions, including storing water, contributing to downslope groundwater discharge, and facilitating localized mixing of groundwater before discharging to the stream. These hydrologic and chemical measurements intersect an existing salmon monitoring weir and rotary screw trap that record both adult salmon upstream migration and juvenile outmigration. This work provides an initial framework for characterizing peatland-stream hydrologic connectivity and establishes a foundation for the next step of the project to develop a statistical model with local management that links water quality and discharge to observed salmon migration patterns.

Working at the confluence of wildfire, hydrology, drinking Water, and human dimensions

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Across the United States, approximately 180 million people rely on drinking water supplies that originate in forested watersheds. When wildfires burn these source watersheds, impacts can be complex, unpredictable, expensive, long-lasting, and far-reaching; affecting downstream water utilities, public health, aquatic ecosystems, and communities while testing their resilience. Because these consequences span ecological, biogeochemical, engineering, and social systems, providing solutions to wildfire threats to water security cannot be achieved by a single discipline or organization. The Wildfire and Water Security (WWS) Project is an interdisciplinary, multi-institutional effort focused on co-producing actionable science to support forest and drinking water managers. The project brings together 65+ members from 13 organizations, including universities, federal agencies, and a non-profit organization, as well as more than 25 external partners and collaborators. Our shared vision is to ensure forest and drinking water managers have enhanced information and tools to safeguard water supplies and reliably deliver clean drinking water to communities from forested watersheds that have been impacted by wildfire. To support this vision, WWS is organized into five research focus areas: source water quality, watershed and ecological processes, drinking water treatment, human dimensions, and integration. Here we share stories and lessons learned from across the project that illustrate both the successes and challenges of coordinating science across disciplines, institutions, and career stages. We also highlight experiences with external partner and science user engagement, internal and external communication strategies, data management, and project leadership.

We reflect on early challenges associated with transdisciplinary science, including rapid funding timelines that limited initial team science planning, time commitments required to create shared understanding of cross-disciplinary consideration, and a willingness and vulnerability to step outside familiar disciplinary silos. These same challenges also revealed effective strategies such as in-person all-hands meetings, informal spaces for cross-team communication, and embracing play and fun in the scientific process. Together, these approaches have supported meaningful progress in addressing the interdisciplinary problem of wildfire impacts on water. By sharing these successes and challenges, we aim to provide practical insights, while continuing to learn from others, about collaborative, co-produced science that can be translated into decision-relevant outcomes.

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Investigating the role of landscape connectivity on dissolved organic matter and nitrogen dynamics in an ice-wedge permafrost landscape

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Many regions of Arctic tundra are characterized by ice wedge polygons, created by the repeated freezing and thawing of the ground surface. Troughs form between polygons, ranging from half a meter to three or more meters across. These troughs create networks of microtopographic drainages, called capillaries here, with varying degrees of complexity and connectivity, as water pools or runs off across the landscape. This lateral export of water and associated carbon remains poorly understood, particularly in the context of changes to permafrost degradation and stabilization. We sampled dissolved organic matter and dissolved nitrogen concentrations in the outlets of capillary trough systems near the Jago River in the Arctic National Wildlife Refuge, on the North Slope of Alaska in July and August of 2024 and 2025. Several capillaries also had associated hydrological data. Additionally, we traced connected capillaries upstream of outlets, sampling at intervals to determine relationships between water chemistry and landscape position within a network. Low flow conditions during the field season in 2025 led to substantial changes in nutrients and carbon in repeatedly sampled capillaries. We also found variation between capillary systems that may be linked to the size and length of the drainage system. Overall, this work fits into a larger effort to characterize how the connectivity of the landscape impacts the export of carbon and water to larger river systems in permafrost polygon regions.

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Announcing the second edition of “Where Rivers Are Born: The Scientific Imperative to Protect Small Streams and Wetlands”

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In 2003, American Rivers published the first edition of *Where Rivers Are Born*, a document that aimed to communicate the scientific evidence for the value of headwaters to general audiences. Written by Dr. Judy Meyer and colleagues, *Where Rivers Are Born* has been widely regarded as a valuable tool for communicating about the ecosystem services of non-perennial streams, isolated wetlands, and other waterbodies that frequently lack legal protection. By the mid-2020s, however, both scientific and policy references of the document were due for an update. At the request of American Rivers, we recently completed a revised version of *Where Rivers Are Born*, which has been jointly published by American Rivers and the University of Georgia's River Basin Center. We retained a few sections of the original prose that resonate today as well as they did when originally published, but most of text is new, along with new scientific illustrations by Emily Underwood and photographs from Freshwaters Illustrated. Like the original edition, our focus is on North America, and we specifically discuss linkages to the United States Clean Water Act, the primary statute protecting water resources in the US. However, the scientific basis for the value and services of headwater streams and wetlands applies equally across the globe, and we hope this new edition proves to be as broadly useful as the original.

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Diversity and Ecological Roles of Pupil Shapes in Freshwater Stream Fishes of the United States

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Pupil shape plays a pivotal role in visual ecology. In taxa such as mammalian herbivores, anurans (frogs), and snakes, previous research has established linkages between pupil shape and ecological niche. However, systematic assessment of pupil shape diversity and their association with niches in freshwater fish is limited. This study investigates interspecific variation in pupil shape and its niche relationship of freshwater stream fish, encompassing over 20 families, 90 genera, 500 species, and approximately 1500 photographic records of individual fish specimens. This study surveyed image-based records from online platforms (e.g.,

iNaturalist, Joel Sartore's Photo Ark, state and federal agency sites), field guides, and primary literature. In addition, feeding group assignments were made using the open source FishTraitsDatabase. For each species, a minimum of three and a maximum of five individual images were assessed. Species-level pupil shape was assigned using a majority rule, whereby a shape was recorded if at least two of three individuals shared the same form. A total of six pupil shapes were identified such as circular, anteriorly elliptical (oval & almond shape), dorso-ventrally elliptical, antero-posteriorly elliptical, etc., with the oval shape occurring at the highest frequency. Stacked bar plots showed greater pupil-shape diversity in species preferring moderate current speeds compared to fast or slow currents. Species exhibiting egg-guarding behavior predominantly had almond-shaped pupils. Pearson's Chi-square and Monte Carlo Fisher's Exact tests confirmed significant associations between pupil shape and trophic ecology, habitat preference, and reproductive ecology. Additionally, Multiple Correspondence Analysis indicated that specific pupil shapes clustered with particular ecological niches. Notably, pupils with vertices in the anterior position (almond shape) tended to be associated with fast current stream predators. This study shows that the pupil shape categories provide a useful framework for characterizing ecological niches in freshwater fish. Results suggest that pupil shape is functionally associated with habitat preference and trophic ecology in fishes.

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Investigating fish assemblage response to wildfire and fire regime in a wilderness river network.

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Wildfires can have large and direct impacts on ecosystems; however, little is known about how fish respond to wildfires, especially in remote wilderness settings. Research has shown that invertebrate productivity can benefit from wildfires in wilderness settings, that this pulse of productivity can extend to fish after fire, and that wildfire is an essential factor in the maintenance of native biodiversity. Big Creek, located in the wilderness of central Idaho, USA, recently experienced a large fire during the summer of 2025. This fire follows several major fires since the late 1980's. Fish abundance surveys have been conducted at various points during this period, preceding the 2025 wildfire. We are taking advantage of this circumstance to repeat these abundance surveys using a Before-After-Impact-Control (BACI) study design. Sites surveyed that burned will be resurveyed to collect post-fire data, and sites that did not burn will also be resampled for comparison. Our aim is to determine if the high-severity fire in 2025 influenced fish populations residing in Big Creek and its tributaries. Further, at a subset of sites that have burned repeatedly over the period of monitoring, we will be able to assess responses to a series of wildfires. Therefore, results from this investigation, combined with long-term monitoring conducted in the area, may not only reveal short-term responses to wildfire, but also the consequences for fish populations as the result of a shifting fire regime.

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Investigating the predator-permanence hypothesis for macroinvertebrates in southeastern lentic habitats

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The predator-permanence hypothesis posits that two transitions – between temporary and permanent habitats and between habitats with and without fish predators – control the overall structure of animal assemblages in lentic freshwater systems. Evaluation of this concept, developed in the northern United States, at broader spatial scales has provided inconsistent results, however. To determine the applicability of this paradigm outside of the region in which it was conceived, we conducted a comparative analysis of macroinvertebrate communities across lentic habitats in Whitehall Forest, Athens, GA, USA, which vary principally in hydroperiod and fish assemblages. In order to characterize these communities, we collected benthic and littoral samples from 14 Whitehall pond and wetland habitats across a gradient of permanency (temporary, permanent) and predation regime (none, small fishes, small and large fishes) and will use non-metric multidimensional scaling (NMDS) to visualize community differences, as well as PERMANOVA for statistical differentiation. Adherence to or deviation from the predator-permanence hypothesis will be determined by comparing overall community structures across habitats and by examining the life history traits of major taxa within those structures. If our results follow those predicted by the hypothesis, we will observe more active, rapidly developing prey taxa and few predators in ephemeral habitats, large invertebrate prey and predators in permanent fishless habitats, and small, inactive invertebrate prey in permanent habitats containing fish. It is possible that some deviations will arise from the hypothesis, which may be due in part to the subtropical environment. As part of a larger collaborative project seeking to more holistically characterize these aquatic communities, this macroinvertebrate-focused study aims to contribute to a better understanding of the applicability of a major paradigm for lentic community structure beyond the region in which it was conceived. Knowledge of how communities structure themselves in response to environmental and ecological stressors is crucial for informed biological control efforts (e.g. fish stocking/removal) and for predicting the effects of climate change on lentic habitats (e.g. shortening hydroperiod).

1. Ali, S., P. K Mishra, A. Islam, and N. M. Alam. 2015. Simulation of Water Temperature in a Small Pond Using Parametric Statistical Models: Implications of Climate Warming. *Journal of Environmental Engineering* 142:04015085.
2. Batzer, D. P., L. B. Epele, and S. Reindl. 2024. Assessing the relative impacts of habitat size, hydrology, and fish occurrence on aquatic invertebrate assemblages in a set of depressional wetlands of Georgia, USA. *Hydrobiologia* 851:2519-2527.
3. Wellborn, G. A., D. K. Skelly, and E. E. Werner. 1996. Mechanisms creating community structure across a freshwater habitat gradient. *Annual Review of Ecology, Evolution, and Systematics* 27:337-363.

Nutrient Enhancement in the Westside Tributaries to the Kootenai River in Northern Idaho: Using Alternative Measures to Increase Productivity

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For some Arctic streams, the only nutrients available come from the decaying carcasses of spawned-out kokanee and salmon. The absence of these kokanee creates a productivity deficit which then cascades into a decline in food availability. The decline and eventual collapse of the kokanee population in the Kootenai River tributaries have resulted in a net loss of productivity for the tributaries due to the disappearance of returning spawning kokanee.

Returning kokanee that spawn and eventually die in the tributaries are a significant source of nitrogen, phosphorus, carbon, and other micronutrients. Collapse of this fishery has negatively affected productivity in both the tributaries and the mainstem Kootenai River. Kokanee are considered a keystone species and are prey for burbot, white sturgeon, and bull trout. We propose adding nutrients to experimentally offset the nutrient deficit from the decline and/or disappearance decaying kokanee. The goal is to stimulate productivity and potentially increase food availability at critical times in both the tributary and the mainstem Kootenai River. Results from this study are expected to inform future scientific decisions regarding stream nutrient management in the Kootenai River tributaries.

This experimental study is designed to investigate the effectiveness of adding limiting nutrients to Kootenai River tributaries with the goal of increasing productivity in tributaries that historically supported kokanee populations.

Quantifying areal excretion and egestion rates of endemic freshwater snail *Elimia boykiniana* in a low-gradient, karst stream

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Animal consumers contribute to fluxes and ambient pools of key nutrients through excretion and egestion, influencing stream biogeochemical processes. Aggregations of consumers can create hotspots of nutrient cycling due to high areal rates of biological activity. Freshwater gastropods have the potential to influence ecological processes due to their high abundance and role as primary consumers, yet few studies have attempted to quantify their contributions to nutrient cycling in streams. To understand the role of freshwater snails in nutrient cycling, we conducted excretion assays with *Elimia boykiniana*, a Pleurocerid occurring in high densities in Ichawaynochaway Creek, GA, USA. We incubated individual snails in 15 mL of filtered stream water for one hour, after which we filtered and analyzed water for ammonium and soluble reactive phosphorus. We quantified egestion rates using the mass of egestion pellets that we filtered out of the sample at the end of the trial, and hourly rates of ammonium and phosphorus excretion by taking the difference of snail treatment and the average of control (no snail) treatment concentrations. We determined mass-specific excretion and egestion rates using a length-mass regression curve developed for *E. boykiniana*, and calculated areal rates using densities from 10 surveyed shoal reaches on Ichawaynochaway Creek. Excretion and egestion rates varied with individual size, and areal rates varied across sites with position in the watershed. Our results shed light into the functional role of an endemic pleurocerid in a low-gradient, coastal plain stream, illustrating the ecological significance of an under-studied group of freshwater organisms.

Evaluating paired air and stream temperatures as a method to monitor process based restoration

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Low-tech process-based restoration (LTPBR) aims to restore physical, chemical, and biological processes in river and floodplain ecosystems and has been widely adopted, particularly in the Intermountain West. These projects often employ small structures designed to slow and spread flow across the landscape, reconnect floodplains, and thereby support chemical and biological processes. While LTPBR shows promise, scientifically rigorous monitoring is frequently limited by funding and time constraints.

Here, I explore whether paired air and stream temperature (PASTA) monitoring may serve as a suitable low-cost method for assessing whether LTPBR projects alter hydrology. PASTA uses paired air and stream temperatures, and particularly the difference between the two in both magnitude and lag, to infer the contribution of groundwater. If LTPBR projects are successful at reconnecting floodplains, then there should be an increase in shallow groundwater flow, which may be detectable through changes in the relationship between air and water temperatures.

This study was conducted in three streams treated with LTPBR in Northern New Mexico, USA on established sites deemed effective by practitioners. This study uses paired upstream-downstream sampling from three treated reaches to test whether changes in groundwater connectivity can be detected through shifts in the relationship between air and water temperatures. Specifically, I evaluate differences in temperature magnitude and temporal lags between air and stream temperature both above and within reaches treated with LTPBR. I hypothesize that treated reaches will see greater lags between changes in air temperature and the subsequent changes in water temperature, due to an increase in shallow groundwater contributions. Further, I hypothesize that these changes will be largest in after high flows when water absorbed by the floodplain is reentering the stream channel. If

these changes are detectable, then our results will suggest that the PASTA method could provide practitioners with a practical, cost-effective tool to evaluate whether restoration projects are influencing hydrological conditions.

Opportunities In Integrated Ecosystem and Water Chemistry Data for Springs in the Valley and Ridge Province of Central Appalachia

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Biological communities can be useful indicators of water quality. Mollusks and crustaceans require adequate alkalinity and mineral content, along with an appropriate pH, to molt, build, and maintain shells and exoskeletons. Calcium is a crucial mineral for these animals. Consequently, they often dominate springs sourced from calcium-rich source rocks. Conversely, insects dominate springs flowing through source rocks with insoluble, Ca-poor minerals. Previous work documenting the link between spring water chemistry and biological communities in the Valley and Ridge Province (V&R) examined end-member springs whose source rock was either sandstone or carbonate. However, this overlooks mixed lithology springs and the variability of water chemistries spatially and temporally. Integrated chemical and biological data are limited and don't account for source rocks or chemical variability.

Our goal is to fill this knowledge gap by examining the chemistries of different spring types. We compared chemical data from 78 springs in the V&R in West Virginia and Pennsylvania (374 samples total). Data included pH, specific conductance (SC), alkalinity, and major dissolved ions. The study focuses on springs sourced from siliciclastic rocks (e.g., sandstone and shale) and carbonates (limestone and dolomite); springs from deeper thermal flow systems were included for comparison.

The spring water chemistry ranged from very dilute (sandstones) to highly mineralized (thermal-mineral waters). The carbonate-fed springs had circumneutral pH and high concentrations of Ca. Chemistry of siliciclastic springs varies by lithology. Shale-sourced springs had comparable pHs to carbonate springs, higher alkalinity, SC, and Ca concentrations than sandstone springs, likely due to small limestone layers and cements associated with the shale strata. Based on these preliminary data, we predict that shale springs will have a higher number of crustaceans and mollusks than sandstone springs.

Future studies include the collection of integrated chemical and biological data for all spring types to better understand the relationship between chemical conditions and biological communities in V&R springs. We hope that this data can be used in the future to develop biological indices for spring water chemistry. Understanding the connection between time-averaged biological communities and sporadic water samples could also be useful for evaluating roadside spring-related public health issues.

Hidden in plain sight: freshwater sponges in eastern Washington

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Freshwater sponges are immobile, filter-feeding invertebrates that can have photosynthetic algal symbionts in their tissues. However, they remain poorly documented relative to other freshwater organisms. So far, no published studies describe the populations of freshwater sponges in eastern Washington. The goal of this study is to document the distribution and diversity of freshwater Porifera in this region.

We conducted an initial census of 27 stream sites in late summer 2021. At each site, we searched the reach for 1 hour and examined at least 100 substrate particles. We measured water quality parameters including temperature, dissolved oxygen, turbidity, pH, oxidative-reductive potential (ORP), and concentrations of nitrogen and phosphorus. We found freshwater sponges in 9 of the 27 sites. The only water quality parameter that was significantly associated with sponge presence was temperature. Sites had warmer water where sponges were present compared to sites without sponges (t-test, $t = 3.43$, $P = 0.003$). We used cytochrome oxidase subunit I gene (COI mtDNA) sequences and compound microscopy of gemmuloscleres and megascleres to identify sponges. We observed 4 common genera (*Heteromeyenia*, *Ephydatia*, *Eunapius*, *Spongilla*), but were unable to identify all specimens.

We will build on this work with additional sampling during spring and summer of 2026, followed by improved genetic and microscopic techniques. Additional species, with different environmental requirements, may be present earlier in the year. Where sponges are present, specimens will be collected for phenotypic and genetic analyses.

Phenotypic traits will be analyzed through scanning electron microscopy (SEM) to examine diagnostic structures such as spicules and gemmuloscleres. Genetic traits will be assessed using optimized DNA extraction and amplification protocols targeting commonly used mitochondrial markers for freshwater sponge identification. We may also include genetic analysis of algal symbionts.

This research aims to improve our knowledge and understanding of freshwater sponge populations in eastern Washington. By combining field sampling, SEM phenotypic classification, and genetic techniques, this research will contribute to a better understanding of these freshwater invertebrates and their symbiotic relationships.

Impact of urbanization on the behavior of freshwater invertebrates

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Urbanization impacts the invertebrate communities in aquatic ecosystems, typically causing species assemblages to be dominated by pollution-tolerant, generalist taxa. Less is known, however, about how urbanization alters intraspecific traits in aquatic organisms. Intraspecific variation in behavioral traits is particularly important because the behavior of aquatic organisms can have significant impacts on ecosystem functioning and trophic relationships. We hypothesized that aquatic invertebrates from urban environments would be more bold, active, and voracious foragers relative to conspecifics from less-impacted systems, since predation pressure, a major selective force on behavioral traits, tends to decrease with increasing anthropogenic impact. To test this hypothesis, we collected peninsula crayfish (*Procambarus paeninsulanus*) from streams with either predominantly developed or forested watershed land use in and around Gainesville, Florida. We put crayfish through three separate laboratory assays to test their latency to emerge into a novel environment (i.e., boldness), foraging voracity, and activity levels. Our preliminary findings suggest crayfish from forested streams may have spent a greater proportion of time active, a lower proportion of time utilizing refuges, and foraged more rapidly than crayfish from urban streams, all of which contradict our hypothesis. Crayfish from both types of streams displayed similar boldness. We plan to expand upon this initial work by collecting and assaying crayfish from urban and forested ponds, and to determine the generality of our findings by conducting a similar experiment on another aquatic invertebrate species in a different order (Odonata). Changes in crayfish behavior that result from urbanization could be ecologically important, as crayfish often represent the dominant benthic consumers in aquatic habitats, and their behavior can have appreciable impacts on the functioning of these ecosystems.

INVURTS + NEON: temperature variability measures for rural and urban streams of the eastern U.S.

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Urban streams promote many ecosystem services in towns and cities, yet these sites are often overlooked in ecological research. This leaves a gap in our understanding about how important processes may vary across streams with different levels of urbanization. Thermal regimes play a central role in shaping stream communities and urbanization is known to impose characteristic thermal signatures on streams (urban stream syndrome). However, the ecological consequences of thermal variability remain poorly understood. Additionally, climate change is increasing variability and potentially reshaping thermal dynamics. Potential changes add urgency to the question of how important processes such as thermal variability may vary across streams with different levels of urbanization. The INVURTS project (Investigations with NEON: variability of macroinvertebrates versus urban and rural temperature dynamics) aims to represent temperature regimes and macroinvertebrate diversity at wadeable stream sites in National Ecological Observatory Network (NEON) domains (1-2, 5-8). NEON project sites are in basins with relatively little development (0-22%) or impervious surface (<1%), while half of INVURTS pilot-project basins are 20-100% developed and 30-40% of these INVURTS sites are urban (>10% impervious surface). This analysis used data from 18-22 sites with in-stream temperature loggers, plus model estimates of climate and land use variables for each basin above the site. We investigated urban-rural differences in annual, seasonal, and average daily temperature variability for specific seasons. We ranked sites and compared streams in basins with >10% versus <10% development using Mann-Whitney U tests. This study's rural NEON sites and rural and urban INVURTS sites represented similar ranges and median values for latitude, mean annual precipitation, mean annual temperature and average stream order. However, average daily temperature range in summer was marginally higher at less developed sites. Also, while annual temperature range (Aug-Feb) was not different between site groups, this pilot study found that more developed sites were significantly warmer in August and marginally warmer in February. The differences observed are consistent with predictions for urban streams. Ultimately, we will identify if measures of temperature dynamics at different time scales affect macroinvertebrate richness and community composition.

Does Raw Sewage Fuel Food Webs in Urban Streams with Failing Infrastructure?

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Urban waterways link human infrastructure and ecological systems, with direct consequences for water quality, biodiversity, and downstream resource management. As cities expand and infrastructure ages, urban streams increasingly receive excess

nutrients like carbon, nitrogen, and phosphorus from wastewater, stormwater runoff, and altered land use. Although urban streams can intercept some nutrient pollution through biological uptake and biogeochemical regulation, chronic nutrient enrichment can destabilize aquatic communities and impair ecosystem functions that regulate additional contaminants. A persistent challenge in urban stream science is understanding how human-derived nutrients move beyond the water column to become incorporated into stream ecosystems, and how land use and infrastructure govern these pathways. This research will examine how urban land use shapes the incorporation of sewage-derived nutrients into stream food webs and influences aquatic community structure. The study will be conducted in eight urban tributaries within the Ohio River Basin in Pittsburgh, Pennsylvania, USA, a region characterized by aging sewer infrastructure, buried headwaters, and highly modified hydrologic regimes. These streams are monitored under an EPA-mandated Consent Decree, providing high-resolution discharge data that will be integrated with biological and chemical measurements. This project will combine field sampling, stable isotope tracers, and watershed-scale geospatial analysis to address two objectives. First, it will quantify the incorporation of sewage-derived nutrients into urban stream food webs using a single, comparable functional feeding group of stream insects (shredders) present across all study sites. Carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) isotopes signatures from insect consumers and basal resources will be used to estimate the relative contribution of human versus natural energy sources, while insect abundance and diversity will be measured to assess community-level responses. Second, food web metrics will be evaluated in relation to watershed-scale indicators of the built environment, including impervious surface cover, housing age, hydrologic intensity, and sewer infrastructure density, to determine how urbanization governs the spatial distribution of enriched nutrients across ecosystems. By explicitly linking land use, infrastructure, and biological nutrient incorporation, this research will advance a mechanistic understanding of ecological exposure to urban nutrient pollution.

1. Chung AH, Elliott EM, Daniel JB, Thomas BF, River M, Nim CJ, et al. Riverine nitrogen source and yield in urban systems. *Front Ecol Environ*. 2023;21.
2. Warter MM, Tetzlaff D, Ring AM, Christopher J, Kissener HL, Funke E, et al. Environmental DNA, hydrochemistry and stable water isotopes as integrative tracers of urban ecohydrology. *Water Res*. 2024;250:121065.
3. Divers MT, Elliott EM, Bain DJ. Constraining nitrogen inputs to urban streams from leaking sewers using inverse modeling: implications for dissolved inorganic nitrogen retention in urban environments. *Environ Sci Technol*. 2013;47(4):1816–23.
4. Bergbusch NT, Hayes NM, Simpson GL, Leavitt PR. Unexpected shift from phytoplankton to periphyton in eutrophic streams due to wastewater influx. *Limnol Oceanogr*. 2021;66:1–17.
5. Grimm NB, Sheibley RW, Crenshaw CL, Dahm CN, Roach WJ. N retention and transformation in urban streams. *J N Am Benthol Soc*. 2005;24(3):626–42.
6. Sterling JL, Rosemond AD, Wenger SJ. Watershed urbanization affects macroinvertebrate community structure and reduces biomass through similar pathways in Piedmont streams, Georgia, USA. *Freshw Sci*. 2016;35(2):676–88.
7. Singer GA, Battin TJ. Anthropogenic subsidies alter stream consumer resource stoichiometry. *Ecol Appl*. 2007;17(2):376–89.
8. Kaushal SS, Belt KT. The urban watershed continuum: evolving spatial and temporal dimensions. *Urban Ecosyst*. 2012;15(2):409–35.
9. United States v. Allegheny County Sanitary Authority, Civ. No. 2:07-cv-00737-NR (W.D. Pa. May 14, 2020). Retrieved from <https://www.alcosan.org/docs/default-source/clean-water-plan-documents/2020-consent-decree.pdf>
10. Morrissey CA, Boldt A, Mapstone A, Newton J, Ormerod SJ. Stable isotopes as indicators of wastewater effects on the macroinvertebrates of urban rivers. *Hydrobiologia*. 2013;700:231–44.
11. Smucker NJ, Kuhn A, Cruz-Quinones CJ, Serbst JR, Lake JL. Stable isotopes of algae and macroinvertebrates in streams respond to watershed urbanization, inform management goals, and indicate food web relationships. *Ecol Indic*. 2018;90:295–304.
12. Ayayee PA, Wesner JS, Ouellette SP. Geography, taxonomy, and ecological guild: factors impacting freshwater macroinvertebrate gut microbiomes. *Ecol Evol*. 2022;12(12):e9663.
13. Huang J, Gergel SE. Effectiveness of landscape indicators for explaining the variability of benthic macroinvertebrates in urban streams. *J Environ Manage*. 2023;342:118130.
14. Forgrave R, Elliott EM, Bain DJ. Event scale hydrograph responses highlight impacts of widespread stream burial and urban infrastructure failures. *Hydrol Process*. 2022;36(5):e14584.
15. Whorley SB, Smucker NJ, Kuhn A, Wehr JD. Urbanisation alters fatty acids in stream food webs. *Freshw Biol*. 2020;64(5):984–96.

Restoration monitoring of large floodplain wetland

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Robert Porter Allen Natural Area (RPANA) is a 227-acre former farmstead in South Williamsport, Pennsylvania, with one of the largest remaining floodplain wetlands on the Susquehanna River's West Branch. This area underwent dredging in 2024 to restore previous hydrologic patterns. Pre-restoration sampling of vegetation and water depth provided baseline data of the biotic and abiotic conditions of the wetland on eight transects established throughout the area. During the summer of 2025, the four transects that were most impacted by restoration practices were resampled to assess recolonization of vegetation communities, measure water depth to track hydrologic changes, and monitor the early stages of restoration impact. In the autumn of 2025, soil nitrate-nitrogen, phosphorous, and pH levels were tested between two unrestored transects and 3 restored transects to provide initial and comparative soil data as restoration progresses. Transects ranged from 144 meters to 405 meters, with sampling occurring every 40 meters for plant species and every 50 meters for soil. A total of 115 vascular plant species were identified, with 71% (82 spp.) native and 29% (33 spp.) introduced. Sixty-six percent of plots showed increased species richness one year after restoration. A Wilcoxon signed-rank test indicated a significant positive change in species richness following restoration ($p < 0.05$), with native, wetland-obligate species dominating. Some plots saw a recolonization of introduced species. Fourteen plots achieved a higher average Floristic Quality Index (FQI) post-restoration, 11 plots with a lower average FQI, and four plots maintained an FQI of zero. The seven plots with the highest FQI scores experienced an increase in water depth post-restoration. A Mann-Whitney U-Test revealed no significant difference in soil pH, phosphorous, or nitrate-nitrogen between restored and unrestored

sites. Monitoring efforts of vegetation, soil, and water actively inform management decisions and will become more important with time, contributing to broader knowledge of post-restoration wetland development. The increasing species richness, native and high FQI species, and water depth are promising signs of future success for this wetland restoration project, and the soil biogeochemical processes should enhance as organic matter accumulates and nutrient cycling improves.

Evaluating the effects of hydrologic variability on the population dynamics of a burrowing crayfish

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The Everglades crayfish, *Procambarus alleni* is a short-lived crayfish that lives in seasonal subtropical wetlands in Florida and is an important diet item for nesting wading birds. The hydroperiod limitations of the populations of *P. alleni* and the mechanisms by which hydrologic variation influence crayfish production are not well understood. To address questions regarding population limitation by hydrologic stress, we studied Everglades crayfish in seasonal wetlands for 3 years. We selected and sampled six marl prairie sites (1 ha each) in two hydroperiod categories; three shorter hydroperiod (3-5 mo.) and three longer hydroperiod (7-9 mo.) sites. Below ground water depths appeared to recede a meter or more below ground in the shorter hydroperiod sites in some years. Aboveground crayfish densities and biomass were quantified with replicated 1-m² throw traps multiple times annually from 2023-2025. During dry seasons, active crayfish burrows were tagged and monitored for two years to calculate burrow densities and emergence success in the wet season. We conducted linear mixed-effects models to identify effects of hydrologic variables on aboveground densities during 5 of the sample seasons when most sites were flooded to > 5 cm. Aboveground crayfish densities did not generally differ between regions during flooded seasons; densities in the longer hydroperiod sites were only significantly higher in October 2025 following 9 months of dry conditions at some short hydroperiod sites. The longer hydroperiod sites had significantly more aboveground biomass in two seasons when some short hydroperiod sites were too dry to sample. Mean biomass did not differ in three seasons when all sites were flooded ($P > 0.1$). Densities of active burrows varied between 2024 and 2025 in a way that suggested adult crayfish redistributed to lower elevation sites in anticipation of a longer 2025 dry season, but there were differences in the emergence rates between regions or years. Crayfish densities were both negatively correlated with long dry periods (days dry) and with deeper average depths over the past 6 months. The results suggest shallow conditions are important for promoting high densities of juveniles, but prolonged dry conditions (> 7 months) limit *P. alleni* densities.

Active vegetation management increases metaphytic mats and shifts aquatic animal communities in degraded wetlands

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The pre-drainage Everglades (FL, USA) was a shallow slowly flowing wetland with a "corrugated ridge and slough" habitat mosaic. Drainage and compartmentalization in the 20th century reduced water levels and degraded the spatial patterning of the landscape. In some places, high densities of sawgrass (*Cladium jamaicense*) and cattails (*Typha* spp.) from the ridges have filled the historically deeper lily and spike rush sloughs and the microtopographic relief has been lost. Over three years we quantified densities of small fishes and large macroinvertebrates with 1-m² enclosure traps in 12 degraded (sawgrass-filled) sloughs and 12 paired (adjacent) remnant sloughs within a 14 km² area of contiguous wetland where most sloughs had been lost to sawgrass encroachment. We found higher densities of fish in the remnant sloughs than the degraded sloughs, but higher densities of crayfish and seasonally higher densities of insects in the degraded sloughs. In September 2023, sawgrass and cattails in 4 of the 12 degraded sloughs were treated with herbicide to re-establish slough-like habitats. We conducted Before-After Control-Impact analyses on the combined live stem density of sawgrass and cattails and the total biovolume of metaphytic microbial mats. We found that six months after spraying, the sawgrass and cattails in the impacted sites were dead. We also conducted similarity-based BACI-type analyses with PERMANOVA of the aquatic animal community similarity looking for compositional changes away from the degraded conditions over the four sample events post-spraying. We found no immediate treatment effect in the year following the application (2024), but observed some significant shifts in composition in 2025. Our results suggest the active marsh improvement efforts are having an effect on the aquatic animal community composition that should become more evident as slough vegetation grows and the sawgrass detritus fully decomposes.